Instructions

Load Commutated Inverter AC Adjustable-Speed Drive Static Power Conversion and Control Equipment

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CHAPTER 1

1-1. BASIC DRIVE CONTROL

The Load Commutated Inverter (LCI) is a static, adjustable frequency drive system that controls a synchronous motor from near zero to rated speed. The basic system consists of a line commutated, phase controlled thyristor converter that feeds a load commutated thyristor converter through a dc link reactor. Figure 1-1 is a simplified one-line diagram of a single channel LCI. The transformer provides isolation from the ac system bus and provides the correct voltage at the terminals of the rectifier. Also, the internal impedance of the transformer limits the magnitude of any downstream bus faults.

The rectifier is a thyristor bridge whose gating is controlled to produce a variable dc voltage at its output. The output of the rectifier is fed through the dc link reactor, whose function is to smooth the current and keep it continuous over the operating range of the system.

The dc link reactor output is then fed into the inverter bridge, which produces variable frequency ac at the stator terminals of the synchronous motor. The inverter bridge and the rectifier bridge use the same power hardware and are controlled by microprocessor-based electronics. Although the bridge are labeled "rectifier" and "inverter," it is possible for their roles and hence power flow to reverse. In this, case the motor would be braked by pumping its energy back into the ac line. A more general nomenclature is to call the line side bridge the source converter, and to call the motor side bridge the load converter.

The synchronous motor field is usually excited by a brushless exciter coupled to the motor shaft. The brushless exciter is a wound rotor induction machine whose rotor voltage is rectified to supply field current to the synchronous motor. The exciter stator phase rotation is connected so that its rotor voltage and thus the synchronous motor field excitation varies with motor speed.

A more detailed diagram of a single channel LCI is shown in Figure 1-2. As shown in this figure, the electronic control receives the following signal inputs:

1. Attenuated line and load bus voltage signals.
2. Attenuated line and load current signals from current transformers.
3. Tachometer speed feedback
4. Communication with a master controller via a data link (when used).

The attenuated bus voltage signals are used by the control to synchronize thyristor firing with the line and load voltages, and to provide voltage sensing for overvoltage protection. The attenuated current signals are used for regulator current feedback, electronic overcurrent detection, and software-implemented fault detection.

The electronic control evaluates the data link commands, process inputs, and internal status signals to determine whether the drive should be in a stopped, started, alarmed, or faulted condition. If in a started condition, the electronics provides gate signals to

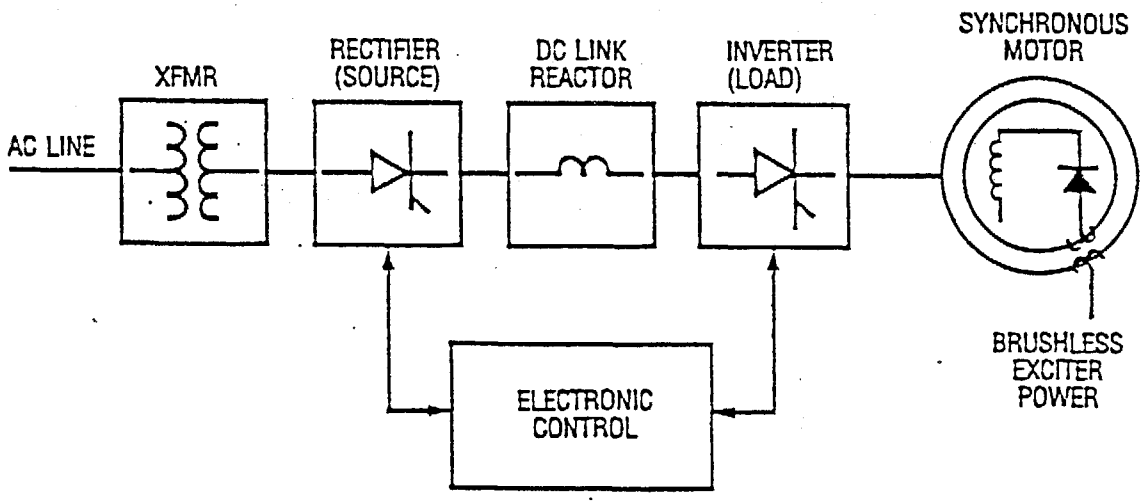


Figure 1-1. Simplified LCI One-Line Diagram

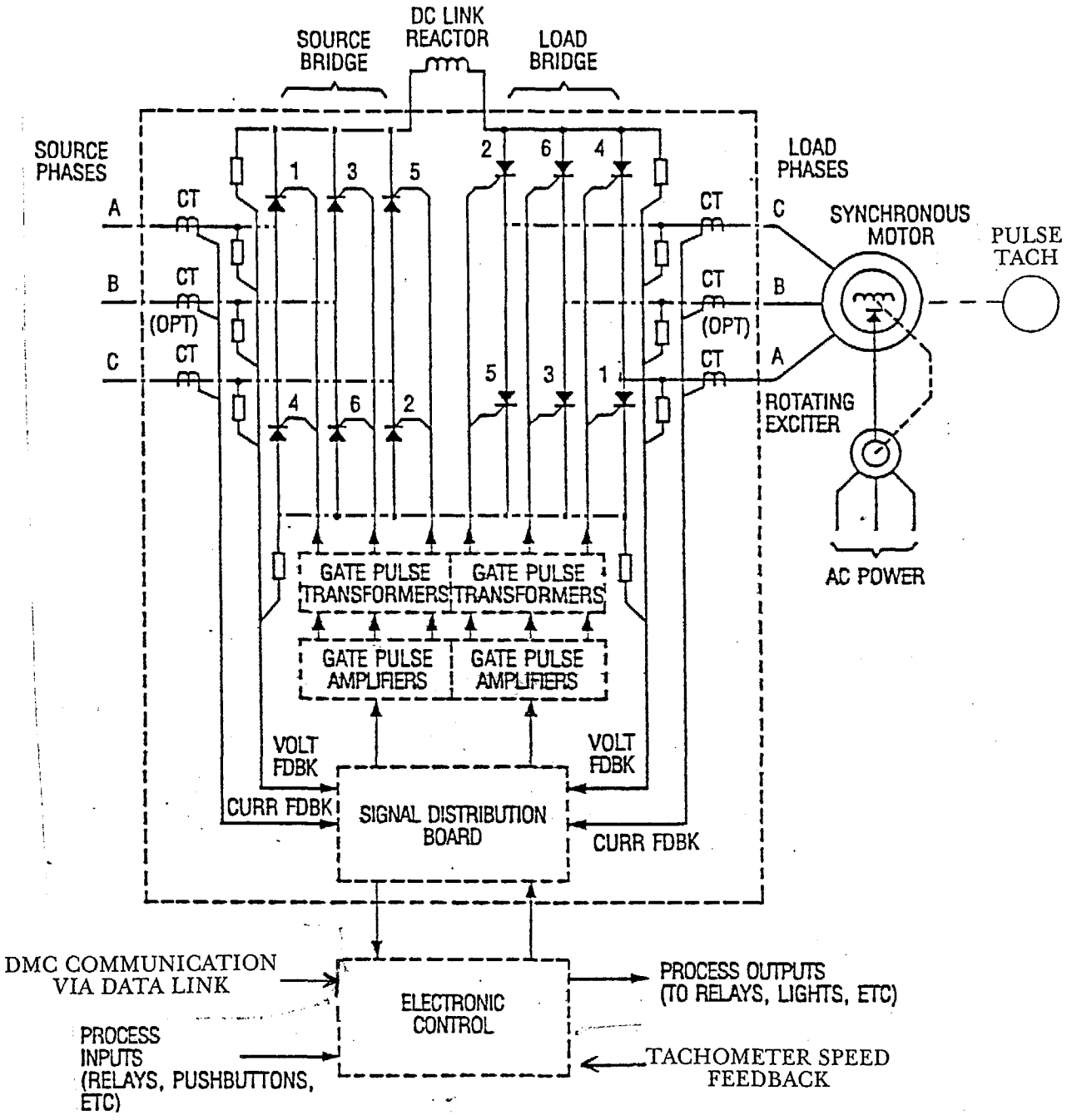


Figure 1-2. Single Channel LCI Drive

control the thyristor bridges. These signals are low level and are conditioned in the power bridge circuitry to provide the necessary isolation and power level.

The power bridges are of the six pulse, double way types and the thyristor bridge legs "fire" in the order that they are numbered in Figure 1-2 and as shown in Figure 1-3.

The source converter current is successfully transferred from one leg to the next by the ac line voltages.Similarly, the inverter current is successfully transferred by the motor stator voltages.

The process of switching motor/inverter current from one leg to the next is illustrated in Figure 1-4. The principles illustrated here apply to a rectifier bridge as well as to an inverter bridge. This phase-controlled switching is accomplished by using the following two thyristor characteristics: (1) When the voltage across the thyristor is positive, it can be triggered into con-

duction, and (2) it will not permit current flow in the reverse direction. Thus, in an alternating voltage circuit, thyristor conduction will cease and reverse voltage will begin to appear when the current becomes zero.

Current transfer must be completed before voltage cross-over, with a margin angle. This angle must be long enough to allow the previously conducting leg thyristors to recover to their blocking state before reverse voltage is applied. This is why the fundamental component of current must lead the voltage, from an inverter/motor viewpoint. From the rectifier/source viewpoint, the fundamental component of current will always lag the voltage. For successful commutation, the firing angle $\alpha = 180° - \beta = 180° - \mu - \gamma$ must always be less than 180°; a practical limit is 155°. Correspondingly, a practical minimum value for β, for the inverter bridge, is 25°. The motor power-factor angle is always less than, but cannot become zero.

The LCI control system must conform to the characteristics of the synchronous motor operating at leading power factor. A phasor diagram for a synchronous machine operating at leading power factor is shown in Figure 1-5.

The motor voltage characteristic is mainly a function of the rotor field-excitation, E_f, and the de-magnetizing action of direct-axis current. This produces the $I_D X_{AD}$ voltage in opposition to the voltage produced by field excitation.

Note that an increase in stator current results in higher direct - axis current, which increases $I_D X_{AD}$. This in turn decreases the machine voltage E_G available for commutation, thereby increasing displacement angle Θ. This increase of angle Θ further increases stator current, etc., until equilibrium is reached at a new operating point.

Typical voltage, current, and torque characteristics are shown in Figure 1-6.

In a typical system, a brushless exciter supplies the motor field excitation. The exciter output voltage varies with motor speed. Field current is exciter output voltage divided by field resistance. Neglecting saturation, field flux is proportional to field current, and flux and volts/hertz are considered synonymous.

The interrelationship of voltage, current, and torque is complex. Typical characteristics for an LCI drive system are depicted in Figure 1-7.

The control functions, which conform to the drive characteristics discussed here, will be further described under Control Electronics.

1-2. POWER CONVERTERS

1-2.1. Basic Operation . When operating in any mode the electronic control must synchronize firing of the source and load converters to the ac line and motor bus voltages, respectively. The primary feedback for accomplishing this is the attenuated bus-to-ground signals for both converters. Attenuated bus-to-ground voltages are brought into the electronic control and combined to produce line-to-line analog voltages for both converters. These line-to-line voltages are then integrated to obtain flux signals. The zero crossings of these signals are used in the synchronization of the phase locked loop for the firing control of both source and load converters. At low speed, before the phase locked loop is effective on the load side, the zero crossing marks are used as a timing reference for firing in force - commutated operation.

The electronic control is designed to operate with two types of speed feedback. The first method employs the zero crossings/phase lock loop feedback discussed in

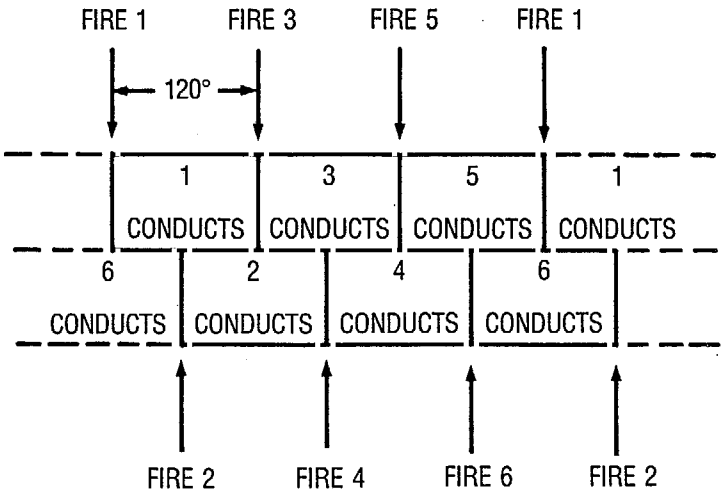


Figure 1-3. Thyristor Bridge Leg Conduction Sequence

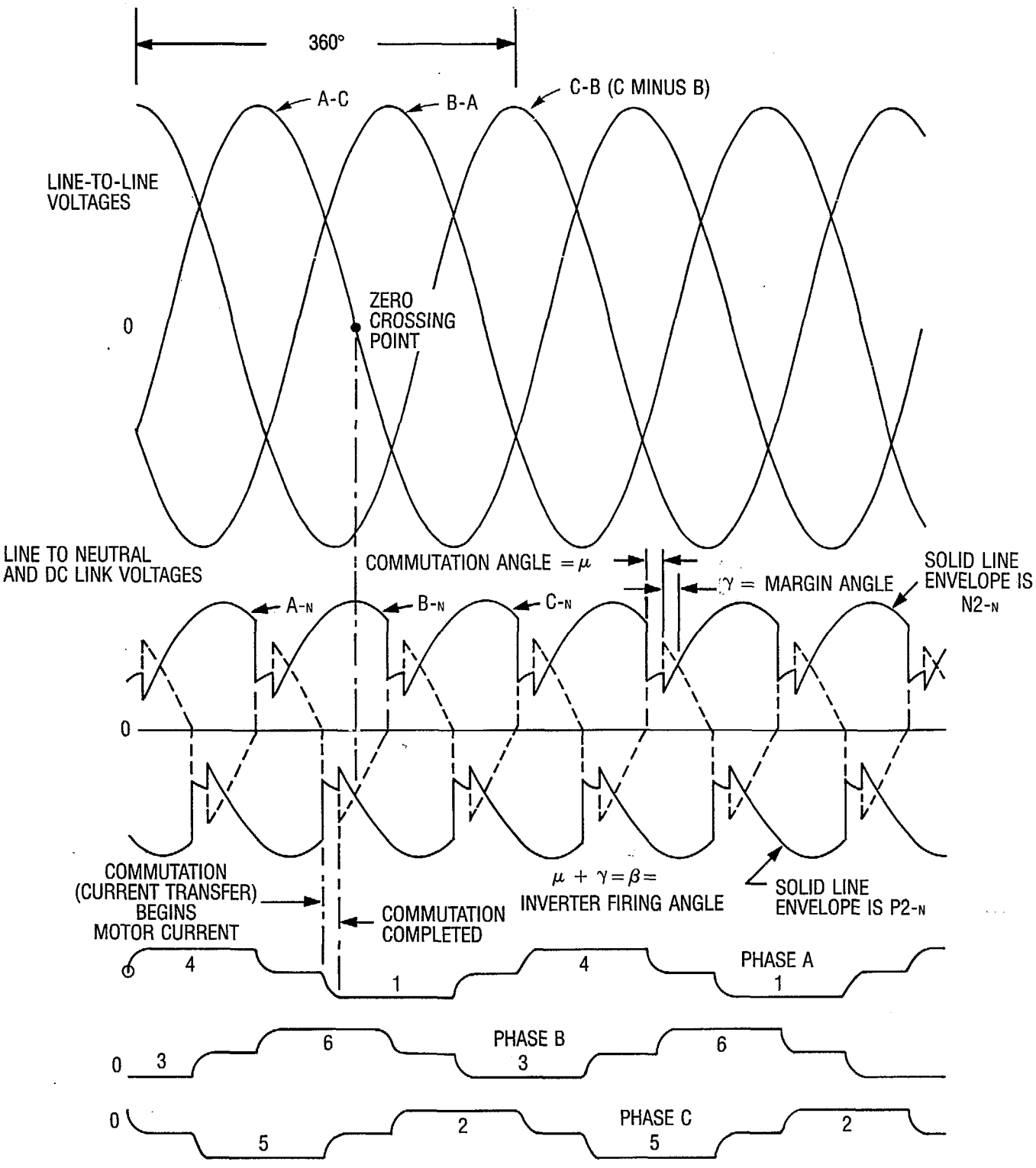
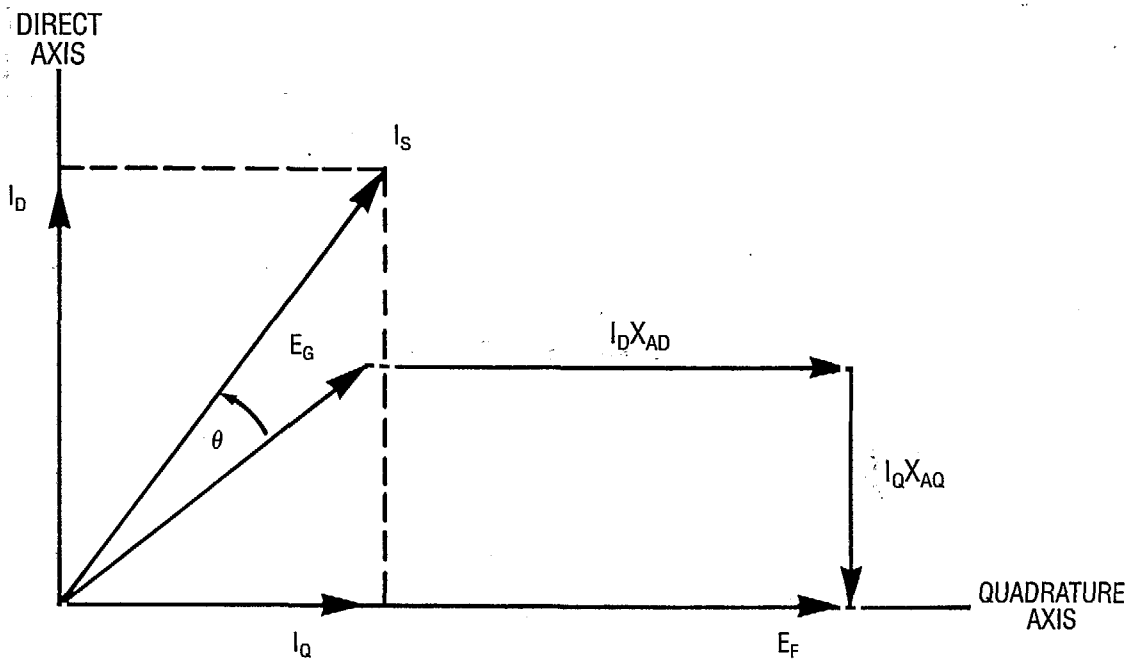


Figure 1-4. Load Converter Voltage and Current



Where: I_D = Direct axis current
I_Q = Quadrature axis current
I_S = Stator current
E_G = Generated voltage, CEMF
X_{AD} = Armature reactance, Quadrature axis
E_F = Rotor fields – excitation voltage
Θ = Displacement angle

Figure 1-5. LCI-Driven Synchronous Motor Diagram; Leading Power Factor

the previous paragraph. The second method relies upon a pulse tachometer to derive it's speed feedback.

The source side converter for the LCI always operates line-commutated; i.e., the ac line provides the means for transferring conduction from one thyristor to the next. The load side converter may operate either load (self) commutated or force-commutated, depending on motor speed and flux level.

As the synchronous motor rotor (field) rotates, the near-sinusoidally shaped field flux cuts the stator windings, producing a set of three sinusoidal voltages in the stator. These sinusoidal voltages are angularly displaced by 120 electrical degrees. The magnitude of this counter emf is proportional to speed and field strength. At low speeds, the induced cemf is insufficient to commutate the thyristors in the load side converter. In this mode, the control must operate in a forced commutated mode.

There are seven possible modes of operation of the control. One is a "Non-Running" mode. Three are concerned with PLL operation and three more with position mode operation. The load side mode selected is one of the following:

- Idle mode (mode 0).
- Segment fire mode (mode 1).
- Forced commutated, closed PLL mode (mode 2).
- PLL self commutated mode (mode 3).
- Forced commutated position mode (mode 7).
- Forced commutated position mode - low speed (mode 8).
- Self commutated position mode (mode 9).

Idle mode. This is the mode that executes when the drive is not running. This is the only interrupt that executes initially following a system reset. The other

interrupts are enabled from this module as a function of presence of source voltage and background sequencing inputs. When the drive receives a "RUN" command, idle mode is exited to one of the other modes, depending on drive operating conditions. If the drive is stationary when this occurs, then the field is energized and the stator pole positions determined before exiting to Segment fire mode.

Segment fire mode. This firing mode is used at motor speeds too low to lock the PLL. After exiting idle mode, the load cell pairs are commutated at fixed time intervals in the motor direction of rotation desired. This continues until it builds sufficient flux levels to begin firing based on load flux crossovers. Motor pole positions are determined within 30 degrees by a combination of observing the negative to positive transitions on the motor flux square waves and reading the flux wave magnitude through the A/D converter. Commutation times are then set by motor pole positions so determined. Operation is transferred to the next mode when the PLL locks.

Forced commutated, closed PLL mode. When the PLL has locked but the motor voltage is too low for self commutation, the load operation switches to this mode. Current is taken to zero at each load commutation but the load angle of firing can be varied linearly. The load power factor can be better than the self commutation since no margin for commutation is necessary.

PLL self commutated mode. This is the normal operating mode on the load side. Self commutation is usually entered at about 6 Hz. The basic strategy is to fire as late as possible maximizing the motor power factor. Commutation margin is calculated from the last flux reading, load current, and commutating impedance.

Forced commutated position mode. This is a position mode requiring the pulse tachometer for operation. The position counter replaces the PLL time counter and the tach pulse train replaces the PLL clock. Fire counter loads are in terms of mechanical degrees.

Forced commutated position mode - Low speed. This is a position mode requiring the pulse tachometer for operation. It differs from the mode above in that the algorithm is called at a fixed 10 millisecond rate and selects cell pairs from a table based on reading present position. It is used to get the motor rolling. It normally operates at speed below 2 Hz.

Self commutated position mode. This mode also requires a tachometer. It is similar in operation to mode 7 described above except that the source force commutations are not necessary.

Three distinct load side control strategies are possible using combinations of the modes defined above. The standard control strategy (no tachometer) uses the following:

Mode 0, 1, 2, 3

When a tachometer is available, the standard position strategy is to use the following modes:

Mode 0, 8, 7, 3

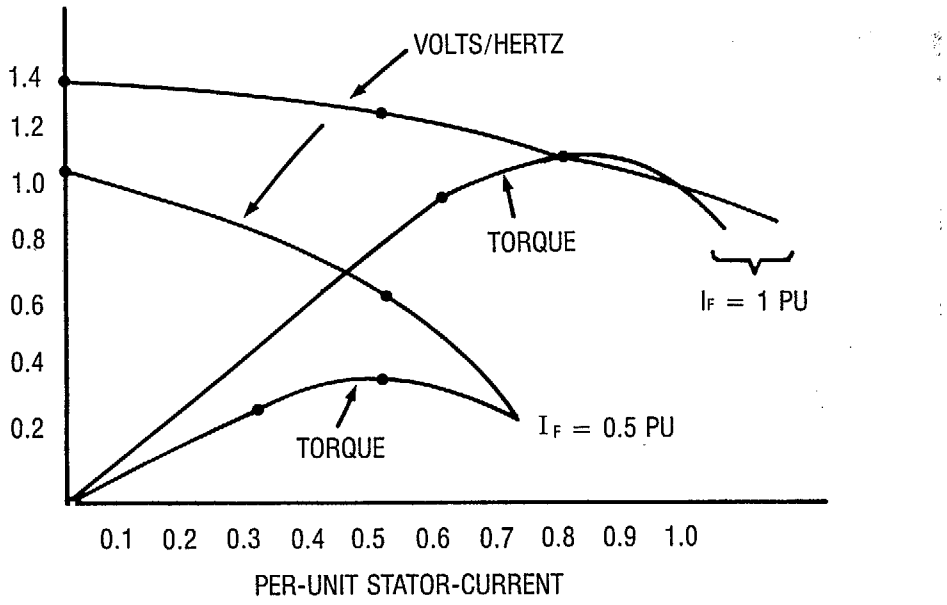


Figure 1-6. Flux and Torque as a Function of Stator Current

Using the PLL mode at higher speeds has the advantage of improved load side firing resolution, tighter margin control, and improved steady-state power factor.

In the self commutated mode, which is the principle operating mode, the motor must be operated at a leading power factor in order to be able to ensure commutation of the load converter. The electronic control acts to keep the motor power factor, and therefore torque per ampere, as high as possible. This is accomplished by firing the load converter as close to inversion limit as possible while maintaining sufficient margin for successful commutation of current from one cell to the next. The commutation time required is a function of motor current, motor (commutating) inductance, and the voltage difference between the lines involved in the attempted commutation.

For a given load current and motor inductance, a corresponding amount of volt-seconds is required for a successful commutation. The control reads the peak volt-seconds of the integrated line-to-line motor voltages and the motor current. The motor commutating inductance is a constant stored in the microprocessor system memory. The microprocessor calculates calculates the current and inductance to determine the amount of commutation volt-seconds required. Using this value of voltseconds and the peak voltseconds of the previous flux wave, the latest possible time to fire is calculated to give a specified margin after commutation is completed. The relationship of the system voltages, currents, and flux waves are illustrated in Figure 1-9.

The commutating "notch" identified in the A-C line-to-line voltage is equal in amplitude to the simultaneous commutating "bump" on the B-C voltage. The corresponding notch in the A-B voltage (A and B are the two lines commutating together at this instant) is twice this amplitude; the notch area is twice the commutating inductance per phase times the current. The voltage at the commutating point, where the lines are temporarily connected by the thyristor legs, is practically zero during commutation; the line-to-line voltage is only the forward voltage drops of the conducting thyristor legs.

At high load on the motor, the apparent power factor "seen" by the power source increases. This is because the source converter firing advances (is reduced) to obtain more current. The harmonics in the current and the resultant harmonics in the voltage decreases, due to commutation "notching".

The fundamental control strategy is: In order to increase motor current in response to a load torque

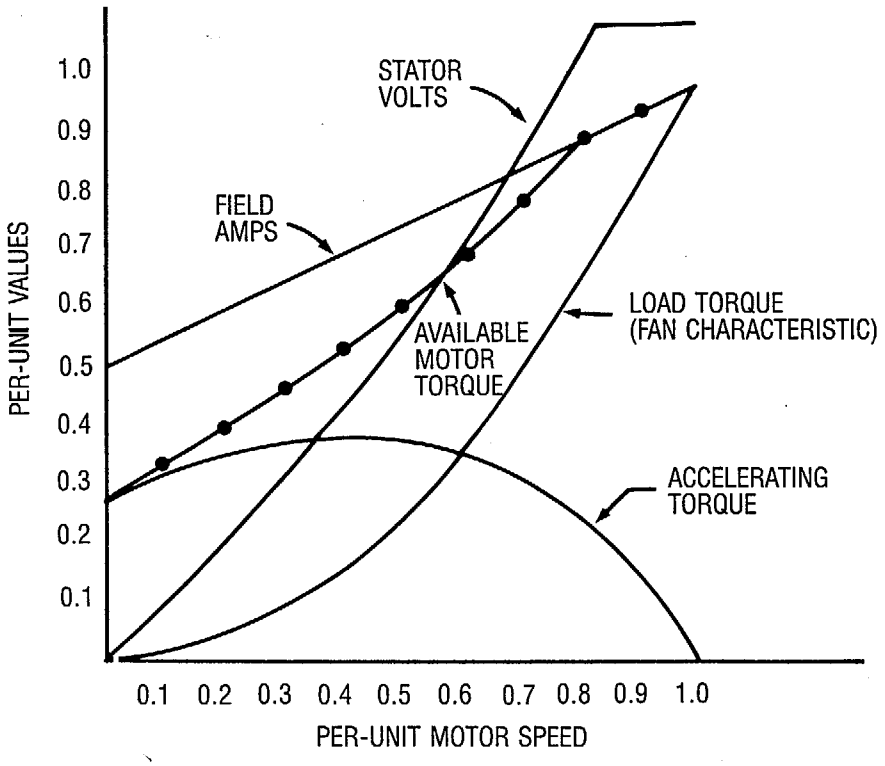
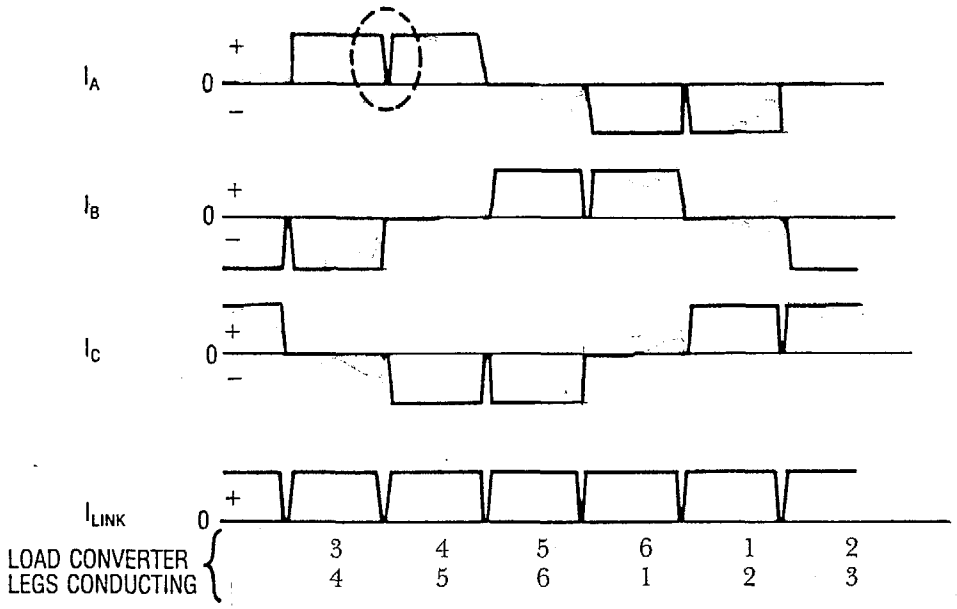


Figure 1-7. Typical LCI Drive Excitation, Voltage, and Torque Variations with Speed



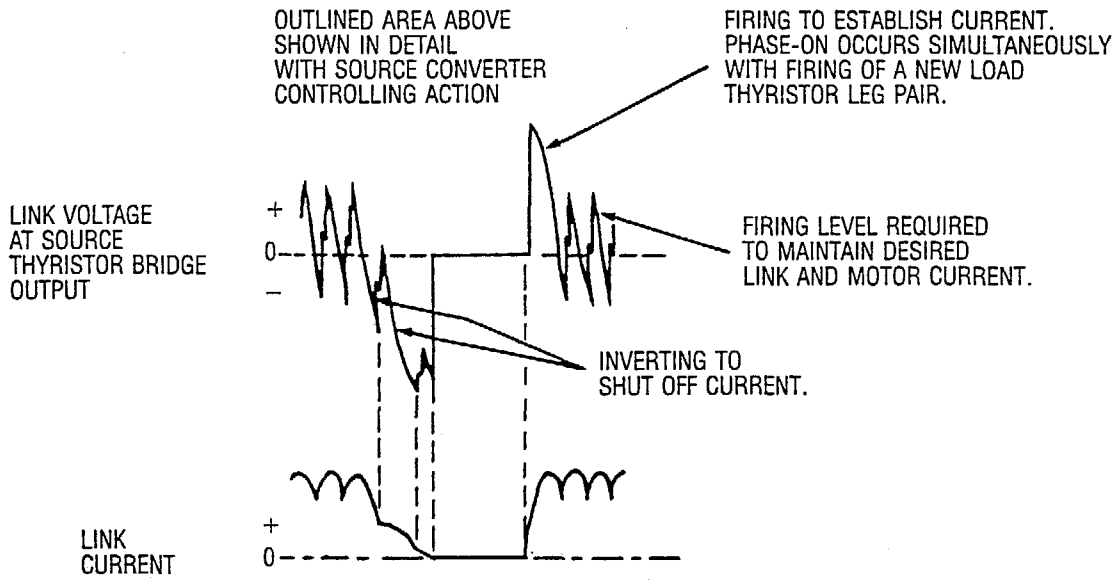


Figure 1-8. Forced Commutation Firing Mode

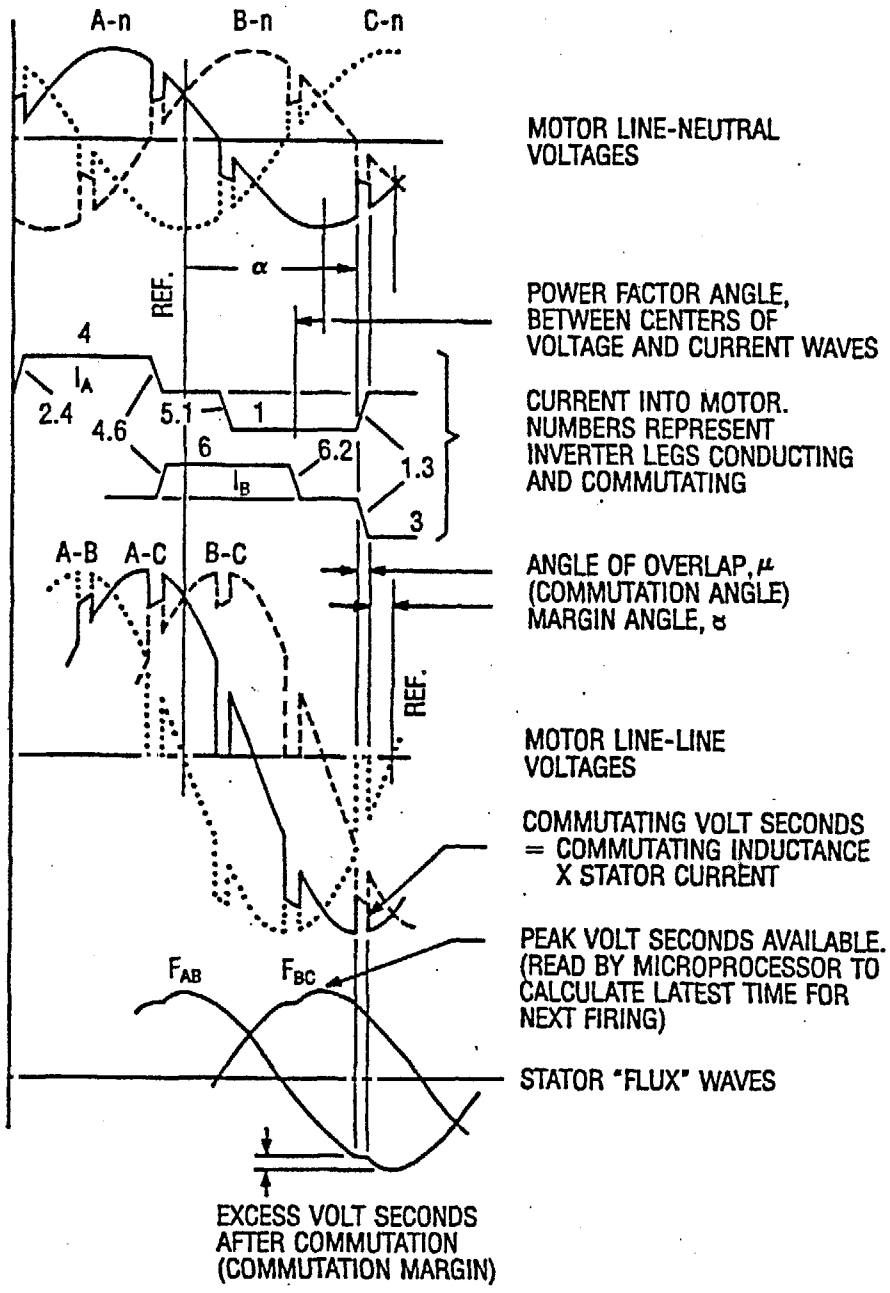


Figure 1-9. Motor Voltage and Current in the Self-Commutated Mode

increase, the dc link voltage, on the source side, is increased, thereby raising motor current so as to keep motor speed constant.

1-2.2. Reversing Operation. The drive has the capability to operate in either direction of rotation as a function of the polarity of the drive speed reference. The tachometer for position mode is required for reverse operation.

1-3. TWELVE-PULSE OPERATION

Twelve-pulse operation is like two identical, separate, six-pulse drives operating from a common source at the same current and firing angle, with the firing reference angles shifted 30 degrees by the isolating transformers.

The two motors are combined into one frame. This reduces the motor cost, including installation, and reduces the torque pulsation amplitude while raising the torque pulsation frequency. This is achieved by separating the stator winding into two identical windings, but isolated and phase-shifted 30 degrees.

This construction, utilizing a common magnetic frame, including a common field, causes the load-side converter voltage to be equal in amplitude and frequency between the two drive channels. The transformer design makes the source-side converter voltage equal, but also 30 degrees apart between the two channels. It is not necessary that source-side and load-side voltages be phase-shifted. The motor windings are shifted to obtain smoother torque for equal current. The transformer windings are shifted to reduce the harmonic distortion on the power system and to raise the harmonic frequencies.

Inter-channel communication allows one channel to be master and the other the slave. This communication allows the two motor winding currents to be balanced. Thus the channels deliver equal power, take equal current, and fire at the same relative firing angle. This best utilizes the motor and drive capabilities and also minimizes the harmonic distortion on the power system by canceling the first two harmonics present in each six-pulse channel cancel (5th and 7th harmonics). Other, higher-order harmonics also cancel, but their amplitudes are much smaller.

A twelve-pulse system also presents the capability of shutting down one channel for maintenance while the motor continues to run on the other channel (with reduced torque and usually reduced speed range). When the out-of-service channel is ready for operation, it may be returned to service without interrupting the drive system.

1-4. DMC MASTER DATA LINK (IF USED)

The LCI drive is designed to communicate with a master controller via a data link. Both discrete logic control and integer variables can be passed to or from the drive. In this mode of operation, several drives can be co-ordinated to perform a given task. See Figure 1-10.

Communication to the drive is through a dual ported ram on the HXMA printed wire board. One side of the ram is connected to the internal L-bus of the LCI drive and the other to a F-BUS. The F-BUS is then interfaced to the C-bus (HXRA). A F-BUS is basically a parallel bus with 8 address bits (256 ADDRESSES MAX) and an 8 bit data bus for data transfer.

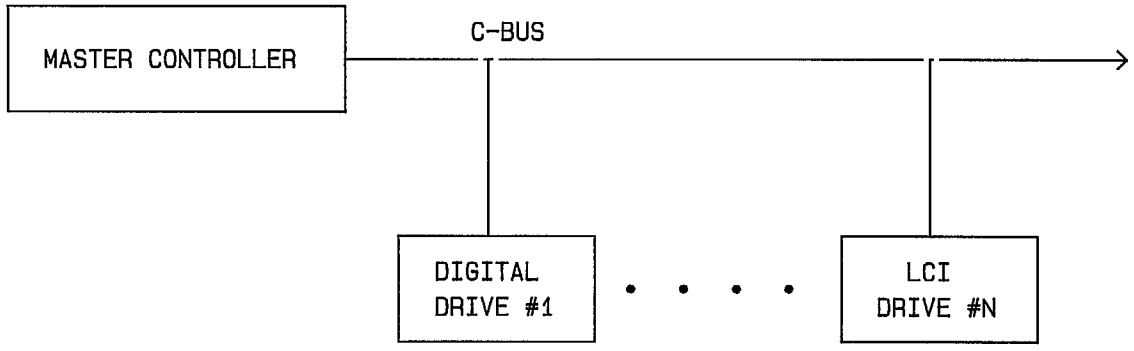


Figure 1-10. Master Controller/Drive Interface

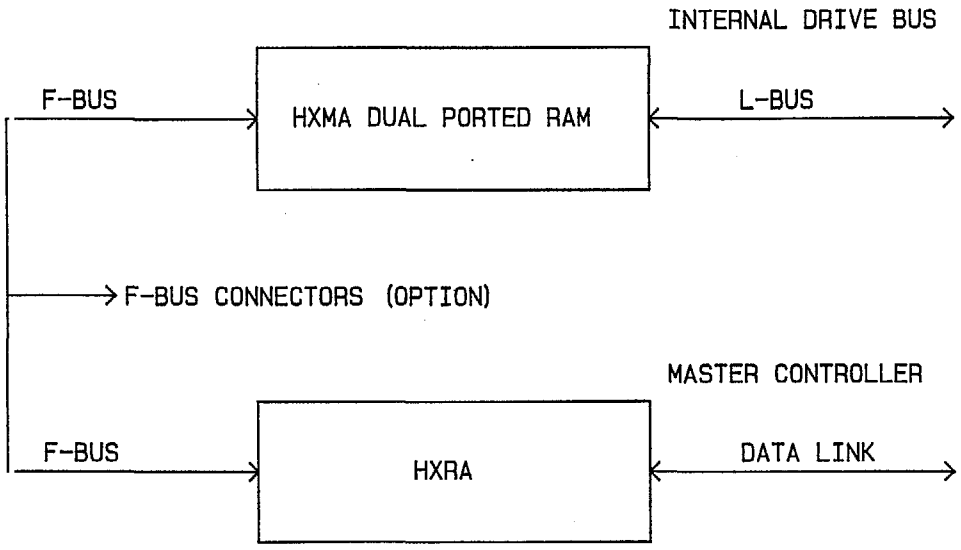


Figure 1-11. HXMA/HXRA Communication Link

CHAPTER 2

POWER EQUIPMENT

2-1. EQUIPMENT OPTIONS

The power converters may be either 2 kV or 4 kV medium voltage bridge construction. The 2 kV MVB uses 3 SCRs in series per leg, and the 4 kV MVB uses 5 SCRs in series per leg.

2-1.1. 2 kV Medium Voltage Bridge . Figure 2-1 shows the arrangement of the power converter silicon controlled rectifier (SCR) stack assemblies. This figure and the elementary diagram show the major power components.

SCR's are stacked horizontally, three in series between four heat sinks. The 18 SCR's between the P-bus (P1, P2) and the ac lines, source and load, are on the left side in the bridge compartment; also CT1, CT3, FU1-4 and FU-6.

The N-bus SCR's are on the right side of the bridge compartment; also CT4, CT6, FU-5, and FU7-10.

The blower compartment contains the blowers, source and load ac and dc filters, MOVs, CT1A, CT3A, CT4A, and CT6A. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay.

2-1.2. 4KV Medium Voltage Bridge . The arrangement of the power converter SCR stack assemblies is shown in Figure 2-2. This figure and the elementary diagram sheets AC, AD, and AE show the major power components.

Thyristors (SCR's) are stacked horizontally, three in series, between four heat sinks. All 30 SCRs between the P-bus (P1, P2) and the ac lines, source and load, are in the left bridge compartment; also CT1, CT3, FU1-4 and FU-6.

All N-bus SCRs are in the right bridge compartment; also CT4, CT6, FU5 and FU7-10.

Leg modules A are on the left side in each compartment; these are the master stacks containing a full complement of cards in the stack (module) insulating box. Leg modules B are the slave modules, and contain only the XPTN card below the cells. The left blower compartment contains blower numbers 1 and 2, dc filters, MOVs and terminal boards. Also, the hinged base in this compartment contains blower starter numbers 1 and 2, the 70V gating power supply, and thyristor fan-out PWB XTFH.

The right blower compartment contains blower numbers 3 and 4, load ac and dc filters. The hinged base in this compartment contains blower starters 3 and 4, and the bottom blower selection relays.

2-2. THYRISTOR MODULES, MEDIUM VOLTAGE BRIDGE (MVB)

Each power converter leg module consists of either two or three thyristors arranged in series in a heatsink assembly.

The master leg module is mounted in an insulated box with associated printed wiring boards in front as shown in Figure 2-1. The master module always has three SCRs.

An R-C snubber network is also contained in the leg module box, above the cells.

Each cell stack assembly consists of either two or three cells sandwiched between heatsinks as shown in Figure 2-4. The stack assemblies are oriented horizontally in the module as shown. Air flow over the heatsinks is from front to back of the bridge enclosure and the cell modules.

The cells are selected, matched, and assembled at the factory under the correct amount of pressure. For these reasons, it is recommended that the entire stack be replaced if a cell failure occurs. The defective stack may then be sent to the factory for repair.

Each cell in the module is paralleled by a voltage-equalizing resistor and by a neon lamp and dropping resistor network to indicate when voltage is present across the cell. These lights, visible through the front doors of the equipment, can be used to detect a shorted cell. The cell light network is mounted on a small card (NTDA) located above each cell on the front of the thyristor modules. Figure 2-3 shows the location of these cards. The NTDA card also contains thyristor gate interface circuitry, as shown in Figure 2-5. Thyristor gating signals are routed from the gating transformers through the NTDA cards to the thyristor gates.

2-3. GATE DRIVER TRANSFORMERS

The cell gates are driven through pulse transformers mounted on the front of the module, as shown in Figure 2-3. The pulse transformers for each module are wired so that all their primary windings are in series. The primary winding is simply a conductor through the center hole of the transformer. Current pulses are applied in this conductor by the gate driver circuitry. This circuit ensures that all thyristors in the string are gated simultaneously and it further provides

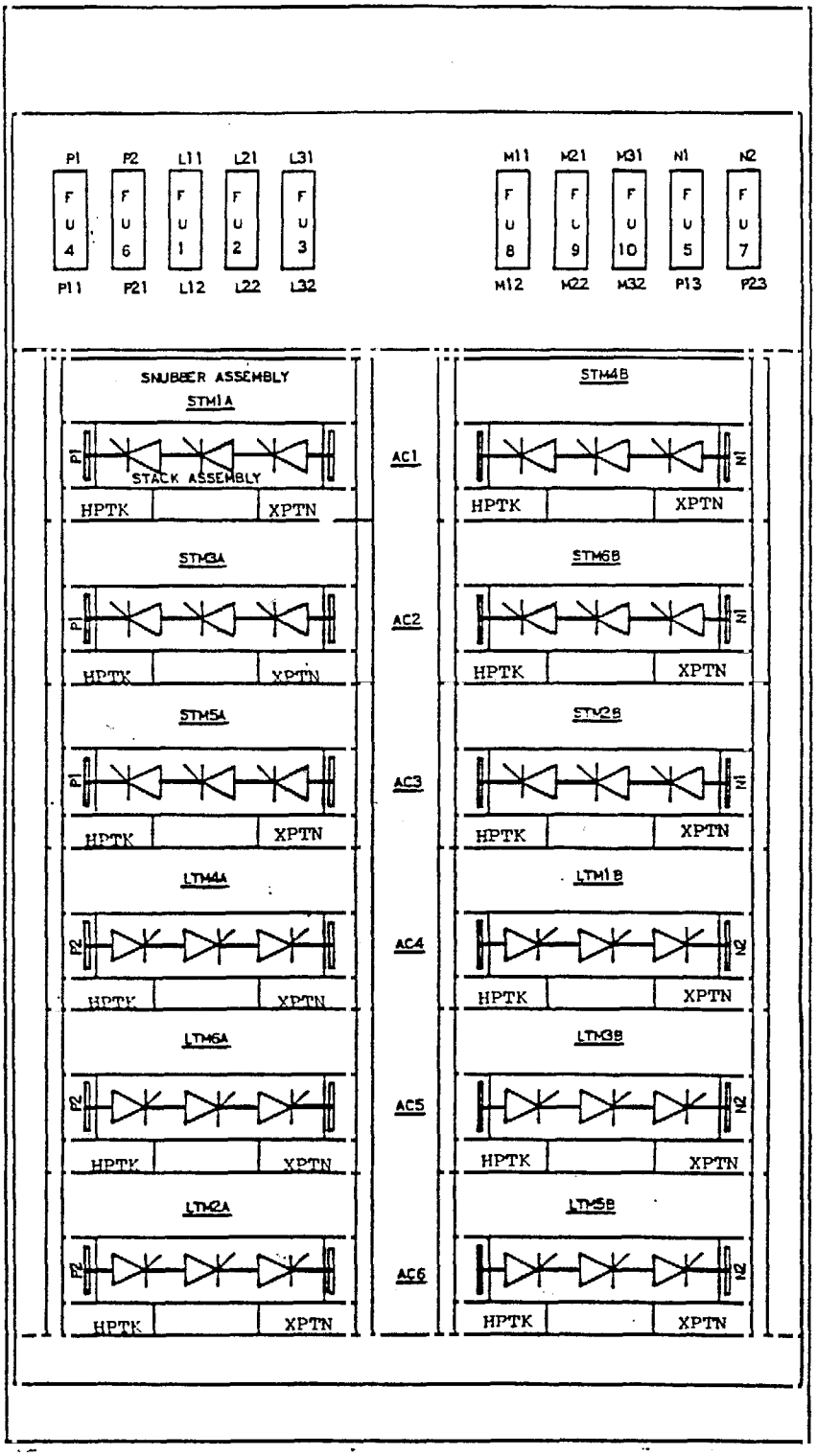


Figure 2-1. Medium Voltage Bridge Assembly, 2 kV

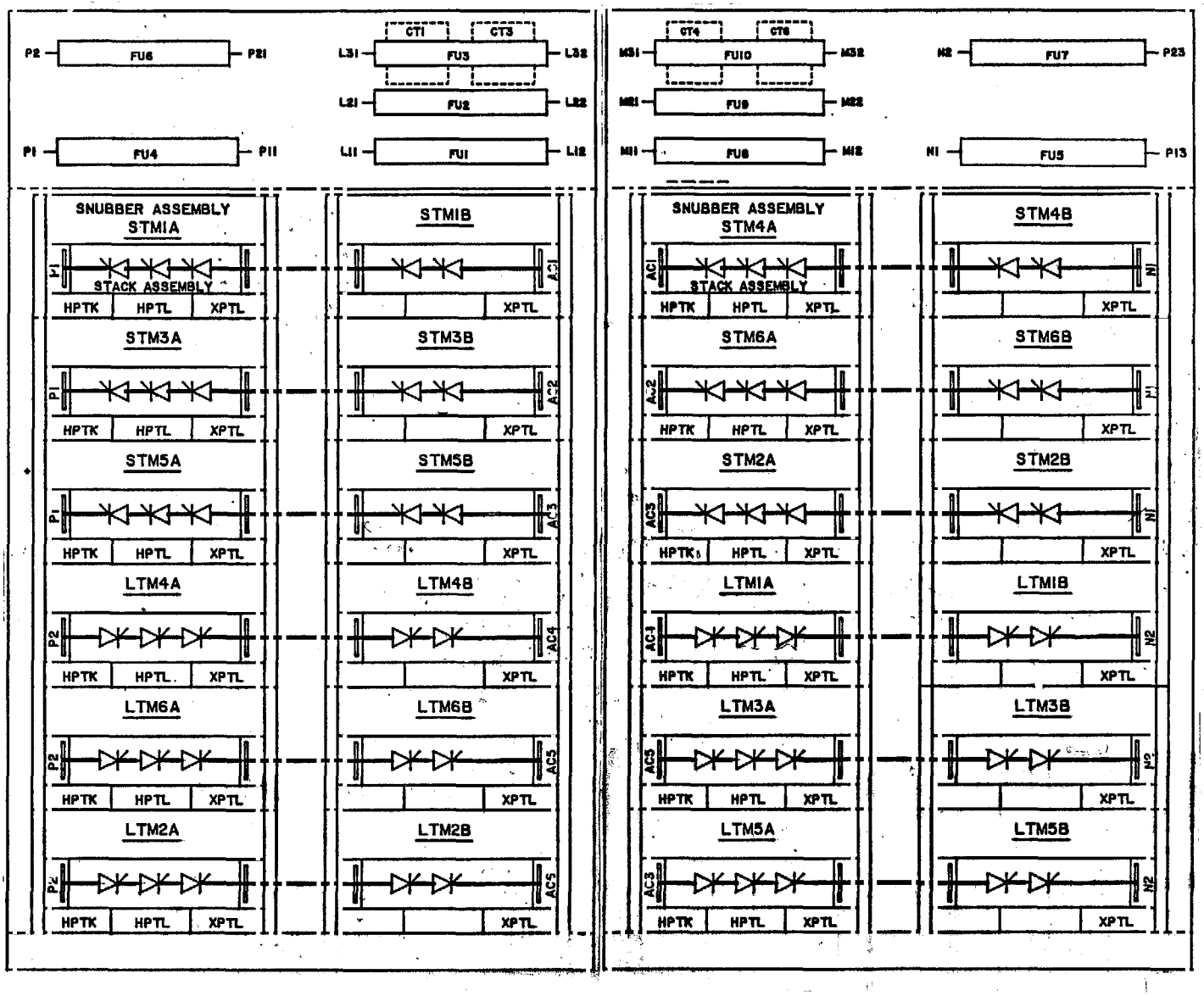


Figure 2-2. Medium Voltage Bridge Assembly, 4 kV

voltage isolation between the thyristor gates and the electronic gate pulse generation circuitry.

2-4. GATE DRIVER CARD (HPTK)

Gate pulses of proper duration and timing are developed in an electronics module located in a control enclosure at the left end of the MVB. These signals do not possess the necessary drive and isolation to interface directly with the gate driver current transformers. This interface is provided by the HPTK gate driver card located in the master thyristor module in each leg. The HPTK receives the low level signals and, after conditioning, passes them through a pulse transformer to provide isolation. The pulse transformer output drives a pulse amplifier circuit, which is powered by a 70 Vdc gating power supply located in the fan and filter compartment of the MVB. The pulse amplifier output drives all gate pulse transformers in the leg, as described above. A red light emitting diode (LED) on the board glows to indicate when gate pulses are present.

A second function of the HPTK card is to detect an undervoltage condition of the 70 Vdc supply. If it detects an undervoltage, the HPTK outputs a logic signal to the electronics module indicating the fault. The electronic control then takes appropriate action to shut down the drive. Additionally, an undervoltage detector on each gating card detects wiring defects in

the gating power. Figure 2-5 is a block diagram of the HPTK card. The pulse driver delivers a current pulse simultaneously to all thyristors in the leg module.

2-5. VOLTAGE FEEDBACK ATTENUATOR CARD (XPTN)

In order to properly control the gating signals to the MVB, the electronics in the control enclosure must have available all ac and dc bus voltages. The XPTN attenuator cards in the thyristor modules provide bus-to-ground voltage signals which are combined in the electronics module to reproduce bus-to-bus signals for each converter bridge

Figure 2-8 shows the overall attenuator connections for developing bus voltage feedback signals for an entire bridge. Note that only five bus-to-ground signals are required. Consult MVB elementary DS5201 for complete details of attenuator connections.

Five signals are derived from the attenuator units as illustrated below:

In Figure 2-8, a series string of thyristors is represented by a single SCR symbol.

Note that the line-to-ground attenuator in leg module 4 is not used.

As explained in the paragraph discussing power converters, the voltage signals are used to control firing of the SCRs and to provide various protective functions.

The XPTN contains a berg jumper that is positioned to provide proper attenuation between the bridge and the electronics. See system elementary for proper setting.

The arrangement of the attenuators for the converter and inverter is identical.

2-6. SIGNAL DISTRIBUTION BOARD (XTFH)

All signal interchange between the drive electronic control and the medium voltage bridge is made through the XTFH signal distribution board located in the fan/filter compartment. A small ribbon cable from each HPTK gating card connects to the XTFH board. These twelve ribbons each contain gating control signals, attenuated bus voltage signals, and 70 volt power supply undervoltage signals. The six undervoltage signals from each bridge are combined in the XTFH card to produce a composite undervoltage signal for each bridge. The gating signals originate in the electronics module in the control compartment, while the remaining signals originate in the medium voltage bridge and must terminate in the electronics module. Signal interchange to the electronics module is made via two large ribbons that connect the XTFH cards with firing cards in the electronics module.

The XTFH contains the burden resistor circuits for the source and motor line current transformers. The CT leads plug into the XTFH board and two small ribbons route these signals to the electronics module in the control cabinet.

2-7. EXCITER VOLTAGE CONTROLLER (EVC)

This installation requires controllable excitation in order to meet requirements such as high performance over a certain speed range, high acceleration/deceleration rates, or a special torque/speed characteristic.

There is a phase-controlled three-phase switch between the source and the brushless exciter. Each of the three legs consists of two parallel thyristors connected in reverse polarity. Refer to Figure 2-7. All six thyristors are phase-delayed identically relative to their own connected source line voltages so that symmetrical, bidirectional current in the three lines is obtained. The effect is to reduce the output three-phase voltage magnitude in response to a reference signal.

The reference signal is programmed to obtain a motor flux vs. speed profile necessary to meet the performance requirements.

The EVC consists of the following major components:

SCR Circuit

DS3800NHVL (PWB)

Six SCR Cells

Six high voltage attenuator circuits

Three line fuses

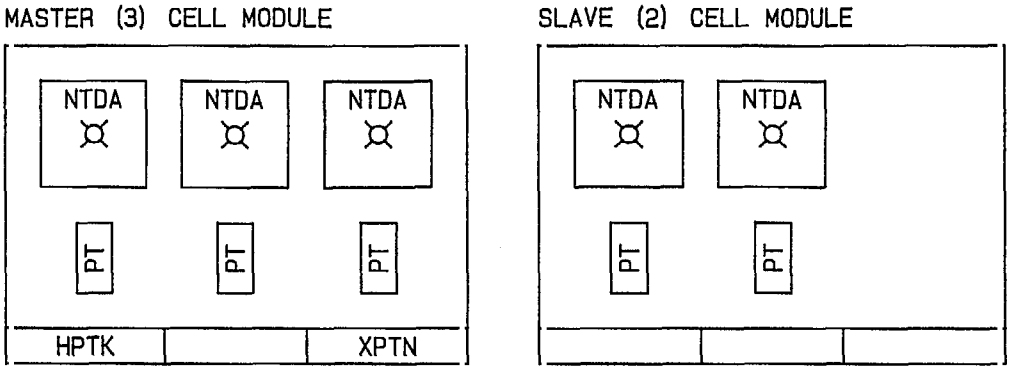
Six pulse transformer firing circuits

Two Current Transformers

CT burden resistor circuit

The NHVL PWB attenuates the incoming and outgoing line voltages of the SFF, 3-phase rectifies and scales (thru burden resistors) the CT signals, and provides the pulse transformer circuits that fire the SCR gates. The attenuated voltages and the scaled current signal are sent back on the same 50-pin ribbon cable that sent over the pulse signals from the electronics module PWB DS3800NSFE. The NSFE provides signal conditioning and interface to the microprocessor control. The microprocessor fires the SFF/SFE SCRs based on a selected phase relation to the source line voltages.

The SFF incoming line must be phased zero degrees, +30°, or -30° to the main power bridge's incoming (source) line voltage.



NTDA - CELL LIGHT AND GATE INTERFACE BOARD
HPTK - GATE DRIVER CARD
XPTN - ATTENUATOR CARD
PT - PULSE TRANSFORMER
NOTE: 2 CELL SLAVE MODULE USED ON 4 KV MVB ONLY

Figure 2-3. MVB Thyristor Module

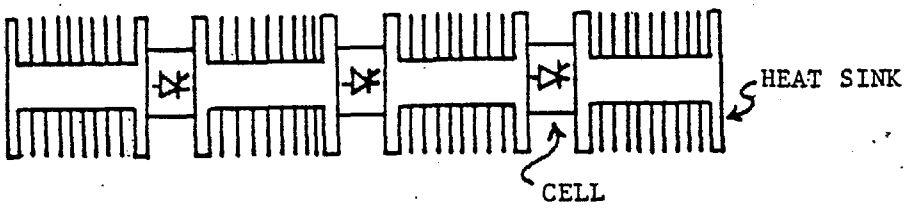


Figure 2-4. MVB Cell Stack Assembly

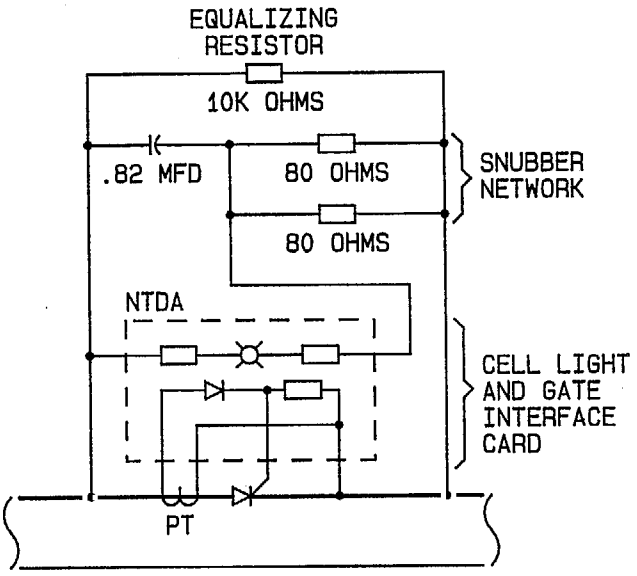


Figure 2-5. Cell with Gating, Display, Snubber, and Voltage Equalizing Circuits

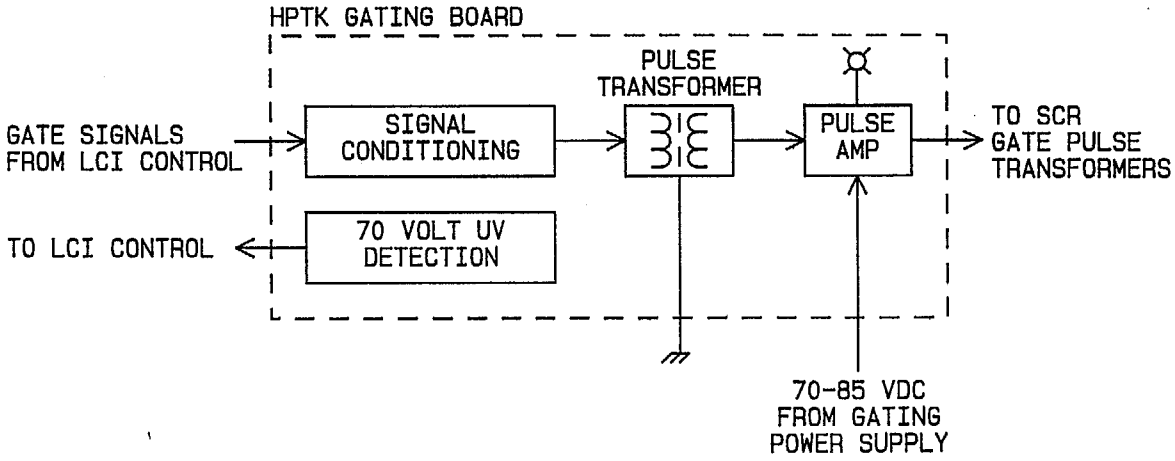


Figure 2-6. HPTK Gate Driver Card Block Diagram

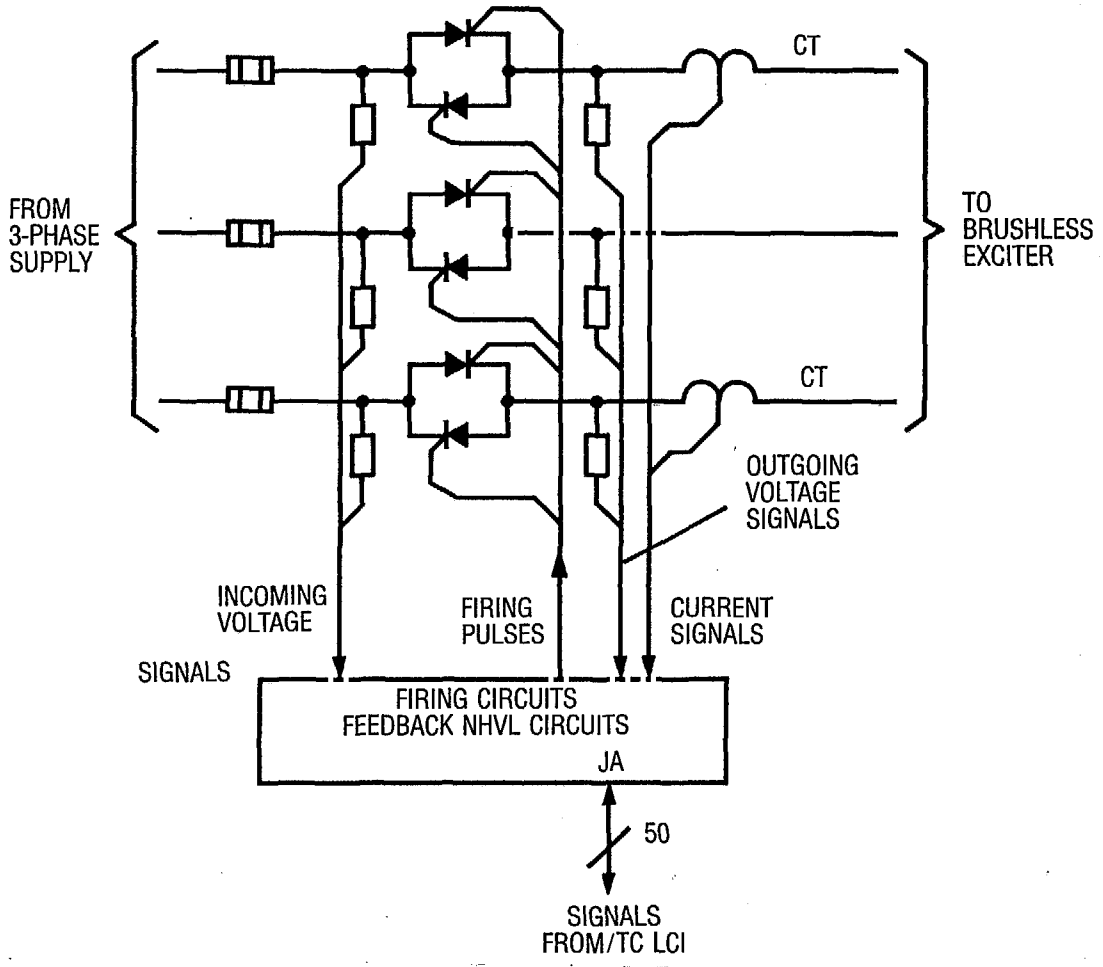


Figure 2-7. Synchronous Field Exciter Module (DS3820SFE)

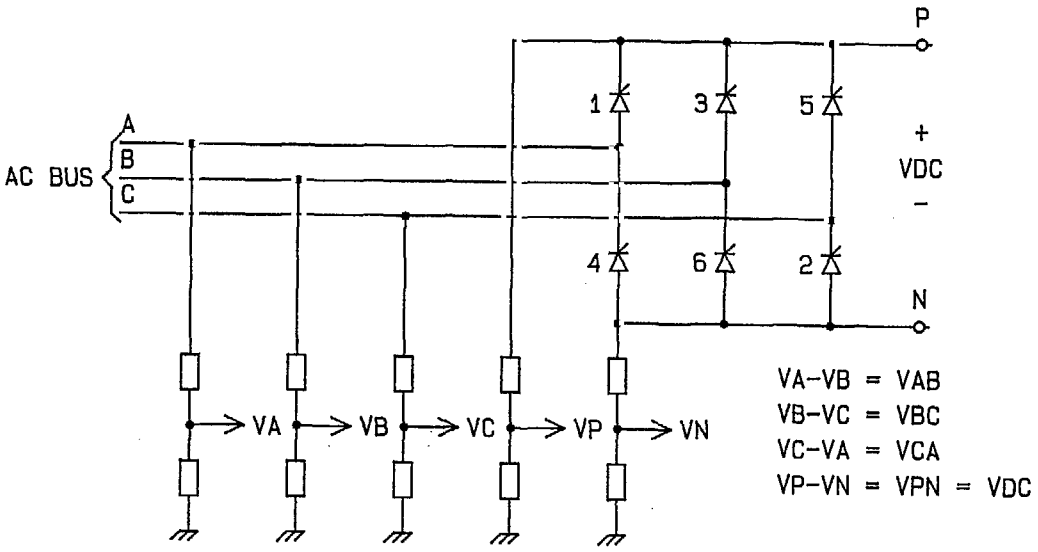


Figure 2-8. Bridge Attenuator Circuits

CHAPTER 3

CONTROL ELECTRONICS PANEL

3-1. INTRODUCTION

The control electronics panel is made-up mainly by various modules interconnected thru ribbon cables and harnessed wiring. The main modules are:

DS3800LCG Main Control electronics
DS3820IOM Inputs/Outputs (I/O)
DS3820SIM Auxiliary interface electronics
(When more than the normal signal is needed)
DS3820SFF Field Exciter (Discussed in Chapter 2)
DS3820LDM Display Module

The modules are typically configured as in Figure 3-0. Following are brief descriptions of the functions of the module. For more specific information the corresponding modules elementary and documentation should be consulted.

3-2. ELECTRONICS MODULE (LCG)

The control electronics module consists of cards in 2-rows x 6-slots. Its primary purpose is to control the firing of the power converters and field exciter module SCRs for driving a motor. It also provides various protective functions for the motor and bridges. Interface between the LCG to the customer is thru the IOM and SIM.

See Figure 3-1 for the physical layout of the module. Connection is made to the electronics module (SEM) through the plugs located at the top, and also through card front connectors.

From a functional standpoint, the electronics module processes signals as represented in Figure 3-2.

The following are the primary functions provided by the module.

1. Pulse train generation and gating signals for 2 ac to dc power converters.
2. Signal conditioning for line-to-line attenuaturs and dc link attenuators for two power converters.
3. Signal conditioning for power converter generated faults and other control signals such as CT's.
4. Hardware sensed faults such as suppression, overvoltage, etc.

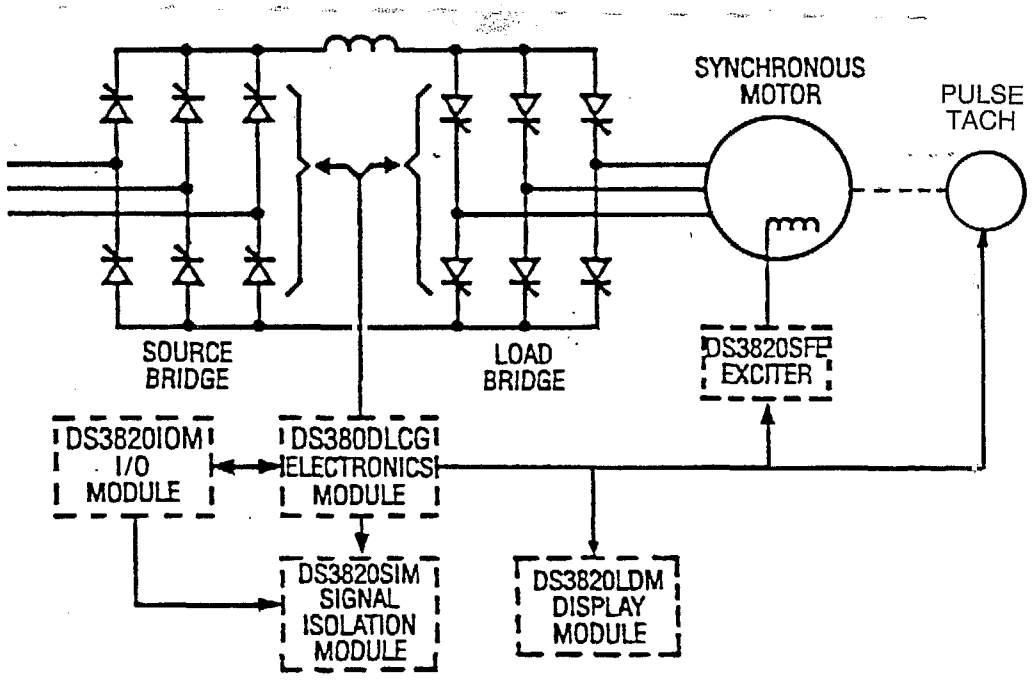


Figure 3-0. Control Electronics Panel

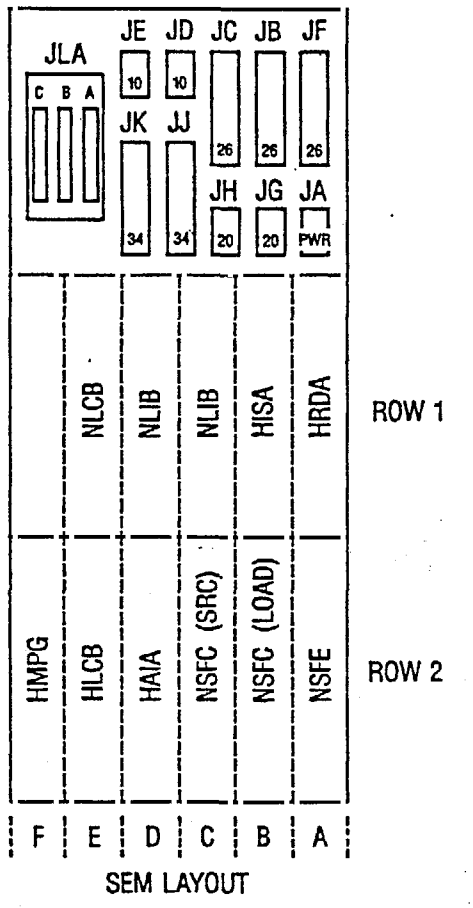


Figure 3-1. Small Electronics Module (SEM) Card and Connector Layout

5. Microcomputer for implementation of control algorithm.
6. Pulse train generation and gating for field excitation module DS3820SFF.
7. I/O conditioning for signals received and transmitted to I/O module DS3820IOMA.
8. Signal conditioning for tachometer interface.

The hardware of the electronics module is represented from a block diagram standpoint in Figure 3-3. There is one microcomputer controlling the firing of both of the power converters and the field exciter.

The microcomputer establishes a local bus (L-bus) structure on which the input/output for it is interfaced. In addition, it communicates to an 8k byte read/write (RAM) memory and a 2k byte electrically alterable memory (EAROM). This memory combines features of read/write memory (RAM) and read only memory (ROM) in that it is electrically alterable and at the same time does not lose its contents on power loss. The primary use of this EAROM memory is to store parameters that must be tailored to each different system. These parameters are field adjustable for drive tuneup.

The local bus has a timer and interrupt card that provides the necessary hardware for control of interrupt priority and timing for the microcomputer.

There are input/output ports on the local bus that interface static logic signals to/from the microcomputer. As an example, thyristor gating commands are written out to one of these ports (on each side) and sent to an analog card to produce the gating signals to the converter bridge. The electronic control contains a number of analog cards that process the voltage feedback, current feedback, meter signals, etc.

Analog to digital converter cards interface the speed reference, voltage feedback current feedback signals, etc., to the microcomputer bus on each side.

I/O cards interface process command and status signals to the microcomputer. These inputs are combined

logically by a sequencing program in the microcomputer to control the relays in the IOM. These outputs are used to control system devices such as panel lights, contactors, etc.

For a dual channel LCI system, each power channel has its own electronic module. To establish load sharing and for communication of other status, it is necessary for information to be interchanged between the two electronic modules. This interchange is accomplished via 12 Bit D/A and A/D converters (± 10V) and some I/Os of the IOM.

3-3. CONTROL OPERATIONAL DESCRIPTION

3-3.1. Basic Principles . Refer to the control system block diagram in Figure 3-4. The speed reference from the process controller is compared to a speed feedback, either derived from the integrated motor voltage (PLL) or from tachometer feedback, to develop a speed error input to the speed regulator. The output of the speed regulator is a torque command. The speed regulator program resides in the load side processor.

The torque command signal is applied to both the source and load side control. As illustrated in Figures 1-6 and 1-7, motor torque is a function of voltage, current, and the angle between them. Therefore, torque can be controlled by adjusting current magnitude at a fixed angle or maintaining a constant current and varying the displacement angle. The current is controlled from the source side, whereas the angle is varied from the load side. At any instant in time, the torque is actively controlled by only one of these means.

The torque command to the source side control is applied to a maximum and minimum current limiter. The minimum current level is set to maintain continuous current in the dc link.

The current level control also affects the load side angle control. When the voltage spillover is not activated, the angle control is in one of its limits. If current is adjusted to control voltage, the angle control gain is decreased and displacement angle Θ is increased, which reduces developed torque to that required by the load.

The current command is compared with current feedback, I_{DC}, and the error is applied to the current regulator. The current regulator controls the firing of the thyristors in the source side converter (rectifier). Thus, the source side control adjusts the dc link voltage as required (within its limits) to produce the motor current needed to drive the load.

The load side control is designed to adjust firing angle so as to maintain the displacement angle Θ as small as possible. This feature is designated "Fire as late as possible" (FALAP).

Referring to Figure 1-4, it is shown that the angle of advance, β is the sum of the commutation angle, μ and the margin angle, Ψ. For maximum efficiency, it is desired to keep the margin angle as small as possible

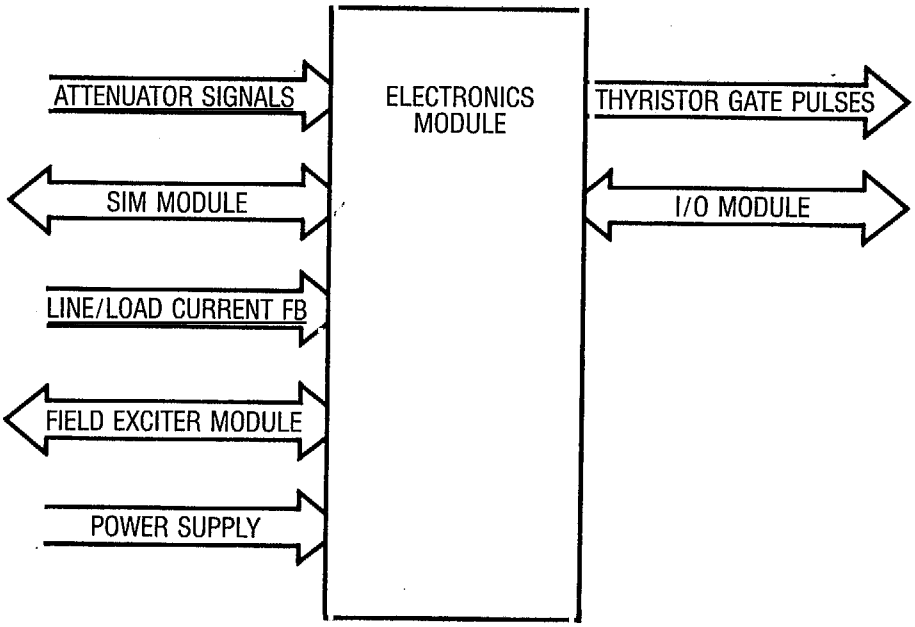


Figure 3-2. SEM Input/Output

without encountering commutation failure. The objective is to determine the latest possible firing point for each oncoming commutation.

When the drive requires regeneration, the load-side thyristors are fired full advance and the source side controls current by reversing the dc voltage to match the rectified motor voltage.

3-3.2. Phase Locked Loop Operation . Since the inverter thyristors are fired in synchronism with the integrated motor stator voltages, a phase locked loop (PLL) counter is used to maintain correspondence between the integrated voltage wave zero crossing (Figure 3-5) and the firing pulses.

When the PLL counter output is precisely in synchronism with the "flux" wave zero crossings, the loop is said to be "locked". If the counter output is lagging (slower than) the flux crossings, the PLL controller acts to increase the counter frequency (by decreasing PLLC LOAD). Similarly, if the counter frequency is leading (faster than) the flux crossings, the controller acts to decrease frequency (by increasing PLLC LOAD).

Although the output of a phase locked loop, when "locked", has the same frequency as the reference signal, there is a phase difference between the two due to the integrating action of the loop low-pass filters; thus, the name "phase locked-loop."

3-3.3. DMC MASTER CONTROLLER DESCRIPTION

3-3 3.1. F-BUS MEMORY SPACE. An LCI DRIVE looks like a block of bytes on the F-BUS. NOTE: The F-BUS is only 5 volt single ended TTL logic with very little noise margin. Thus it is not recommended to bus the F-BUS to other drives.

3-3.3.2. TIMING CONSIDERATIONS. The time between data link accesses from the master controller should not be faster than 20 milliseconds; otherwise it is not be accepted by the LCI Drive. The transfer of information must also occur in a burst of reads and writes because of the HXMA hardware characteristics.

3-3.3.3. SIGNAL FLOW. The first 6 bytes of the memory space are reserved for logic control and status, thus providing 48 control signals into the drive and 48 status signals from the drive. The actual signal word/bit assignments can be found in another section of this document. Also, 9 integer variables can be sent to the drive as references, and the same number of status feedbacks can be returned to the master controller.

3-3.3.4. SPECIAL REQUIREMENTS. The 'DLUNSU' and 'DLSIC' signals must be sent to the drive as complementary pairs (note under the bit assignments, signals 30 and 31). The true state is signal 31 and the complement is signal 30.

3-3.4. Microprocessor = HMPG/DMPB. The DS3800HMPG is a single board computer with the following features.

1. 80286 microprocessor
2. programmable communications interface
3. programmable interval timer
4. on-board RAM memory (16k bytes)
5. on-board PROM memory (64k bytes)
6. mapped I/O
7. bus hog timer
8. stall timer

The configuration chosen gives a powerful 16 bit processor.

Connectors.

A 50 pin ribbon connector is included on the DMPB to interface the Nonvolatile RAM storage.

A 25 pin male connector is present to serve as an RS232C communications port. A "null modem" is supplied to allow a standard data terminal interface cable to be used with the communications port. The connector is normally reserved for use with the drive monitor via a standard data terminal during startup or trouble shooting of the drive.

Table 3-1A, RS232C Connector Signals

Pin#	Name	Signal	Direction	Function
JB-2	BA	TXD	output	transmit data
JB-3	BB	RXD	input	receive data
JB-4	CA	RTS	output	request to send
JB-5	CB	CTS	input	clear to send
JB-6	CL	DSR	input	data set ready
JB-7	AB	DCOM	input	signal ground
JB-9	CD	DTR	output	data terminal ready

Table 3-1B. HMPG Berg Jumpers, Testpoints, LEDs

Berg Jumper	Function	Position
J1	selects type of PROM	8/16
J2	selects size of PROM	16k
J3	selects bootup method (on/off bd)	DGN (not)
J4	selects 8 or 16 address segments on L bus	8
J5	ground for RS232	GND

Testpoint	Function
RST	Reset. Jumper to common to reset processor.
DCOM	DCOM. Digital common.
I/S	Inhibit stall. jumper to common to activate.
TNMI	Test nonmaskable interrupt. jumper to common to activate.

LED	Function
BHOG	Bushog. On when non-existent address is called by software.
TST	Software controlled and defined. On during initialization self test.

3-3.5. Electrically-alterable ROM = DS3800DMPB.. This is a non-volatile memory board containing 8k bytes of memory. This memory is used to store drive tuneup parameters, which may be customized on a per drive basis. Physically, the board is configured as a daughter board and mounted on the HMPG

Constants stored in the board's memory are normally changed by first enabling board writes with switch "WREN" and then supplying the data through the drive monitor.

In the Drive Systems documentation procedures, PROMS for DMPB socket U2,U3 and PROMS for HMPG sockets U22,U23,U24,U25 are ordered under the prom set number shown on sheet EA of the system elementary.

3-3.5.1. TOGGLE FLAG. A complementary pair of status signals are sent back to the master controller and are toggled whenever the drive accepts the data. 'DLTOG', signals 6 and 7, can be used by the master controller to determine if the drive is alive. Figure 3-6 shows the suggested sequencing that can be put into the master controller to determine data health and if the drive is alive.

Table 3-1C. Electrically-Alterable ROM - LEDs and Switches

LED	Function
IMOK	Software controlled as a function of self diagnostics. Normally on, is off if selftest is not passed.
WREN	Write enable. On when write capability to board is enabled with switch WREN.

Switch	Function
WREN	Activates the board to allow alteration of the drive turnup parameters.

3-3.6. LCI processor extender board = DS3800HLCB. The HLCB board extends the functions provided by HMPG to include those required for an LCI system. The following functions are provided.

1. Address decoding for on-board functions, NSFC, and NSFE.
2. Interrupt controller.
   - source crossover
   - load crossover
   - source angle control
   - load angle control
   - field excitation control
3. 5 bit general purpose input port used for:
   - ground fault
   - I/O module input ribbon cable continuity
   - I/O module signal cable continuity
   - I/O module output ribbon cable continuity
   - door module ribbon cable continuity
4. Zero crossing detection. Circuitry is included to derive source and load crossover interrupts from square wave information originating on the NLIB boards. A single clock pulse interrupt is obtained at each crossover.
5. Two programmable timers provide the functions of
   - PLL clock (source and load)
   - TIME counter (source and load)
   - SPEED calculation
   - CLOCK for serial communications port
6. RS-232 compatible port. Only the transmit function is implemented at the prot is intended for driving a printer located in the door module. The control line, DSR (data set ready), is also monitored to determine the status of the remote printer.
7. Digital to analog converters. Three D/A converters are supplied. D/A 1 and D/A 2 are 8-bit non-committed outputs available for transmission to the signal isolation module (SIM) if supplied. The 12 bit D/A is committed to supplying the torque reference between 6 pulse converter controls when used in a 12 pulse redundant system. System level documentation defines its use (if any) in other systems.

Table 3-2 LCI Extender - LED's and Testpoints

LED	Function
IMOK	Software controlled as a function of selftest. Normally on, is off if selftest is not passed.

Testpoint	Function
ACOM	Analog common.
DA1	Digital to analog output 1. Normally uncommitted.
DA2	Digital to analog output 2. Normally uncommitted.
DA3	Digital to analog output 3. Normally used for horsepower output.
DA4	Digital to analog output 4. Normally used for torque command to other channel (if supplied).
OTESTINO	Tie to common at initialization to download EEPROM values
OTESTIN1	not used
SFT1 *	Execution flag for source crossover and angle control routine.
SFT2 *	Execution flag for load crossover and angle control routine.
SFT3 *	Execution flag for field angle control interrupt.
SFT4 *	Execution flag for background program.
VA1 *	Phase A signal for source power.
VA2 *	Phase B signal for motor voltage.
DCOM	Digital common.

* See figure 3-5 for typical waveforms

3-3.7. Analog = Digital interface = DS3800HAIA

The HAIA board accepts 16 analog input signals, multiplexes them to one line, and converts to a two byte digital word. The A/D provides 11 bits of magnitude resolution. Also provided are three, 8-bit digital input ports, three 8-bit output ports and one 4-bit digital port.

3-3.8. FLUX INTEGRATORS = DS3800NLIB. The NLIB contains various analog signal conditioning and protective functions. Two identical boards are used - one for the source side and one for the load. The following functions are provided:

1. Voltage integrators
2. Line/Motor current transformer receivers
3. Voltmeter driver
4. Overvoltage protection
5. Undervoltage sensing
6. Suppression
7. Variable reference comparator
8. Current meter driver

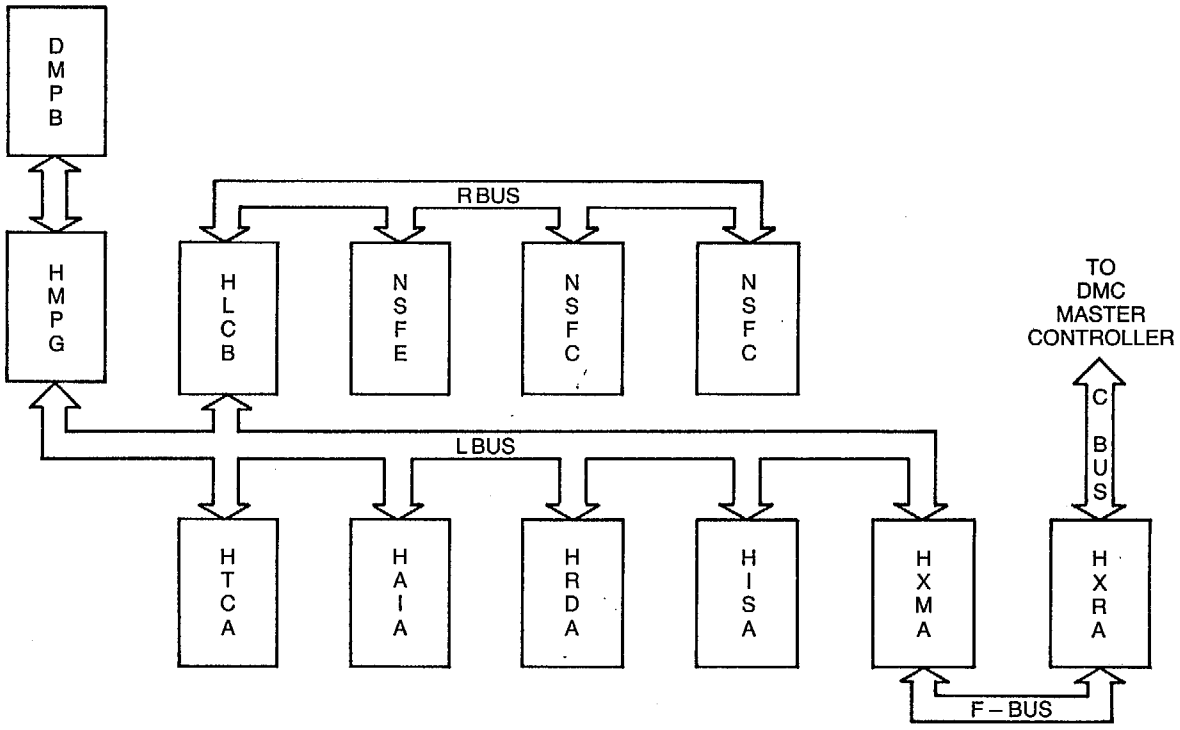


Figure 3-3. Electronics Module Block Diagram

3-3.8.1. VOLTAGE INTEGRATORS. A signal proportional to each line-toline voltage is obtained from the NSFC board. This signal is applied to an analog integrator to obtain a replica of motor flux according to the equation:

$$V = K * dPhi/dT$$

or

$$Phi = V * dT$$

Components of line currents are used to compensate for drops due to motor resistance and commutating impedance so that the flux represents the motor internal counter-emf or the so called voltage behind the commutating reactance. The analog sine wave corresponding to flux is then applied to a zero crossing comparator where the digitized output is then available to the HLCB for phasing information. The analog flux signals are also read by the microprocessor based software for use in the control algorithms.

3-3.8.2. LINE/MOTOR CURRENT TRANSFORMER RECEIVERS. Two signal conditioning differential receivers are included for line currents A and C. Line current B is calculated from A and C assuming a balanced load. Delta currents are also determined for use in the flux reconstruction.

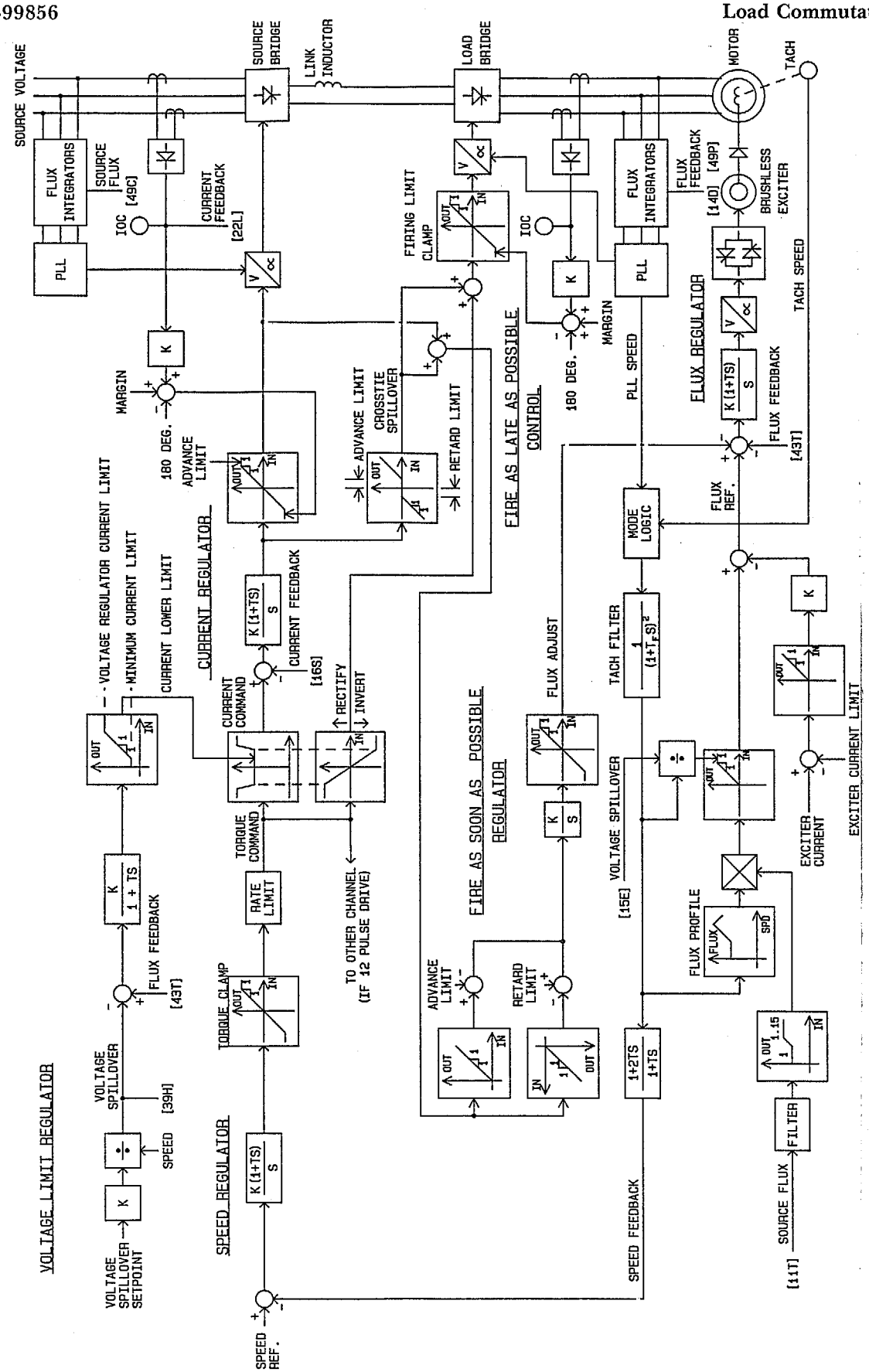


Figure 3-4. LCI System Digital Regulator Block Diagram

Table 3-3 Digital Interface. Berg Jumpers and Testpoints

Berg jumper	Function	Position
J1	Selects either a fixed +7 .5 V v input to channel 15 or a variable input thru pin 12. The variable input thru pin 12. The variable input is selected.	A to P
J2	Selects either a fixed +7 .5 V input to channel 14 or a variable input thru pin 12. The variable input is selected.	A to N (SRC) A (LD)
J3-J6	Selects L bus address for board.	F
J7	Selects L bus address for board.	T
J8	Selects L bus address for board.	F
J9	Selects filter for analog side of A/D converter. Filter is selected.	0

Testpoint	Function
ACOM	Power supply analog common.
FOG	Test input. Tie to common to latch internal card data bus.
AN	Analog signal at input to A/D conversion.
DB	Data buffer direction. High transmits to microprocessor and low receives.
DV	"0" data valid signal.
BSY	"1" busy signal.
ES	"1" enable switch.
CLK	card clock - 1.152 mHz

3-3.8.3. VOLTMETER DRIVER. The magnitude of the line-to-line voltage is determined by a 3 phase absolute value network. The absolute value is then scaled by an operational amplifier for use as a nominal line voltage signal.

3-3.8.4. OVERVOLTAGE PROTECTION. The absolute value of line/motor volts obtained above is compared with an adjustable reference for a time determined by daughter board component, C37, then the overvoltage latch is set. The signal is wired to and I/O port for fault logging by the control software.

3-3.8.5. UNDERVOLTAGE SENSING. The absolute value of line volts is compared with an adjustable reference.If the voltage is less than the reference for time delay set by daughter board component, C17, then signal "1LV" becomes true. This signal is used by the software system sequencing to determine low source volts condition. The function is not used on the load side.

3-3.8.6. SUPPRESSION. The absolute value sum of the line currents is taken and applied to a comparator. If the rectified current exceeds the reference threshold for 26ms, a fault latch is set. The time delay is included since the primary protection is in software, which includes an on-line recovery. The function implemented here is strictly backup.

3-3.8.7. CURRENT METER DRIVER. The absolute value of current is scaled and available from either source or load for meter driving.

Table 3-4 Flux Intergrators. Jumpers, Adjustments, LEDs and Testpoints

Jumper	Function	Position
BJ1-BJ3	Selects voltage integrator scaling for 60 Hz or 120 Hz	120 Hz or 60 Hz (var)

Adjustment	Function
FAB, FBC, FCA	Scale flux integrators that calculate flux. Scaling is 4.75 volts RMS corresponds to 1 PU.
OV	Set overvoltage protection trip point.
UV	Set undervoltage protection trip point. (not used on load side.)
SUPP	Set trip point for overcurrent backup fault.
COMP	Not used.
VDC	Not used.

LED	Function
OV	On when overvoltage fault latch set.
SUPP	On when overcurrent fault latch set.

IMOK	On if microprocessor based self test is passed and no saturation conditions are present.

Testpoint	Function
OVR	Analog dc voltage proportional to setpoint of overvoltage fault latch.
UVR	Analog dc voltage proportional to setpoint of source undervoltage sensor.
SUPP	Analog dc voltage proportional to setpoint of overcurrent fault latch.
FAB, FBC, FCA	Analog voltage proportional to motor/line flux reconstructed. See Fig. 3 for typical wave forms.
FVAR, FCBR	Analog voltage proportional tomotor/line flux reconstructed for IX drop. See Fig. 3 for typical
FACR	waveform.
OIA, OIB, OIC	Analog waveform representative of inverted line current. See Fig. 3-6 for typical waveform.

3-3.9. GATING REFERENCE BOARD = DS3800NSFC. The NSFC's primary task is to convert information from the microprocessor based control to cell pair and timing for firing 6 thyristors configured into a 3 phase - double way bridge. In performing this task, the following functions are used:

1. R bus interface
2. Firing mask latch
3. cell state sensors
4. Line-to-line voltage signal conditioners
5. Pulse train generation
6. Gate drivers
7. Attenuator buffers
8. D-C voltmeter driver

Quantity 2 NSFC's are used assigned as follows:

1. Source converter
2. Load converter

3-3.9.1. R BUS INTERFACE. Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected. All information exchange with host processor takes place through the R bus interface.

3-3.9.2. FIRING MASK LATCH. The particular cell pair to be fired is selected in software and communicated to the NSFC over the R bus. The mask

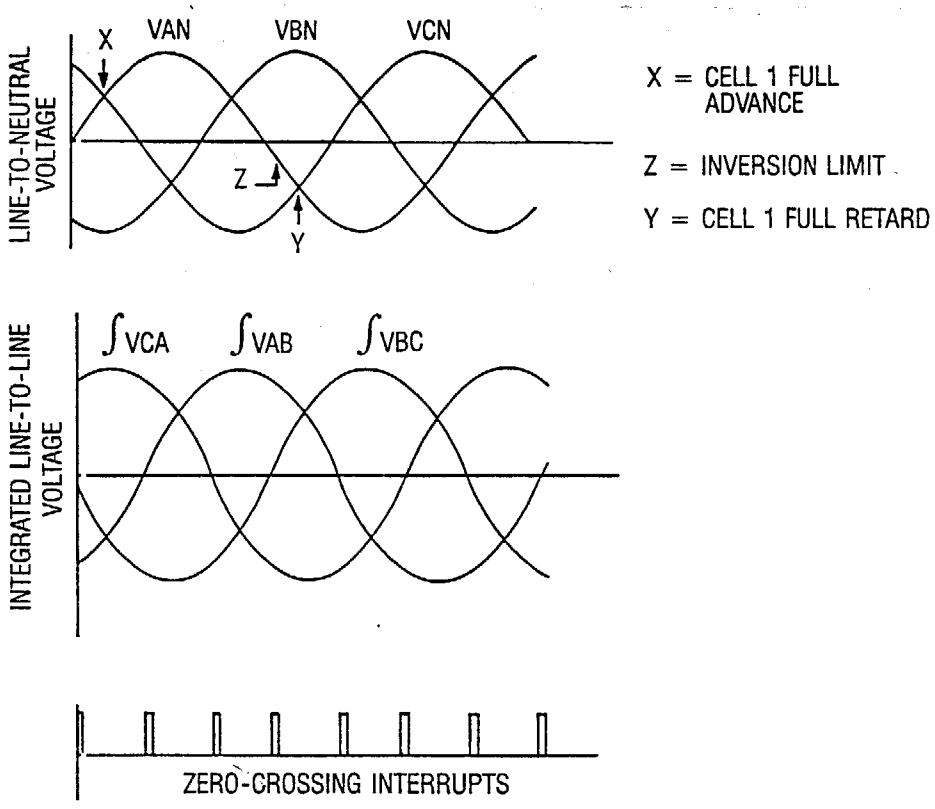


Figure 3-5. Flux Wave Zero-Crossings

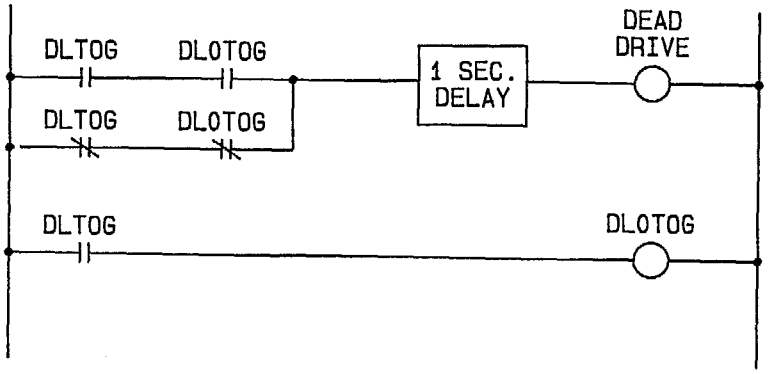


Figure 3-6. Drive Alive Logic

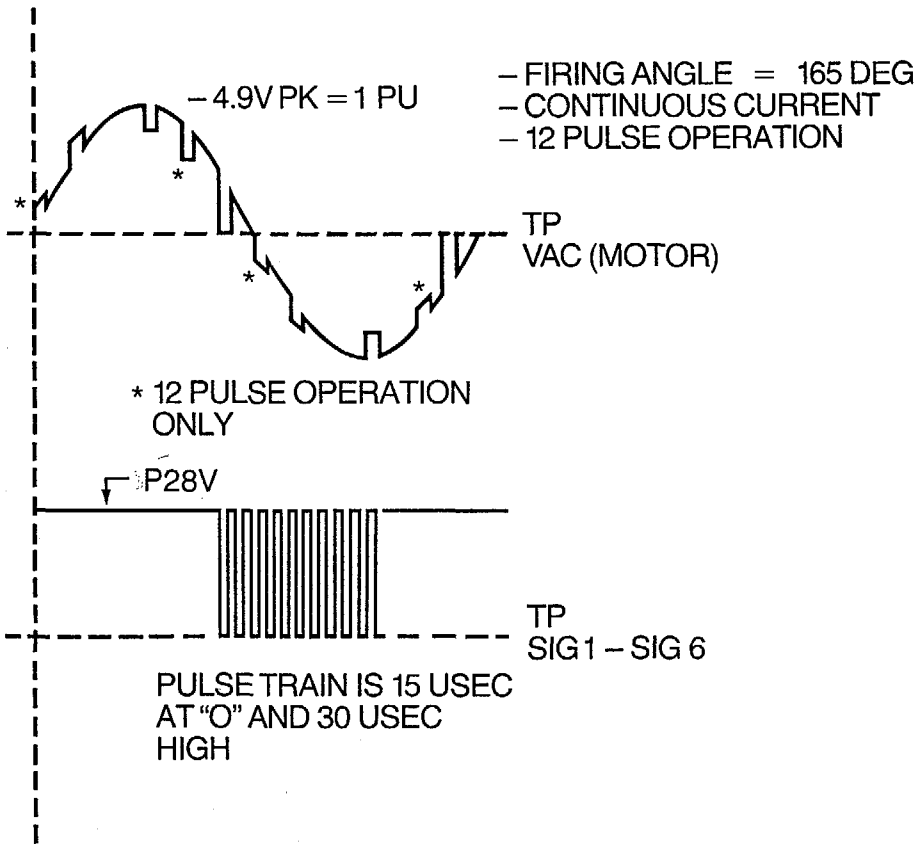


Figure 3-7. Cell Firing Pulse

information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed. This information is routed to a timer. At the end of the delay an interrupt is generated and the cell mask is clocked into the final latch.

3-3.9.3. GATE DRIVERS. Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 25 VDC levels.

3-3.9.4. PULSE TRAIN GENERATION. The gate pulse reference driver signals consist of a "0" true pulse train nominal 120 degrees long during the conduction period. The reference pulse train is generated from a crystal clock and shaped for a duty cycle of 15 microseconds on and 30 microseconds off by a timer and control logic.

3-3.9.5. BUFFERS. Attenuator buffers are provided to receive each attenuator string connected to phase voltages 1,2,3 and also the positive and negative bus. These buffered signals are then used for analog line-to-line voltage signals as well as cell state sensor inputs.

3-3.9.6. LINE-TO-LINE VOLTAGE SIGNAL CONDITIONERS. The buffered phase to neutral converter voltage signals are differentially combined to recreate line-to-line voltages. These are used after further processing (on NLIB) as information for the software control algorithms.

3-3.9.7. CELL STATE SENSORS. From a cursory check of the standard three phase double wav bridge configuration, it can be seen that the voltage across each cell is a combination of one phase voltage and one D-C bus voltage. Since each of these voltages is available from an attenuator, the voltage across each cell or "on state" maybe uniquely determined. Circuitry is provided for reading the "state" of all cells at a particular instant and for making the information available to the microprocessor via the L bus.

3-3.9.8. D-C LINK VOLTS. The positive and negative D-C link bus voltages are combined differentially to obtain a dc link volts signal suitable for driving a remote analog meter with approximatedly 10 Vdc maximum scale.

Table 3-5 Gating Reference - Berg Jumpers, Adjustments, LEDs and Testpoints

Berg jumper	Function	Position
J1	Determines if new mask should be clocked into final latch at termination of count down timer.	NINT

Adjustment	Function
VDC	Sets remote meter full scale to corresponding dc link volts.

LED	Function
IMOK	Board hardware status indication. On normally. Off if self diagnostics determines a board hardware failure.

Testpoint	Function
SIG 1-SIG 6	Gate pulse command. "0" true. Steady state length of 120 degrees relative to operating frequency. Each gate pulse is shifted 60 degrees relative to lower numbered ones. SIG_ numbers correspond to thyristor leg being gated.
VDC	Output of dc link voltage meter driver.
VAC,VCB, VBA	Analog output proportional to motor/source line-to-line voltage.
ACOM	Power supply analog common.

3-3.10. Field Excitation control board = DS3800NSFE. The NSFE's primary task is to convert information received from the microprocessor based control to cell pair and timing for firing 6 thyristors configured in a 3 phase switch configuration. It also provides analog interface and diagnostic interface for the external power converter.

The following functions are provided.

1. R bus interface
2. Firing mask latch
3. Cell state sensors
4. Attenuator signal conditioning
5. Pulse train generation
6. Gate drivers
7. Current feedback conditioning
8. Current unbalance detector

3-3.10.1. R BUS INTERFACE. . Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected. All information exchange with the host processor takes place through the R bus.

3-3.10.2. FIRING MASK LATCH. The particular cell pair to be fired is selected in software and communicated to the NSFE over the R bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from software is a time delay corresponding to the number of degrees the cell pair firing is to be delayed from the present time. This information is routed to a timer. At the end of the delay an interrupt is generated and the cell mask is clocked into the final latch to gate the selected cell pair.

3-3.10.3. GATE DRIVERS. Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 Vdc levels.

3-3.10.4. CURRENT FEEDBACK CONDITIONING. A differential amplifier is used to receive the field current signal from the field exciter module (remote from SEM). The scaling is arranged for 2v corresponds to 1 per unit field on the input and +5.0 volts equals 1 per unit at EI. An additional signal (EIMTR) is also available for meter driving with a pot adjustable gain range of 2:1.

To detect fault conditions within the motor exciter, the field current is applied to a filter tuned to the excitation fundamental frequency. It can be shown that conditions of unbalance cause a dramatic increase in current feedback's harmonics. The condition of the sensor is available to the system processor through the data bus.

Table 3-6. Field Excitation. Berg Jumpers. Adjustments, Testpoints

Adjustment	Function
EIMTR	Sets full scale reading on external field current meter (if supplied).

Testpoint	Function
CIP	Cable in place. When the cable from the NSFE board front is properly installed in the external module, voltage at CIP is less than .5 V. If the cable is not in place, the voltage is be 15 V.
VABS	Signal proportional to the absolute value of exciter output voltage. Always positive and scaled for approximately 10 volts at 1 per unit output.
LVAC	Signal showing the exciter module input waveform on lines A-C. The scaling is approximately 4.5 volts peak corresponds to peak line volts.
FUN	Output of fundamental bandpass filter monitoring field current. Is near zero volts under normal conditions.
EIMTR	Output of field current meter driver. Range 0 - 10 volts.
IFB	Field current signal with 5 volts corresponding to 1 per unit.
S1 - S6	Gating signals to external thyristors. Nomenclature corresponds to the firing order. Each signal is at a 28 volt level and is 0 when true.
ACOM	Analog common.
CONT	Continuity. signal for checking continuity of ribbon cable to external excitation module.

3-3.11. ANALOG LCI BOARD = NLCB. The NLCB provides various miscellaneous signal conditioning and test functions required for the structured LCI. These features include:

1. Speed feedback generation from motor voltage zero crossing signals with overspeed protection and indication.
2. External reference signal conditioning.
3. Test reference signal conditioning.
4. Torque reference signal conditioning.
5. Selftest features.
6. 4-20 MA signal drivers.
7. Ground fault detector.
8 power applied reset.
9. Fault relay drivers.
10. Horsepower meter driver.
11. Continuity detection circuits.
12. Test switch.
13. Restart upon power loss.

3-3.11.1. SPEED FEEDBACK. A pulse rate signal proportional to drive speed is obtained from the zero crossing detection circuitry on the HLCB. A pulse rate to analog converter then produces a corresponding dc level proportional to speed suitable for driving a meter and protective functions.

The overspeed function is implemented by comparing an overspeed setting reference to the analog speed voltage. Transient conditions will be sealed. A red LED indicates the overspeed condition until reset.

3-3.11.2. EXTERNAL REFERENCE CONDITIONING. A differential amplifier committed to accepting the remote speed reference is supplied. Berg jumpers are used to configure the circuit to accept a 0 to 10 volt signal or a 4 - 20 mA signal. In either case, the output voltage is transmitted to one channel of the HAIA analog to digital converter for reading by the system processor.

3-3.11.3. TEST SPEED REFERENCE. A differential amplifier is included to condition a voltage source signal from a potentiometer located on the I/O module. This is also known as the local speed reference.

3-3.11.4. TORQUE REFERENCE SIGNAL CONDITIONING. A differential amplifier for receiving the master torque reference from the other six pulse drive (in a 12 pulse system) is also supplied.

3-3.11.5. SELFTEST FEATURES. An array of static switches and voltages sources is included for use in testing the analog circuitry of this and other boards in the module during the module selftest mode.

A sinewave oscillator is used to simulate line and motor voltage during "gate test mode". This is to allow the power converter electronics to be tested without the necessity of applying high voltage to the associated bridges.

3-3.11.6. 4 - 20 MA SIGNAL DRIVERS. Two voltage to current converters provide capability of remote driving system information to customer interfaces. The signals selected for transmission are motor current and motor speed. In each case, 4 ma corresponds to 0.

3-3.11.7. GROUND FAULT DETECTOR. The ground fault detector sums line-to neutral voltages Va, Vb, and Vc. It operates on the principal that when all impedances from phase to neutral are balanced that the summation of the phase-to-neutral voltages will be zero. Any unbalanced impedance will cause an error to be detected and a ground fault noted by the system processor.

3-3.11.8. POWER APPLIED RESET. A power applied reset is generated here and distributed throughout the electronics module. It is used to prevent false operations of relay drivers and fault latches during the process of powering up the module.

3-3.11.9. FAULT RELAY DRIVERS. Relay driver no. 1 is generally used as a method of tripping the source side drive switchgear from both hardware and software sensed fault conditions. It is inhibited from false operations during the power up phase by the power applied reset.

Relay driver no. 2 is generally used as a method of tripping the motor side isolator (if used) from both hardware and software sensed fault conditions. It is also inhibited from false operations during the power up phase.

3-3.11.10. HORSEPOWER METER DRIVER. An adjustable meter driver for powering a door mounted meter indicating horsepower is also included. The horsepower signal is calculated in software and converted to analog on the HLCB board.

3-3.11.11. CONTINUITY DETECTION CIRCUITS . Four circuits are included to monitor the continuity of various ribbon cables connected to the electronics module when it is integrated in to a system control panel. These signals are only monitored during the panel selftest. An appropriate diagnostic message is printed out if a cable is not in the correct location.

3-3.11.12. TEST SWITCH. The test switch on the board front edge is used to initiate the panel selftest mode. When the switch is on, and a reset is performed, the electronics module begins to execute a selftest routine, which extends to other components of the control panel. Upon completion of the selftest, the control enters gatetest mode using the three phase test oscillator as a source of synchronous waveforms.

3-3.11.13. RESTART UPON POWER LOSS. If the drive has been previously running and the control power is lost for a time (determined by setting of RTD potentiometer), a pulse at signal 1RSRT automatically restarts the drive when power returns. Note that this is an option, and the I/O module must be properly configured.

Table 3-7. Analog LCI Board. Berg Jumpers, Adjustments, Testpoints, LEDs

Berg jumper	Function	Position
LOAD	Selects remote speed reference for either current input or voltage input. Should be coordinated with GAIN	variable
GAIN	Selects gain necessary for remote speed when used as current input or voltage input.	variable

Adjustment	Function
SPD	Adjusts analog speed feedback circuit for 10 volts at 60 Hz.
SPM	Adjusts door mounted speed meter for proper full scale reading.
OSP	Hardware overspeed reference.
RTD	Adjusts allowable time for auto restart after loss of control power.

Testpoint	Function
SPD	Output of analog speed feedback calculation
OSP	Reference to hardware overspeed detection circuit. Scaled same as speed i.e. 10 volts = 60 Hz.
IOA	Output of channel A current source.
IOB	Output of channel B current source.
REF	Output of remote speed reference. With a voltage input will be 0 to 10 volts. with a current source input, will be 2 to 10 volts.
TREF	Output of local test reference. Varys from 0 to 8.3 v.
12P	Output of torque reference obtained from other channel.
NTL	Neutral. Summation of line to neutral voltages. Normally near 0 volts.
RTD	Run time delay. Sets allowable time for loss of control power while still allowing an auto restart. Scaled 0 to 10 volts.
PHA	Test oscillator, phase A. Shows a 55 Hz sine wave.
PHB	Test oscillator, phase B. Shows a 55 Hz sine wave lagging phase A by 120 degrees.

LED	Function
OSP	Red LED indicating that the hardware overspeed latch is set.
RSRT	Green LED that is pulsed when the auto restart circuit is activated.
FTl	Green LED indicating the status of fault relay driver NO. 1. Will be on when the relay driver is energized.

LED	Function
FT2	Green LED indicating the status of fault relay driver NO. 2. Will be on when the relay driver is energized.
TEST	Green LED that is luminous when the Test switch is closed and the system software indicates gate test mode.
IMOK	Yellow indicating light will be on normally. An off condition indicates a possible hardware problem with the board. It can be turned off either locally or from the processor as a function of selftest.

3-4. RELAY DRIVER BOARD = DS3800HRDA

The HRDA is relay driver board with 16 open-collector outputs. Output voltages can range from 5 to 50 volts. In the LCI system, it is applied at a nominal voltage of 28 volts d-c. The board may be connected to the computer R-bus for accessibility by the system control.

The relay driver board is used principally as an interface between the system software control and the power devices located in the I/O module.

Table 3-8. Relay Driver Testpoints and LEDs

Testpoint	Function
TPl	Reset. Connecting testpoint l to testpoint 2 resets only this board. This function isn't useful in the LCI system.
TP2	DCOM.
LED	**Function**
O - F	A red LED for each of the 16 relay drivers indicates the energized status of each circuit.
ADR	Yellow LED. Turns on when the board is addressed; stays on if the board is addressed at least every 200 microseconds.
WPOK	Parity check. Not used in this system will always be on.

3-5. INPUT SIGNAL CONDITIONING – DS3800HISA

HISA is designed to work in conjunction with HRIA (in I/O module) to condition thirty high level (105 Vdc) signals to logic levels and allow access to them through the processor R bus. Each signal is separately isolated by a photocoupler and indicated by a bar LED. Two relay drivers included are only used in the I/O module's hardware selftest.

Table 3-9. Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LEDs

LED	Function
I00 - I1D	Green bar LED's. Each indicates the status of its associated input. Luminous when the input is energized.
IMOK	Yellow LED. On during normal operation. will be off only as the result of module selftest.

Electronics Module Options
The following functions are optional and may not be supplied. Refer to system documentation to determine presence.

3-6. TACHOMETER INTERFACE = DS3800HTCA

The HTCA accepts a two-phase pulse tachometer input to calculate electrical position of the synchronous motor shaft. Additional circuitry is included to allow a high accuracy calculation of motor speed.

Table 3-10. Tach Interface Berg Jumpers, Adjustments, Testpoints, LEDs.

Testpoint	Function
A	Tachometer signal - channel A
B	Tachometer signal - channel B
MK	Marker pulse
P12	Tachometer power supply.

LED	Function
IMOK	Yellow LED. On during normal operation. will be off only as a result of selftest.

3-6.1. I/O Addressing .

The following addresses are used to obtain the speed and position information.

Address	Function
811E	Upcommitted counter No. 2
811C	Speed time counter No. 2
811A	Speed pulse counter No. 2
8118	Speed counter 2 status register
8116	Uncommitted counter No. 1
8114	Speed time counter No. 1
8112	Speed pulse counter No. 1
8110	Speed counter 1 status register
8102	Position counter high byte
8100	Position counter low byte

3-6.2. Signal Interface.

JA - 10 pin ribbon connector for signal interface.

Pin	Nomenclature	I/O	Description
1	A	I	Tachometer input A
2	ARET	I	Tachometer input A return
3	B	I	Tachometer input B
4	BRET	I	Tachometer input B return
5	CIP	I	Cable in place input
6	N5	O	Cable in place -5 volt reference output
7	M	I	Tachometer marker
8	MRET	I	Tachometer marker return
9	ACOM	PWR	12 volt power supply return
10	P12	PWR	12 volt power supply output

3-6.3. Tachometer Power Supply. A +12 volt regulated power supply is provided with outputs at the 10-pin JA connector at the front edge. This supply will source loads demanding up to 150 mA.

3-6.4. Tachometer Input Conditioning. Three differential amplifiers are available for conditioning tachometer signals. Two circuits are dedicated to conditioning channels A and B, the two phase shifted frequency signals originating at the tachometer. The third may be used to buffer either a marker pulse or a direction signal. For all of the circuits, inputs are received differentially and filtered (5 microsecond time constant).

The two frequency signals are applied to a counter control circuit, constituted by a state-machine, which instructs the 12-bit position counter to count up, count down, reset, or reload, depending on tachometer state changes. This state machine is responsible for maintaining a position count which truly reflects the position of the tachometer.

3-6.5. Position Counter. The 12-bit position counter is allowed to attain values between zero and

some maximum count depending on tachometer activity. Each electrical cycle of the tachometer change the state of the counter. The information held by the position counter's output is stable during any read position.

3-6.6. High Accuracy Speed Circuit. The high accuracy speed circuit, when combined with appropriate software, can calculate rotational speed with sufficient accuracy to yield a 0.05% speed regulator. The high accuracy speed circuit makes use of two banks of counters, each constituted by a counter to accumulate tach pulses and a counter to accumulate 1 micro second clock pulses. Speed is determined by dividing the accumulated tachometer pulses by the accumulated clock pulses.

3-7. DEDICATED DUAL PORT MEMORY = DS3800HXMA (IF USED)

The HXMA allows communication between the L-bus and the F-bus through 32 bytes of onboard random access memory (RAM), LCI application uses lower 16 bytes. Both busses access this RAM via directed read and directed write. Berg jumpers select memory addresses and boundaries, and internal or external busswap control.

Table 3-11. Dual Ported Memory Berg Jumpers, Testpoints and LEDs

Berg jumper	Function	Position
LAB5	Address line 6 RAM select - L bus	False
LAB6	Address line 7 RAM select - L bus	False
LAB7	Address line 8 RAM select - L bus	False
LAB8	Address line 9 RAM select - L bus	True
FAB5	Address line 5 RAM select - F bus	False
FAB6	Address line 7 RAM select - F bus	False
LS1	Memory block 1 select - L bus	True
LS2	Memory block 2 select - L bus	True
LS3	Memory block 3 select - L bus	True
LS4	Memory block 4 select - L bus	True
FS1	Memory block 1 select - F bus	True
FS2	Memory block 2 select - F bus	True
FS3	Memory block 3 select - F bus	True
FS4	Memory block 4 select - F bus	True
TIML	L-bus swap control	Int
TIMF	F-bus swap control	Int

Testpoint	Function
DCOM	TP1 - Digital common

LED	Function
SWAP (CR1)	Indicates activity on the L-bus or the F-bus

3-7.1. RAM Access. Two RAM chips on-board the HXMA contain 32 bytes of memory space divided into four blocks of eight bytes. The L-bus and the F-bus can access both RAM chips in that memory space, but each bus is limited to one chip at a time. Since one RAM chip is always available to a bus, the two busses can access memory simultaneously without contention problems. Berg jumpers provide options for RAM addressing and memory block select.

3-8. S/6 RCVR = DS3800HXRA/DS3800DXRA (IF USED)

The DS3800HXRA S/6 RCVR with daughter board DS3800DXRA allow communication between the Series Six (S/6) parallel bus and the DIRECTO-MATIC™ Plus full bus (F-bus). These boards covert the S/6's differential signals into signals used by the F-bus. The HXRA and DXRA, which function only when tied together as a unit, create write and read strobes along with address and data, enabling transfer of information to I/O boards along the F-bus. The HXRA/DXRA also converts F-bus signals into differential signals and transfers them to the Series Six CPU.

Table 3-12 S/6 RCVR Berg Jumpers, Testpoints, LEDs and Switches

DXRA BERG JUMPER CONNECTIONS

Berg jumper	Function	Position
J1	Allows demand input	DMD
J2	Connects oscillator frequency to board	RUN
J3	Page address bit 7	True
J4	Page address bit 6	True
J5	Page address bit 5	True
J6	Page address bit 4	True
J7	Page address bit 3	True
J8	Page address bit 2	True
J9	Page address bit 1	True
J10	Page address bit 0	True

HXRA BERG JUMPER CONNECTIONS

J1	Causes chain fault when Sl is "off"	STR
J2	External reset	ST
J3	Bus cable shield	GND
J4	Chain start (-FIN)	CS
J5	Chain start (+FIN)	CS

LEDs (DXRA)

LED	Function
CHAIN PRTY	(CR1) Off when parity error in chain
CHAIN OK	(CR2) Off when chain broken
STA. ACT.	(CR3) On when station in use
LOCAL PRTY	(CR4) Off when local parity error
BUS ACT.	(CR5) On when S/6 bus active

Switches (DSXRA)

Switches (DXRA)

Switch	Function
S1 (STATION)	Disables HXRA/DXRA when in "off" position

3-9. DMC/LCI SIGNAL DEFINITIONS

3-9.1. DMC Outputs/LCI INPUTS.

DLBRMP	-Command to bypass ramp conditioning (main reference only)
DLCRP	-Command to apply creep reference
DLFRMP	-Command to select the fast ramp conditioning
DLFWD	-Command to apply forward reference
DLHCL	-Command to select hi current limit
DLJOGF	-Command to jog forward
DLJOGR	-Command to jog reverse
DLMCL	-Command to close motor contactor
DLRESET	-Command to reset the drive faults/alarms
DLREVREF	-Command to apply reverse reference
DLRPER	-Command indicating it is permissive to run (OCOAST stop)
DLRUN	-Command to apply run reference (metals mode)
DLSPD2	-Binary encoded command selecting speed reference 1 or 2
DLSPD3	-Binary encoded command selecting speed reference 3

DLSTRM	-Command to add DLSSPEEDSTRIM to the speed set point
DLTOG	-Bit that is toggled every access of the drive
DLOTOG	-Complement of DLTOG
DLUNSU	-Command to unsuicide the drive
DLSUIC	-Complement of unsuicide

3-9.2. LCI Outputs/DMC Inputs.

DLESTP	-Indicates that a emergency stop is in force
DLXFLT	-Input that indicates a external fault
DLFLOS	-Input that indicates a field loss
DLFOC	-Field over current
DLGFLT	-Indicates that a ground fault exists
DLLOCL	-Indicates that the drive is offline and in the local mode
DLMOC	-Input that indicates a drive overcurrent
DLMOS	-Input that indicates a drive overspeed
DLMTL	-Indicates a motor tach loss
DLMOV	-Input that indicates a drive overvoltage
DLNF	-Indicates there are no faults in the drive
DLCOC	-Indicates a power converter module overcurrent
DLCACF	-Indicates a power converter module ac fault
DLCDCF	-Indicates a power converter module dc fault
DLRR	-Indicates the drive is ready to run
DLRSTO	-Command to reset drive
DLRUNR	-Indicates the run reference is applied
DLZS	-Indicates the drive is at zero speed
DLFLDON	-Active when the field thyristors are being gated
DLMTUN	-Indicates that the drive is in the tune up mode
DLON	-Active when the armature thyristors are being gated

3-9.3. Digital Integer Signals. Inputs from master controller — All inputs are 16 bit 2's complement signed integers.

DLCREF
-This is the current reference to the current regulator or an additive current reference or the value of the positive and/or negative clamp of the speed regulator depending on the option selected.
-Scaling current (1 PU) = 1024 cts
-Min/max = 2048 cts

DLSREF
-This is the speed reference to the speed regulator
-Scaling 1 p.u. (speed) = 28000 (top speed)
-Min/max +- 32767

DLSTM
-This is the speed trim reference to the speed regulator
-Scaling 1 p.u. (speed)= 28000 (top speed)
-Min/max ± 32767

Outputs to master controller
All outputs are 16 bit 2's complement signed integers.

DLFIA
-This is the armature current feedback.
-Scaling current 1 PU) = 1024 cts
-Min/max = 2048 cts

DLMTV
-This is the filtered motor terminal volts feedback.
-Scaling 1 p.u. (motor volts) = 128

DLSPDFB
-This is the speed feedback.
-Scaling 1 p.u. (speed) = 28000
-Min/max +- 1.17 p.u. = +- 32767

DLSRROUT
-This is the unclamped speed regulator output (reference to current regulator).
-Scaling current (1 PU) = 1024 cts
-Min/max = 2048 cts

DLKWATT
-Calculated value of instantaneous motor power.
-Scaling (1 p.u.)= 1024 cts

3-10. DISPLAY MODULE (LDM)

The DS3820LDM is the display module for the structured LCI control panel. The display provides fault, alarm and diagnostic information. The display module comes in two basic layouts. Layout Option 1 provides the following features.

-3 indicator lights (RUN, ALARM and FAULT)

-3 1/2 digit motor speed meter (RPM)

-24 column printer with real-time clock and layout Option 2 provides the following features:

-4 push-to-test indicator lights (RUN, ALARM, FAULT and READY)

-2 push buttons (RESET and DRIVE STATUS)

-4 DB40 type meters (motor speed, motor current, motor voltage and horsepower)

All lights are on a 28 V circuit. The functions and color codes for each light are:

Red for the motor RUN light

Red for a system FAULT light

Amber for a system ALARM light

Green for the system READY light

The push buttons are connected to a 28 V circuit and direct a 28V signal to the output when the button is depressed. The RESET push button is used to reset latched faults in either software or hardware (NLIB and NLCB latched faults - overspeed, suppression, overvoltage, etc.). The DRIVE STATUS push button; when depressed, provides information on the motor voltage, current, speed, etc. via the printer.

The digital meter is a dc voltage measurement panel type. The conversion factor for the meter is 2 volts correspond to 1999 cnts. The panel meter is powered by the 5 volt bus via the 26-pin ribbon cable.

The four DB40 type analog meters are 10 volt movement meters with +/- 10% adjustment rheostats (back).

The printer included in the display module is a Digitec Model 6610 Panel mount printer. The 24 column printer inputs data and control signals via a serial RS232 connection and uses an impact print head mechanism to output the data. The printer also supplies a real-time clock that will be used to log the time of status information and system failures.

The LDM interfaces to the LCG thru a 26-pin ribbon cable JK.

3-11. SIGNAL ISOLATION MODULE (SIM)

The DS3820SIM is used to provide optional functions to and LCI control panel. In particular, it provides four impedance isolated signal outputs for use in a purchaser's external control system. The four signals may be selected as 0 to 10 volts or 4 to 20 mA outputs by header connection jumpers at JM. Each circuit may be selected independently. Inputs to the module (TEM) are received from the LCI control SEM via 26-pin ribbon cable (JM). The four inputs to the isolation circuits may be chosen by header connector jumpers at JB from the following list of signals: Source volts, source current, SPD reference, motor SPD, horsepower, DA1, DA2, exciter current, motor volts.

A DS3800NVAA provides the functions of voltage to current source transformation and isolation. A differential amplifier accepts the signal from the remote control SEM. The signal is isolated from the NVAA on-board ground by approximately 100 K ohms of resistance. The isolated signal is then applied as a reference to a current source with the gains weighted such that a 0 to 10 volt signal provides a delta output of 16 milliamperes. An additional bias is provided to give the 4 milliamperes at 0 volts input. A chopper isolated power supply is present on the NVAA board to power the output stages of the signal isolators. The output of each of the four circuits is referenced to XCOM, the on-board power supply common. It is intended that the XCOM connection be tied to the customer's external circuit common. While it is not necessary for the signals used as inputs to the isolators be referenced to the customer's common or XCOM, the difference between ACOM and XCOM must not exceed 3 volts steady-state or transiently. The voltage developed across any external impedance in series with a current source must a not exceed 10 volts.

The output signals are also available as voltage followers (0 to 10 V). Maximum output load is 1 mA.

An operational amplifier monitor (or saturation detector) is provided for monitoring the output stages of the current driver circuits and LED, "IMOK", is luminous as long as no fault condition exists. In addition, if FUSE 1 in the on-board power supply is blown, then the LED will be non-luminous.

3-12. I/O MODULE (IOM)

The purpose for the DS3820IOMA (I/O) module is to provide signal conditioning and connection for interpanel and purchaser wiring in a load commutated inverter (LCI) control panel. The following functions are provided.

1. Signal conditioning and isolation for interface wiring at the 105Vdc level.
2. Isolated signal interlocks (form C) to remote locations as a function of programmed system requirements. Rated .25 A, 115 Vac (Resistive).
3. Fused 105 Vdc power supply to provide a source of high level power for interface.
4. Power relay control for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10 A, 600 Vac (Inductive).
5. Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).
6. Terminal boards for terminating interconnecting wiring.
7. Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits and feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

Table 3-13 Signal Isolation Module (SIM)

Testpoint	Function
TP1	Voltage output of channel 1 relative to XCOM.
TP2	Voltage output of channel 2 relative to XCOM.
TP3	-2.5 Vdc bias for V-1 circuits.
TP4	Voltage output of circuit 3 relative to XCOM.
TP5	-2.5 Vdc bias voltage for I-V Circuit.
TP6	XCOM, isolated supply ground.
TP7	Voltage output of channel 4 relative to SCOM.
TP8	Voltage output of 4-20 mA to voltage converter.
TP9	N15X, isolated supply.
TP10	P15X, isolated supply.
TP11	ACOM, analog power supply ground.

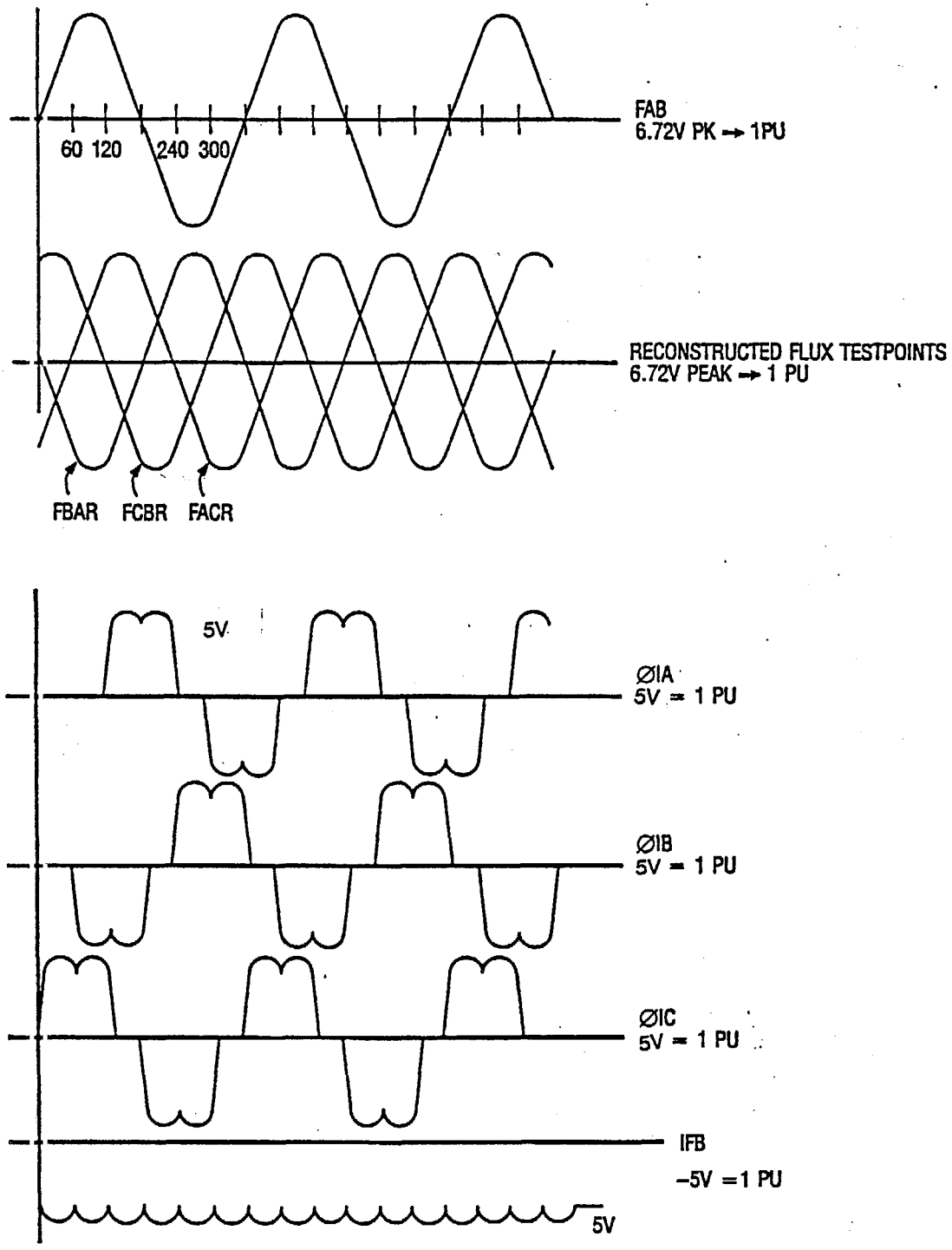


Figure 3-8. Analog Typical Waveforms

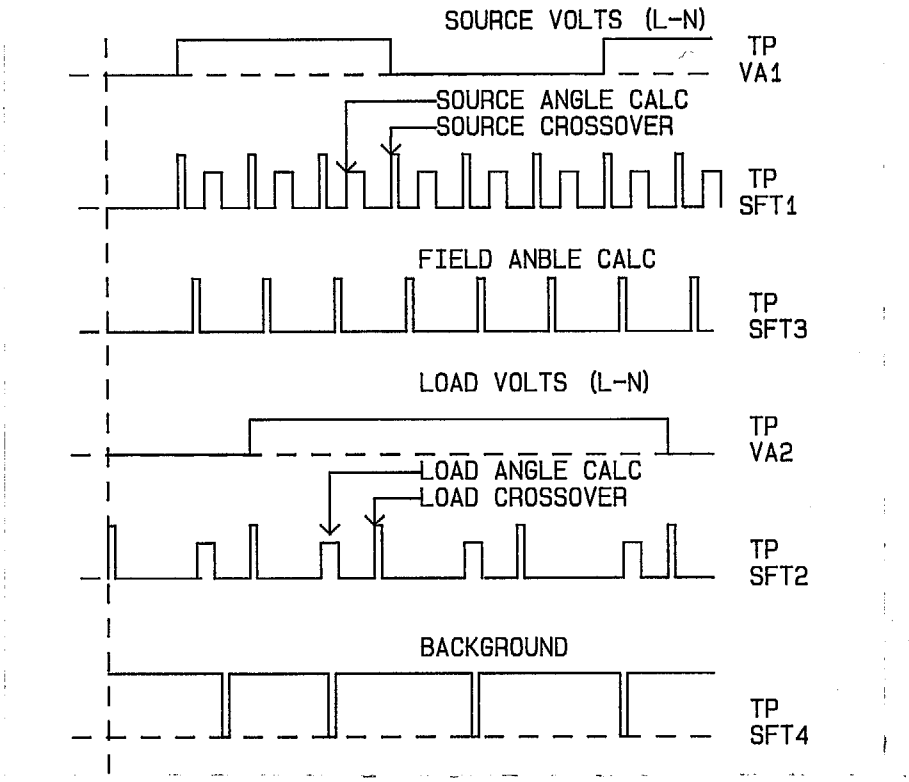


Figure 3-9. Interrupt/Angle Control Routine

CHAPTER 4

FIELD ADJUSTABLE PARAMETERS

SECTION A - EEPROM (FIELD) VARIABLES

EEPROM Memory is a non-volatile memory space used to hold critical drive constants that may be system dependent. The use of EEPROM Memory allows the drive constants to be changed in the field in order to fine tune the system performance. EEPROM Memory space resides on the DMPB card (HMPG daughter board) and is configured to contain 1024 bytes (1K x 8).

Default values for the drive constants are burned into the ROMs residing on the HMPG (the ROMs also contain the executable software to run the drive). These default values are intended to provide a reasonable starting base to from when tuning the drive. They may be loaded into the EEPROM Memory by turning the DMPB write enable switch "on", jumpering test-point TINO on the HLCB pwb to DCOM and then executing a hardware SEM reset. In order to change a value in the EEPROM memory, turn the DMPB write enable switch "on" and use the monitor modify command.

After power up or hardware reset, the microprocessor on the HMPG makes a local copy (shadow) in RAM of the EEPROM Memory. During normal running, the microprocessor accesses the drive constants from this local RAM shadow.

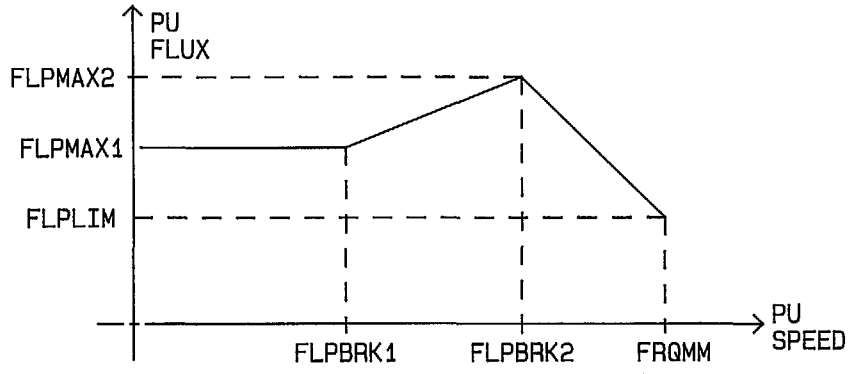
Table 4-1. Description of Variables

Name	Group	Function
CRALPH1	(CR)	Variable used in current regulator to improve transient response during segment fire mode. The variable sets the initial value of LINE COM (firing angle) such that the desired value of current is obtained after the first firing. The following value is recommended. CRALPH1 = 90 degrees Resolution is 1 degree and the range is 60 to 90 degrees. The associated software variable and its relationship to CRALPH1 is given below. CUR__ REG_ FIXED_ ALPH1 = 16384 * COS(CRALPH1)
CRALPH2	(CR)	Variable used in current regulator to improve transient response during segment fire mode. The action of the variable is to precondition the regulator integrator to command a firing angle necessary to obtain forced commutation current limit value. CRALPH2 = ARCCOS[(4NE+ IR) + (.075FPU*VM)+ (XTpu*Ipu/2)]/(1.35*Ell) where E = cell drop (1.25V) N = number of cells/leg FPU = per unit field at .O75 per unit speed I = current(amps) corresponding to current limit value Ipu = value of current in pu R = system resistance (mostly reactor) VM = motor L-L volts (RMS) Ell = transformer sec volts (rms) XTpu = transformer comm reactance Resolution is 1 degree and range is 60 to 90 degrees. The associated software variable and its relationship to CRALPH2 below. CUR__ REG__ FIXED__ ALPH2 = 16384 * COS(CRALPH2)

Name	Group	Function
CRGAIN	(CR)	Source side current regulator gain. The variable's relationship to system parameters is given below. CRGAIN = (759*IM*WC*L)/(VL-L) IM = fund rms rated motor current V_{L-L} = rated rms L-L source volts WC = desired regulator crossover (250RAD) L = inductance in conducting path = reactor + 2 (motor comm. reactance) + 2 (transf. comm. reactance). F = source side frequency Resolution is units and range is 10 to 1000. The associated software variable and its relationship to CRGAIN is given below. CUR__ REG__ G2 = 16*CRGAIN/(6*FRQS)
CRMIN	(CR)	Variable setting the minimum value of dc link current. Normally set to maintain continuous current. The link reactor is usually picked to maintain continuous current at .2 per unit. CRMIN = per unit current Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to CRMIN is given below. CUR REG MIN LMT = CRMIN * 1024
CRSTART	(CR)	Value of current used in fixed fire for breaking away the motor. Also the current command in speed regulator open test mode. The value of CRSTART should be that which would give maximum torque at the stand still field excitation. CRSTART = PU FLD (AT STAND STILL) Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to CRSTART is given below. CUR__ REG__REF__ START = CRSTART * 1024
CRTC	(CR)	Time constant used to stabilize the source side current regulator. CRTC = time constant (in seconds) Resolution is .001 seconds and the range is .001 to .500 seconds. The associated software variable and its relationship to CRTC is given below. CUR__ REG__ G1 = 4*CRGAIN*CRTC
FANGLAG	(FC)	Variable used to compensate field firing algorithm for phase shift between source side and field power. The variable is positive for lagging and negative leading. Resolution is 1 degree and range is ± 30°. Associated software variables and its relationship to FANGLAG is given below. FIELD__ PHASE__ ZERO__ E = (FANGLAG/360) *1024
FCDELAY	(FC)	Timer controlling the delay to allow field to build up before attempting a drive start FCDELAY = delay time in seconds Resolution is .1 and the range is .1 to 10 seconds. The associated software variable and its relationship to FCDELAY given below. FLD__ DELAY__ LIMIT = FCDELAY * FRQS

Name	Group	Function
FCFTIME	(FC)	Controls number of consecutive field overcurrents allowed. A penalty function is implemented such that 6 good firings following an overcurrent will cancel an overcurrent firing. This allows recovery from non-repetitive faults. FCFTIME = number of consecutive overcurrents. The resolution is units and the range is 1 to 6. The associated software variable and its relationship to FCFTIME is given below. FLD__ FAULT__ PERIOD = (FCFTIME - 1) * 6
FCGAIN	(FC)	Field current limit function gain. The gain is set to allow a 2% field current limit with 30% overdriving flux. A value of 100% is typical. FCGAIN = required regulator gain Resolution is 1 and range is 0 to 200 The associated software variable and its relationship to FCGAIN is given below. FLD__CUR__GAIN = FCGAIN
FCLIM	(FC)	Sets the trip level for the field overcurrent function. The spillover field current limit value is set at 94% of this value. FCLIM = desired per unit field where 1 per unit field is on the base of the field exciter. Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to FCLIM is given below. FLD__CUR__LMT = FCLIM * 1024
FCLOSS	(FC)	Field loss setting for field loss fault function. FCLOSS = desired per unit field loss setting Note that one per unit field is on the base of the field exciter rating. Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to FCLOSS is given below. FIELD__LOSS__SETTING = FCLOSS * 1024
FCMAX	(FC)	Field thermal overload function. Scales minimum field thermal rating at zero speed. Must be entered on the base of the field exciter. Resolution is .01 and the range is .05 to 1.2. The associated software variable and its relationship to FCMAX is given below. MAX__FLD__LIM__E = FCMAX * 1024
FCMIN	(FC)	Field thermal overload function. Scales minimum field thermal rating at zero speed. Must be entered on the base of the field exciter. Resolution is .01 and the range is .05 to 1.2. The associated software variable and its relationship to the FCMIN is given below. MAX__FLD__LIM__E = FCMIN * 1024
FCOLEN	(FC)	Field thermal overload function. Used to enable or disable the field thermal overload function. Possible values for FCOLEN are "T"rue or "F"alse.

Name	Group	Function
FCSLOP	(FC)	Variable used in field thermal overload function to define cooling effect as a function of speed. FCSLOP = DELTA PU field current. Note the per unit base is the base rating of the field exciter. Resolution is .01 and the range is .01 to 1.2 per unit. The associated software variable and its relationship to FCSLOP is given below. FLD__SPD__FACTOR__ E = 16 * [FRQM/60]/FCSLOP
FCSTART	(FC)	Sets minimum field current for recovering from field overcurrent during forced commutation. Must be coordinated with value obtained from field precondition. FCSTART = per unit field current where 1 per unit field is on the base of the field exciter. Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to FCSTART is given below. FLD__CUR__STRT__LMT = FCSTART * 1024
FLAGAIN	(FL)	Variable used with the flux regulator routine to set gain of flux adjust integrator to allow source fire-as-early-as-possible. FLAGAIN = required gain The recommended value is 1. The range is 0.01 to 2 with a resolution of .01 The associated software variable and its relationship to FLAGAIN is given below. FLUX__REG__ADJ__G = FLAGAIN * 100
FLALEN	(FL)	Variable used to control enabling/disabling of the flux adjust integrator to allow fire as early as possible on the source side. FLALEN — REGULATOR CONDITION F — disabled T — enabled Only values of F or T are allowed. The associated software variable and its relationship to FLALEN is given below. FLUX__ ADJ__MODE__E = FLASEN or SHL(FLALEN)
FLALIM	(FL)	Variable used with the flux regulator controlled from source fire as early as possible regulator. Function is to limit the range of flux adjustment. FLALIM = desired per unit flux Resolution is .01 units and the range is 0 to .50 The associated software variable and its relationship to FLALIM is given below. FLUX__REG__ADJ__LMT = FLALIM * 8254
FLASEN	(FL)	Variable used to control enabling/disabling of the flux adjust as a function of source voltage variations. FLASEN — REGULATOR CONDITION F — disabled T — enabled Only values of F or T are allowed. The associated software variable and its relationship to FLASEN is given below. FLUX__ADJ__MODE__E = FLASEN or SHL(FLALEN)

Name	Group	Function
FLDPU	(PT)	Variable used in the "print status" routine to define the value of 1 per unit exciter current. The value should be entered in terms of actual exciter current rating (in amps). FLDPU = rated exciter current Resolution is 1.0 and the range is 0 to 100.
FLEARLY	(FL)	Enables source fire-as-early-as-possible regulator as a function of drive speed. FLEARLY = desired per unit speed Resolution is .001 units range is .1 to 2.0 per unit speed. The associated software variable and its relation ship to FLEARLY is given below. SFAEAPR__SPD__ENABLE = FLEARLY* (1024/60)*FRQMM
FLPBRK1	(FL)	Variable used in "FLUX SETPOINT VS SPEED" function in flux regulator. Sets beginning of first slope. See variable FLPMAX1 for profile. FLPBRK1 = desired per unit speed Resolution is .001 and the range is 0 to 2.000 per unit. The associated software variable and its relationship to FLPBRK1 is given below. FLUX__PROFILE__SPD__BRK1 = FLPBRK1*(1024/60)*FRQMM
FLPBRK2	(FL)	Variable used in "FLUX SETPOINT VS SPEED" function in flux regulator. Sets takeover for second slope See variable FLPMAX1 for profile. FLPBRK2 = desired per unit speed Resolution is .001 and the range is 0 to 2.000 per unit The associated software variable and its relationship to FLPBRK2 is given below. FLUX__PROFILE__SPD__BRK2 = FLPBRK2*(1024/60)*FRQMM
FLPLIM	(FL)	Variable specifying flux level at rated speed. See variable FLPMAX1 for profile. FLPLIM = desired per unit flux Resolution is .01 units and range is 0 to 1.5 The associated software variable and its relationship to FLPLIM is given below FLUX __PROFILE__FINAL = FLPLIM * 8254
FLPMAX1	(FL)	Variable used in "FLUX SETPOINT VS SPEED" function in flux regulator. Sets initial per unit flux. FLPMAX1 = desired per unit flux  Resolution is .01 units and the range is 0 to 1.50 The associated software variable and its relationship to FLPMAX1 is given below. FLUX__ PROFILE__ MAX1 = FLPMAX1 * 8254

Name	Group	Function
FLPMAX2	(FL)	Variable used in "FLUX SETPOINT VS SPEED" function in flux regulator. Sets peak flux. See variable, FLPMAX1 for profile. FLPMAX2 = desired per unit flux Resolution is .01 units and the range is 0 to 1.50 The associated software variable and its relation ship to FLPMAX1 is given below. FLUX__PROFILE__MAX2 = FLPMAX1 * 8254
FLUXS	(MI)	Used in the starting routine from zero speed. Sets the amount of flux required to transition to segment firing mode. FLUXS = desired per unit flux Resolution is .01 and the range is 0 to 2.0 PU. The associated software variable and its relationship to FLUXS is given below. START__FLUX__E = FLUXS * 2752
FRCLMP	(FR)	Sets the maximum advance angle of the exciter voltage controller. Generally will be set to 1 per unit exciter volts. When set to some other value the maximum RMS exciter voltage will be approximately proportional to the value of this variable. FRCLMP = desired per unit output volts Resolution is .01 per unit and the range is 0 to 1.2. The associated software variable and its relation ship to FRCLMP is given below. FLUX__ REG__OUT__CLMP = FRCLMP * 255
FRGAIN	(FR)	Variable used in the flux regulator to set the regulator integral gain for a nominal 5 radian regulator. The following equation holds. FRGAIN = 1/(63.68 * Geq * MEV) where Geq = motor equivalent transfer function from brushless exciter input flux. MEV = maximum exciter volts Resolution is .01 and the range is .01 to 2.00 The associated software variable and its relationship to FRGAIN is given below. FLUX__REG__G2 = FRGAIN * 16384/ FRQS
FRQS	(MI)	Source side nominal frequency. The variable is used in various regulator gain and time constant calculations. This variable only goes into effect after resetting the processor. Resolution is 1 Hz and range is 45 to 90 Hz.
FRQM	(MI)	Load side nominal frequency. The variable issued in various regulator gain and time constant calculations. This variable will only affect the drive tune-up after resetting the processor. Resolution is 1 Hz and range is 40 to 90 Hz.
FRQMM	(MI)	Motor top speed frequency should correspond to motor frequency at drive RPM entered for variable MRPM but should no be less than FRQM. When the variable is changed, the value should be saved and the processor should be reset before the desired effect will be obtained.

Name	Group	Function
FRSTART	(FR)	Variable used in the flux regulator used to initialize the flux regulator integrator to call for a value of exciter volts which gives approximately the correct flux value. Also sets the value of flux in forced commutation when the flux regulator is not active. FRSTART = desired per unit output volts Resolution is .01 per unit and the range is 0 to 1.0. The associated software variable and its relation ship to FRSTART is given below. FLUX __REG__PRECONDITION__ E = (FRSTART * 510 - 255)*8
FRTC	(FR)	Variable selecting the flux regulator stability time constant. Normally the value is selected equal to the motor field time constant. FRTC = desired time constant (seconds) Resolution is .1 and the range is .2 to 10 seconds. The associated software variable and its relation ship to FRTC is given below. FLUX __REG__G1 = 2048 * FRGAIN * FRTC
HPPU	(PT)	Variable used in the "print status" routine to define the value of 1 per unit drive horsepower. Generally, the per channel horsepower is used as the base in 12 pulse systems. HPPU = rated channel horsepower. Resolution is 1.0 and the range is 0 to 10,000.
IDCPU	(PT)	Variable used in the "print status" routine to define the value of 1 per unit link current in amperes. 1DCPU = rated link current (amperes) Resolution is 1.0 and the range is 0 to 2,000.
INTOSTOL	(MI)	Variable used in test to determine that flux integrators are balanced before enabling the load side phase- lock- loop. INTOSTOL = desired setting (in per unit flux) Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to INTOSTOL is given below. INTEGRATOR OFFSET TOL = INTOSTOL * 1376
LIMARG	(MI)	Variable scaling load-side commutating angle to motor commutating impedance. Used to compensate margin angle for commutating volt-seconds. When operating in position mode, LIMARG is also used to scale the power angle calculation. LIMARG = per unit commutating reactance Resolution is .01 per unit and the range is .03 to .30 per unit. - The associated software variable and its relationship to LIMARG is given below. ALPH__LD__IMP__MARG = 180.15 * LIMARG
LKLIMD	(PR)	Sets allowable difference between source and load side measurements of link current. If the difference is exceeded then Commutation failure fault is declared. LKLIMD = per unit current difference allowed Resolution is .01 per unit and the range is 0 to 2.0 per unit. The associated software variable and its relationship to LKLIMD is given below. IDC __DIFFERENCE __LIMIT = LDLIMD * 1024

Name	Group	Function
LKLIML	(PR)	Sets level of load current which is declared an overcurrent fault. LKLIML = desired per unit current setting Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to LKLIMS is given below. IDC__ LINK__LMT__LD = LKLIML * 1024
LKLIMS	(PR)	Sets level of source current which is declared an overcurrent fault. LKLIMS = desired per unit current setting Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to LKLIMS is given below. IDC__ LINK__ LMT__ SRC = LKLIMS * 1024
LKOCTIM	(PR)	Controls number of consecutive source overcurrents allowed. A penalty function is implemented such that 6 good firings following a failure will cancel a fault firing. This allows recovery from non-repetitive faults. LKOCTIM = number of consecutive overcurrents. The resolution is units and the range is 1 to 6. The associated software variable and its relationship to LKOCTIM is given below. IDC__LINK__LMT__PERIOD = (LKOCTIM - 1) * 6
LMARG	(MI)	Variable used to control the fixed margin angle for firing the load side. A fixed margin angle of 15 degrees at zero load is normally used. LMARG = desired margin angle in degrees Resolution is 1 degree and the range is 0 to 30 degrees. The associated software variable and its relationship to LMARG is given below. ALPH__FIXED__MARG__LD = 1376*[1+COS(180-LMARG)]
LPLLTIM	(PR)	Sets time delay before a load phase lock loop fault is declared following an error determination LPLLTIM = required time in seconds Resolution is .01 seconds and the range is 0 to 1.5 seconds. The time above will be at rated motor frequency with the time delay inversely proportional to speed otherwise. The associated software variable and its relationship to LPPTIM PLL__OUT__OF__TOL__LD__PERIOD = LPLLTIM * 12 * FRQM
MKPHASE	(MI)	Position mode variable. Scaled in electrical degrees, this variable defines the number of degrees that the motor A—N voltage (- to + transition) lags the tachometer marker pulse when the motor power angle is 0 degrees. Resolution is 1 degree and the range is 0 to 360 degrees. The associated software variable and its relationship to LMARG is given below: MK__PHASE__ E = MKPHASE
MRPM	(PT)	Variable used in the "print status" routine to define the value of 1 per unit motor RPM. Also defines max motor RPM for scaling the tachometer. MRPM = rated motor RPM Resolution is 1.0 and the range is 0 to 5000.

Name	Group	Function
MTRFMIN	(MI)	Variable used in test to determine if the magnitude of motor flux is adequate to enable the load side phase lock loop. MTRFMIN = desired setting (in per unit flux) Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to MTRFMIN is given below. MTR__FLUX__MIN__TOL = 2*1376*MTRFMIN
MVPU	(PT)	Variable used in the "print status" routine to define the value of 1 per unit motor volts. MVPU = rated motor volts Resolution is 1.0 and the range is 0 to 5000.
OVCLIM	(OV)	Variable controlling the maximum current limit called for by the overvoltage regulator. A setting is 1.5 per unit current. OVCLIM = required per unit stator current Resolution is .01 and the range is 0 to 2.0 per unit. The associated software variable and its relationship to OVCLIM is given below. CUR__REG__LMT__OVVOLT = OVCLIM * 1024
OVGAIN	(OV)	Variable controlling the gain of the overvoltage regulator. Presently the overvoltage regulator is selected for a crossover of 10 radians. This results in OVGAIN = 2.4 Resolution is .1 units and range is 0 to 20 The associated software variable and its relation ship to OVGAIN is given below. OVVOLT__REG__GAIN = 256*OVGAIN
OVSPIL	OV	Variable controlling the setpoint of the voltage regulator which is configured in a spillover mode. OVSPIL = per unit motor volts Resolution is .01 per unit and range is 0 to 2.0 per unit. The associated software variable and its relationship to OVSPIL is given below. OVVOLT__ REG__SPLOVR = 1.146 * FRQM * OVSPIL where F = rated motor frequency Vl-lpu = motor voltage (P.U.) where voltage regulator first cuts in.

<table>
<tr><th>Name</th><th>Group</th><th>Function</th></tr>
<tr><td>OVTC</td><td>OV</td><td>Variable controlling an adjustable lag in the forward loop of the overvoltage regulator. The filter is in the forward loop of the voltage regulator and therefore, affects stability. The following table shows the values available

<table>
<tr><th>OVTC</th><th>T(SECONDS) (60Hz)</th><th>T (SECONDS) (50Hz)</th></tr>
<tr><td>0H</td><td>.01662</td><td>.0199</td></tr>
<tr><td>1H</td><td>.050</td><td>.06</td></tr>
<tr><td>2H</td><td>.1166</td><td>.140</td></tr>
<tr><td>3H</td><td>.245</td><td>.30</td></tr>
<tr><td>4H</td><td>.5166</td><td>.62</td></tr>
<tr><td>5H</td><td>1.05</td><td>1.26</td></tr>
<tr><td>6H</td><td>2.116</td><td>2.54</td></tr>
<tr><td>7H</td><td>4.25</td><td>5.07</td></tr>
<tr><td>8H</td><td>8.52</td><td>10.22</td></tr>
<tr><td>9H</td><td>17.05</td><td>20.46</td></tr>
</table>

The recommended value is OVTC = 6
The associated software variable and its relationship to OVTC is given below.
OVVOLT_REG_TC = OVTC</td></tr>
<tr><td>PLLMODE</td><td>MI</td><td>Variable used with the load phase lock loop regulator to determine the speed at which the phase lock loop switches from 6 to 3 calculations per cycle.
PLLMODE = desired (Hz)
Resolution is 1 Hz and the range is 1 to 90 Hz.
The associated software varirable and its relationship to PLLMODE is given below
PLL_SPD_MODE_3_LD_ =7200/PLLMODE</td></tr>
<tr><td>PLLTOL</td><td>PR</td><td>Sets maximum allowable error in load phase lock loop. If error is exceeded then a phase lock loop fault is declared.
PLLTOL = allowable error (in degrees)
Resolution is units and the range is 0 to 20 degrees.
The associated software variaable and its relationship to PLLTOL is given below.
PLL_ERROR_TOL_LD = PLLTOL * (1024/360)</td></tr>
</table>

Name	Group	Function
PO1 PO2 PO3 PO4 PO5 PO6	(PT)	Variables used in the print status routine to select the variables from the following list to be printed on the door mounted printer. The number of the information to be printed is entered as data into each of the variables. If no data for a particular entry is required, then it should be set to zero. Note that the print-out will be formatted in the order of the PO-variables with PO1 first. 0 - No print out 1 - Motor RPM 2 - Speed reference (RPM) 3 - Motor line current 4 - Field current 5 - Motor Horsepower 6 - DC link volts 7 - DC link amperes 8 - Motor line volts 9 - Source amperes 10- Source line volts 11- Source firing angle 12- Load firing angle 13- Motor power factor 14- Source power factor 15- Source kVA 16- Current limit setpoint 17- Voltage limit setpoint 18- Status DCL, volt limit & turning gear
POE	(PT)	Variable used in the "print status" routine to enable the auto print feature. In this mode, the printer prints the selected information at time intervals selected by the printer time interval adjustment.. POE = "T"rue enables function = "F"alse disables function
POSF	(MI)	Variable used in the load side position mode calculations. Sets a minimum fixed margin even if current reading is near zero. POSF = D Where D is the number of electrical degrees of fixed margin. The resolution is units and the range is -15 to +60 degrees. A suggested setting is 10 degrees. POS_E = 19.4 * POSF

Name	Group	Function
RR6PA0 1	(RR)	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input range is 0.0001 to .250 per unit/sec. RR6PA( ) = Per unit speed change/ second The related software variable array is SPD__ RATE__ ACCL1__ 6P(0-1) and is related to the corresponding tuneup variable according to the following equation: ACCL__ 6P ( ) = (RR6PA( ))* 28000*64/FRQS
RR12PA0 1	(RR)	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input is 0.0001 to .250 per unit/sec. RR12PA( ) = Per unit speed change/ second The related software variable array is SPD__ RATE__ ACCL__12P(0-1) and is related to the corresponding tuneup variable according to the following equation: ACCL__ 12P( ) = (RR12PA( ))* 28000 * 64/FRQS
RR6PD0 1	(RR)	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input range is 0.0001 to .250 per unit/sec. RR6PD( ) = Per unit speed change/ second The related software variable array is SPD__ RATE__ DECEL__ 6P(0-1) and is related to the corresponding tuneup variable according to the following equation: __DECEL__ 6P( ) = (RR6P( ))* 28000*64/FRQS
RR12PD0 1	(RR)	Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult the drive ladder diagram to determine mode. Resolution is 0.01% and the input is 0.0001 to .250 per unit/sec. RR12PD( ) = Per unit speed change/ second The related software variable array is SPD__ RATE__ DECEL-12P(0-1) and is related to the corresponding tuneup variable according to the following equation: __DECEL__ 12P( ) = (RR12D( ))*28000*64)/FRQS
RRBP0	(RR)	Selects regions of drive speed where different acceleration/deceleration rates may be selected. The resolution is .1% and the range is 0 to 2.0 per unit speed. RRBP0 = DESIRED per unit SPEED The related software variable is SPD__ REF__ BREAK__ POINT) The relationship between the variables is defined below: __BREAK__ POINT0 = RRBP0*1024* (FRQMM/60) * 1024

Name	Group	Function
RRCREEP	RR	Data link mode variable. Selects creep speed reference. Resolution is .1% and the range is 0 to 1 per unit speed. The associated software variable and its relationship to RRCREEP is RRCREEP__E = RRCREEP * 28000
RRFS1 2 3	RR	Data link mode variables. Select fixed forward speed setpoints. Refer to ladder diagram for method of selecting. Resolution is .1% and the range is 0 to 1 per unit speed. The associated software variables and their relationship is shown below. RRFS__ E = RRFS__ * 28000
RRJOGF	RR	Data link mode variable. Sets jog forward speed. Refer to the ladder diagram for the method of selecting. Resolution is .1% and the range is 0 to 1 per unit. The associated software variable and its relationship is shown below. RRJOGF__E = RRJOGF * 28000
RRJOGR	RR	Data link mode variable. Sets jog reverse speed reference. Resolution is .1% and the range is 0 to 1 per unit. The associated software variable and its relationship is shown below. RRJOGR__ E = RRJOGR * 28000
RRMAX6P	(SR)	Maximum speed reference clamp active only in 6 pulse operation. RRMAX6P = Desired per unit speed, where 1 per unit speed is the motor rated spd Resolution is .1% and the range is 0 to 2.0 per unit. The associated software variable is SPD__ REG__ REF__ MAX__ 6P__ E The relationship between the variables is REF__ MAX__ 6P__ E = (RRMAX6P) * 28000
RRMAXL	(RR)	Variable controlling maximum speed reference scaling in the following speed modes: Mode — Definition 3 — Local volts ref - fwd operation only RRMAXL = Desired per unit speed, where 1 per unit speed is motor rated speed Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD__ REF__ MAX__ LMT This is related to the subject variable by __MAX__LMT = RRMAXL * 28000

Name	Group	Function
RRMAXR	(RR)	Variable controlling maximum speed reference scaling in the following speed modes: Mode — Definition 0 — Remote volts ref - fwd operation only 1 — Remote 4-20 mA reference 2 — Remote volts ref - reversing operation 4 — Up-down contact closure mode 8 — Data link communications RRMAXR = Desired per unit speed, where 1 per unit speed is the motor rated speed Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD__ REF__ PROCESS__ MAX This is related to the subject variable by __PROCESS__ MAX = RRMAXR * 28000
RRMINL	(RR)	Variable controlling minimum speed reference scaling in the following speed modes: Mode — Definition 3 — Local volts ref - fwd operation only RRMINL = Desired per unit speed, where 1 per unit speed is the motor rated spd Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD__ REF__ MIN__ LMT This is related to the subject variable by __MIN__ LMT = RRMINL * 28000
RRMINR	(RR)	Variable controlling minimum speed reference scaling in the following speed modes: Mode — Definition 0 — Remote volts ref - fwd operation only 1 — Remote 4-20 mA reference 2 — Remote volts ref - reversing operation 4 — Up-down contact closure mode 8 — Data link communications mode RRMINR = Desired per unit speed where 1 per unit speed is the motor rated speed Resolution is .1% and the range is 0 to 2.0 per unit speed. The related software variable is SPD__ REF__ PROCESS__ MIN This is related to the subject variable by __PROCESS__ MIN = RRMINR * 28000

Name	Group	Function
RRMODE	(RR)	Variable selecting the speed reference for normal operation. That is , the mode which will be selected if the drive is not in local, turning gear, or in emergency stop. The value of the RRMODE determines the mode according to the following RRMODE = 0 0 TO 10V unipolar reference proportional to speed thru the remote speed reference input - voltage source. RRMODE = 1 4-20 mA customer reference through the remote reference input. This is a unidirectional mode. RRMODE = 2 Bipolar 10 volt reference through remote speed reference input. Allows reversing operation. RRMODE = 3 Selects local speed reference and non-reversing operation only. RRMODE = 04 Allows control of speed reference by raise/ lower interlocks via I/O module. RRMODE = 05 Selects the turning gear speed reference set by RRTG. RRMODE = 06 Selects the reference set by variable RRMINR. RRMODE = 08 Selects speed references from the data link communications.
RRRS1 2 3	RR	Fixed speed point references which are selected by background sequencing as function of variables received over the data link. Resolution is .1% and the range is 1.0 per unit to 0. The associated software variables and their relationship is shown below. RRRS__ E = 28000 * RRRS__
RRTG	(RR)	Variable setting motor fixed speed when speed set point mode 05H is selected. 05H is normally used for turning gear function. RRTG = Desired per unit speed where 1 per unit speed is the motor rated speed Resolution is .1 % and range is 0 to 1.2 per unit speed. The related software variable is SPD__ REF__ TURNING__GEAR The following relationship holds __TURNING__ GEAR = RRTG * 28000
SIMARG	(MI)	Variable controlling the source-side retard limit as a function of current. Compensates the firing as a function of link current for source impedance. SIMARG = per unit commutating impedance Resolution is .01 per unit and the range is .01 to .25 per unit. The associated software variable and its relationship to SIMARG is given below. ALPH__ SRC__ IMP__ MARG = 67 * (XPU)

Name	Group	Function
SMARG	(MI)	Variable controlling the fixed source side margin when inverting. Normal margin is 20 degrees. SMARG = desired fixed margin in degrees. Resolution is 1 degree and the range is 0 to 30 degrees. The associated software variable and its relationship to SMARG is given below. ALPH __FIXED__ MARG__ SRC = 256[1 + COS(180 degrees-SMARG)]
SPLLTIM	(PR)	Sets time delay before a source phase- lockloop fault is declared following an error determination SPLLTIM = required time in seconds Resolution is .01 seconds and the range is 0 to 1.5 seconds. The associated software variable and its relationship to SPLLTIM is given below. PLL__ OUT__ OF__ TOL__ SRC__ PERIOD = SPLLTIM * 6 * FRQS
SQCB	(SQ)	Test variable to allow operation of the source current regulator by diametrically firing the load side cells in a controlled fashion such that the motor load is not required. SQCB = "F"alse (normal operation) = "T"rue (crowbar mode enabled)
SQDL	(SQ)	Enables data link communications and modifies the drive sequencing for operation over the data link. SQDL = "T"rue (data link operation) = "F" alse (disables data link)
SQETST	(SQ)	Test variable used to disable the field exciter selftest feature. This is normally disabled on 12 pulse drives to avoid closing in on field in operation. SQETST = "F"alse disable field exciter test = "T"rue enable field exciter test.
SQFE	(SQ)	Test variable to allow operation of the exciter alone. Source power must be available to provide synchronizing signals. SQFE = "F"alse (normal operation) = "T"rue (exciter only)
SQIOTST	(SQ)	Variable controlling the enabling of microprocessor interrupts. Normally all interrupts are enabled. When SQIOTST is set, all interrupt programs are disabled. SQIOTST = "F"alse (normal operation) = "T"rue (interrupts disabled)
SQMCH	(SQ)	Selects the master channel when two channels are connected in a 12 pulse master-slave configuration. SQMCH = "F"alse (selects channel as slave) = "T"rue (selects channel as master)

Name	Group	Function
SQOPT1	(SQ)	This is a family of flags which control the enabling of various options in the ladder logic sequencing program. The flags are encoded into SQOPT1 as shown below. Background Variable / Bit / Function SQMOVTEMP 0 -Mask MOVTEMP input SQBFDIFSW 1 -Mask BFDIFSW input SQMB1ST 2 -Mask MB1ST input SQMB2ST 3 -Mask MB2ST SQMB3ST 4 -Mask MB3ST input SQMB4ST 5 -Mask MB4ST input SQUSERA 6 -Mask USERA input SQTTRBLE 7 -Mask TTRBLE input SQLTRBLE 8 -Mask LTRBLE input SQPCROT 9 -Mask PCROT input SQBFILF 10 -Mask BFILF input SQRUNPM 11 -Mask RUNPM input SQSTARTPM 12 -Mask STARTPM input SQNBLF 13 -Mask NBLF in system fault SQFSP 14 -Enable flying spare logic. Setting a variable to the "1" state masks out the function so that ladder diagram logic does not affect the sequencing. Setting the "0" allows the function to affect the sequencing. Refer to the ladder diagram logic. The value of SQOPT1 is obtained by setting 1s and 0s in the bit positions and connecting the binary number obtained to decimal.
SQOPT2	(SQ)	This is a a family of flags which control the enabling of various options in the ladder logic sequencing program. The flags are encoded into SQOPT2 as shown below Background Variable / Bit / Function SQBFNDR 0 -Mask BFNDR input SQRDY 1 -READY option SQTACH 2 -Selects tachometer feedback for speed reg in load PLL mode. Setting a bit to the "I" state masks out the function so that ladder diagram logic does not affect the sequencing. Setting the "0" state allows the function to affect the sequencing. Refer to the ladder diagram logic. The value of SQOPT2 is obtained by setting 1s in bit positions required and converting the binary number obtained to decimal.

Name	Group	Function
SQPLTIM	(SQ)	Variable setting the time delay after a loss of source power is sensed and before a power loss fault is declared. The time delay is based on background program executions and is approximately 3 milliseconds per pass. SQPLTIM = (time delay)/3 E-03 The resolution is 1 count and the range is 0 to 20,000. The associated software variable and its relationship to SQPLTIM is given below. POWER__ LOSS__ TIMER = SQPLTIM
SQPOS	SQ	Selects load side firing algorithm. When true position mode is selected for all modes. When false normally PLL mode is used. However low speed modes may use position mode if SQOPT2 set (see above). SQPOS = "T"rue (selects position mode) = "F"alse (selects PLL mode)
SQRA	(SQ)	Selects exciter mode of operation for slave converter in 12 pulse system. SQRA = "F"alse slave exciter does not operate = "T"rue slave exciter operates
SQRT	(SQ)	Test variable used to disable the relay outputs selftest feature. This is normally disabled on 12 pulse drives to avoid interfering with a drive in operation/ SQRT = "F"alse disables relay testing "T"rue enables relay testing
SQTMR1 SQTMR2 SQTMR3 SQTMR4 SQTMR5 SQTMR6	(SQ)	Timers used in the drive sequencing module to implement ladder logic functions. The definition of each timer's use is given in the background program. Note that the time base for the counter is the source-side crossover interrupts. SQTMR- = setting (in seconds) The resolution is .1 seconds and the range is 0 to 20 seconds. The associated software variable and its relationship to SQTMR - is given below. BKGND__ TIMER( ) __LIMIT = SQTMR- * FRQS/10
SQZSPD	(SQ)	Provides a setpoint for drive zero speed indication. Resolution is .1% and the range is 0 to .5 per unit speed.associated software variable and it relationship is shown below. SQZSPD__E = 1024 * FRQMM/60
SRCLFC	(SR)	Speed regulator torque command limit when drive operation is in one of the forced-commutated regions. A typical value is .2 PU torque. SRCLFC = Desired per unit current Resolution is 1% and the range is 0 to 2.0 per unit. The associated software variable is SPD__ REG__ CUR__ LMT__ 1 This is related to the subject variable by CUR__ LMT__ 1 = SRCLFC * 1024

Name	Group	Function
SRCLSC	(SR)	Speed regulator torque command limit when drive operation is in self-commutation. A typical value is 1.0 PU torque. SRCLSC = Desired per unit current Resolution is 1% and the range is 0 to 2.0 per unit. The associated software variable is SPD__ REG__ CUR__ LMT__ 2 This is related to the subject variable by CUR__ LMT__ 2 = SRCLSC * 1024
SRETOL	(SR)	Variable setting the allowable speed error (difference between reference and feedback). If this error is exceeded, a signal to initiate an "EXCESSIVE SPEED ERROR" alarm is generated. SRETOL = Desired % speed error, where 1 per unit speed is speed at 60Hz Resolution is .1% and the range is 0 to 1.2 per unit speed. The associated software variable is SPD__ REG__ DELTA__ TOL The relationship between the variables is DELTA__ TOL = SRETOL * 1024 * (FRQM/60)
SRGAIN	(SR)	Variable setting the speed regulator gain. The following equation defines the required gain assuming a critically damped response. $__GAIN = \frac{(.00000002264)X(N^2)X(WK^2)}{(HP)\ X\ (Tr^2)}$ N = rated motor RPM Tr = regulator response time WK^2 = total drive WK^2 at motor shaft HP = drive HP/channel Resolution is .1 and the range is 0 to 32
SRLLTC	(SR)	Sets the time constant of the lead/lag compensation in the speed feedback path. Normally SRLLTC equals SRTC if critical damping is required. Resolution is .001 seconds and the range is 0 to 3 seconds.
SRMAXR	(SR)	Maximum speed reference clamp. SRMAXR = Desired per unit speed, where 1 per unit speed is the motor rated speed Resolution is .1% and the range is 0 to 1.2 per unit. The associated software variable is SPD__ REG__ REF__ MAX__ LMT The relationship between the variables is __REF__ MAX__ LMT = SRMAXR * 28000

Name	Group	Function
SRMINR	(SR)	Minimum speed reference clamp. SRMINR = Desired per unit speed, where 1 per unit speed is the speed at 60 Hz The associated software variable is SPD__ REG__ REF__ MIN__ LMT Resolution is .1% and the range is 0 to 1.2 per unit. The relationship between the variables is REF__ MIN__ LMT = SRMINR * 28000
SRNFTC	SR	Sets the time constant of the double pole noise filter in the speed feedback path used to remove noise and smooth coarsely calculated data at low speeds. Resolution is .0001 second and lk, the range is .0084 to 1 second.
SRTACH	SR	Scales drive control for the number of tach pulses for one complete mechanical resolution. Resolution is units and the range is 0 to 1800.
SRTC	(SR)	Variable setting the location of the speed regulator "zero" for stability compensation. SRTC = (Desired time constant) where the time constant is specified in seconds. Resolution is .001 seconds and the range is 0 to 3 seconds.
STHRUTIM	(PR)	Controls number of consecutive commutation failures allowed. A penalty function is implemented such that 6 good firings following a failure will cancel an fault firing. This allows recovery from non-repetitive faults. STHRUTIM = number of consecutive overcurrents. The resolution is units and the range is 1 to 6. The associated software variable and its relationship to FCCOUNT is given below. SHOOT__ THRU__ PERIOD__ = (STHRUTIM - 1) * 6
STIM	(MI)	Used in motor starting strategy to set the time for one cycle. Resolution is .1 second and the range is .1 to 6 seconds. The relationship to the software variable is given below. TRANS__ START__FREQ__E = 181/STIM takes on values equivalent to SRLLADJ and is directly in the software
SVPU	(PT)	Defines the value of 1 per unit source voltage. Used in "print status" routine. Resolution is units and the range is 0 to 500 volts.
SWINDTOL	(PR)	Controls the allowable tolerance on motor flux wave unbalance before a shorted winding fault is declared. SWINDTOL = desired per unit flux error Resolution is .01 and the range is 0 to 1.2 per unit. The associated software variable and its relationship to SWINDTOL is given below. SHORTED__ WINDING__ FLUX__ TOL = SWINDTOL * 1361

Name	Group	Function
TRSPD	(MI)	Sets the transition speed for entering self-commutation mode. TRSPD = desired per unit speed Resolution is .001 per unit and the range is 0 to 1.2 per unit. The associated software variable and their relationship to TRSPD is given below. LAC_MODE3_TRAN_SPD = TRSPD * 1024*(FRQM/60) LAC_MODE9_TRAN_TACH = TRSPD * 28000
T8CNT	MI	Variable used for transitioning from load-side firing mode 8 to 7 (open loop speed to closed loop position mode). Provides a minimum number of firing before allowing transition. Typically set to 10. Resolution is units and the range is 0 to 100. MODE8TO7_TRANS_COUNT_E =T8CNT
T8SPD	MI	Variable setting the transition speed from load side firing mode 8 to 7 (open loop speed to closed loop position mode). The variable controls the speed in Hertz. Resolution is .1 Hz and the range is 0 to 6.0 Hz TSPD8TO7 = HZ (normally 2) MODE_8TO7_SPD_E = TSPD8TO7*28000/(FRQMM*10)
XQ	(MI)	Represents the sum of the motor quadrature axis leakage reactance and magnetizing reactance (X1 + Xaq (in per unit) Resolution is .01 and the range is 0 to 1.6 per unit. The associated software variable and its relationship to XQ is given below. XQ_E = XQ * 47

DESCRIPTION OF PUBLIC VARIABLES

SECTION B

Many variables calculated in the software routines and relating to drive operation may be accessed through the monitor by typing in the variable name. These are important drive parameters such as speed, current, and firing angle. The monitor will respond with the latest value for the selected variable. A table of variable resides in the system elementary beginning onsheet DK.

The data given in the table is arranged as follows. The first entry gives the equivalent variable which would be found in the software listing or in a link and locate map. The second entry defines the variable type as byte, word, or integer. The third entry gives the variable scaling. This is defined as counts corresponding to some value of engineering units if the variable is an integer. If the entry is bit coded, the third option is that the variable may be boolean and take on values of true or false only. The last entry gives the variable definition.

CHAPTER 5

SECTION A
DIAGNOSTIC MONITOR SPECIFICATIONS

5-1. INTRODUCTION

The diagnostic monitor for the Structured LCI product provides the user with a variety of functions. Via an RS-232 terminal, the user may display regulator and logic variables and system fault and status information. The monitor also provides the ability to modify system variables and regulator constants, which facilitates troubleshooting and tuning procedures.

Communication with the monitor is handled through the RS232 port on the drive control. Hence, the communication device may be any RS232 compatible terminal, including handheld, hard copy and CRT types Provisions are made to specially format the monitor output when a handheld terminal is being used.

All communication via the RS232 port are 7-bit data, even parity, with one stop bit. Acceptable baud rates are 300, 1200, 2400, 4800, and 9600 bits per second (baud). Upon power up, the monitor searches for carriage return characters to lock onto the baud rate used by the terminal device.

Monitor functions are accessed by entering a valid command (from the monitor's instruction set) at the diagnostic terminal. Many of the commands are called directly from the MONITOR level, '*>' prompt, and control is returned to this level after execution of the requested process. The ENGINEERING and TUNEUP functions, however, are actually extended modes of operation that have their own instruction sets. After execution of one of the ENGINEERING mode commands, for example, control is returned to the ENGINEERING mode, 'E>' prompt. To return to the MONITOR level again, the QUIT command must be invoked -- this is also true for the TUNEUP mode.

In sections 5-3 and 5-4, the purpose and syntax of each monitor command is described in detail. Table 5-1 defines the conventions used in specifying the command formats.

TABLE 5-1 Definitions for Command Syntax Notation

Symbol		Definition
::=		"Is defined to be"
[ ]		Items between brackets are optional
|		Separates items in a list - choose one
{ }		Braces delimit items for the purpose of minimizing ambiguity
( )		Optional characters in a command or parameter are in parentheses
<decimal no>	::=	<1-5 decimal digits> T
<hex no>	::=	<1-4 hex digits> [H]
<new value>	::=	[+>-]<1-5 decimal digits w/ or w/o decimal pt.>
<pointer>	::=	[{CS | DS | <hex no>} :] <hex no>
<register>	::=	[CS | DS]
CS	::=	Code Segment
DS	::=	Data Segment

5-2. OPERATING INSTRUCTIONS

5-2.1. General Information. To initiate communication with the Monitor, the user should enter several "carriage return" characters at the terminal keyboard. The terminal will then respond with a banner message and the monitor prompt character (*>) to indicate that it is ready to accept command inputs.At this point, after passing the PASSCODES the operator may enter any of the MONITOR level commands or select one of the extended modes of operation (for example TUNEUP) as described in detail in sections 3 and 4.

When finished using the diagnostic monitor, the terminal session may be terminated by entering QUIT at the terminal keyboard (see section 5-3.9). The logoff message is then printed at the terminal, indicating that the session is over, and the monitor will idle, waiting for the carriage return characters that indicate the start of another terminal session.

Terminal Format:

The Diagnostic Monitor is designed to work with either a handheld terminal or a standard keyboard type. The default (at the start of the terminal session) is the handheld format. In this mode of operation, the output messages are sent one line at a time to accommodate the limited display length of the handheld terminal (16 characters max per line). Entering a carriage return causes the next line of output to be sent to the terminal. The TERMINAL command (section 5-3.4) may be used to switch to an 80 column listing format with no pauses for use with a standard CRT or hardcopy terminal.

Power-up Status Messages:

When power is first applied to the drive control, a sequence of self-tests and initialization procedures is performed. Any failure status of these tests is automatically sent to the panel printer. Should the printer fail, diagnostic messages will be rerouted to the diagnostic terminal.

5-2.2. Editing Functions. The diagnostic monitor provides text editing functions for correcting errors entered within a command string. Each function is listed below with a brief description.

*** NOTE ***

In the examples below, the symbol *> is the diagnostic monitor prompt (not typed by the user), and the symbol [CUR] represents the cursor position at the end of the sequence shown.

DEL/RUBOUT -
Removes the previous character from the message and echoes the deleted character to the terminal.
Example: Typed - *>
HEKP<DEL><DEL>LP
Echoed - *> HEKP /K P \ LP [CUR]

CONTROL-R
Causes all characters currently in the message to be transmitted to the terminal on a new line. This function is convenient after using DEL/RUBOUT because it shows the state of the message after the editing.
Example:
Typed - *>
HEK<DEL>LP<CONTROL-R>
Echoed - *> HEK /K\ LP
HELP [CUR]

*** NOTE ***

Typing subsequent consecutive CONTROL-R keys will invoke the monitor recall function. This feature calls up previous commands that were typed at the terminal, starting with the most recent and progressing to the oldest command remembered. A total of 20 command lines are remembered. Using the recall feature helps the user avoid retyping a command that he wishes to execute repetitively.

CONTROL-X

Deletes the entire message that has been entered so far.

NOTE: This has the same effect as using the DEL/RUBOUT function to remove all the characters from the message.

Example:
Typed - *> HEKL<CONTROL-X>HELP
Echoed - *> HEKL
HELP [CUR]

CONTROL-Y

Aborts the current command sequence and returns the monitor to the current mode of operation (e.g., ENGINEERING, TUNEUP or the main level).

** *NOTE* **

CONTROL-Y or CONTROL-C may be entered at any time to abort the current process--even during the listing of a message at the terminal.

Example:

Typed - T> HE<CONTROL-Y>

Echoed - T> HE^ Y

T> [CUR]

CONTROL-C

Aborts the current command sequence and returns the monitor to the outermost (main) level of operation. When a CONTROL-C is received, the monitor also displays a message identifying the product as a LCI Drive as in the login sequence.

Example:

Typed - T> HE<CONTROL-C>
Echoed - T> HE^ C
-- IGTO M0NITOR --
*> [CUR]

CONTROL-S

Causes the current output to be halted until it is restarted using the CONTROL-Q switch.

CONTROL-Q

Causes the current output to be restarted after having been stopped using the CONTROL-S switch.

5-2.3. Command Syntax. The general syntax for a monitor command is shown below:

<command> ::= <instruction>
[<delimiter> <parameter> ...] <terminator>

where <instruction> is one of the valid monitor instructions (defined in section 5-3) and <terminator> is a carriage return (<cr>). The <delimiter> and <parameter> entries, if necessary, depend on the particular instruction being called. Each instruction has its own set of valid delimiters and parameters. Also, some commands may require several <delimiter> <parameter> pairs, while some require no parameters at all.

As an aid to the operator, and to facilitate handheld terminal use, all the <delimiter> <parameter> pairs need not be entered on the same line as the instruction -- each needed parameter will be requested by the monitor. The syntax for this type of sequence is:

<command> ::= <instruction> <terminator>
<parameter> <terminator>.

After each <terminator> entry, the monitor displays an appropriate prompt that asks the user for the necessary parameter.

The use of abbreviated commands is another feature designed to aid the handheld user. Rather than typing the complete instruction word, only the first one or two letters need to be entered (see the syntax definition tables in sections 3 and 4 for the minimum acceptable abbreviations).

5-2.4. Protected Commands. Some monitor commands allow the user to modify system variables and change tuning parameters. These functions, because of their potentially destructive nature, are available only after executing a privilege enabling sequence. This is so the user will not inadvertently make modifications that could inhibit the proper operation of the LCI Drive control.

The privileges are enabled by entering a password at the monitor terminal. Privileges are automatically disabled when: (1) the monitor is logged off (QUIT).

TABLE 5-2. Syntax Definitions for MONITOR Level Commands

ASSIGN:	A(SSIGN) {LIST| DA}
DISPLAY:	{D(ISPLAY) { <varname> [C] | STATUS | LIST | FAULTS }| DS | DL | DF } <cr>
HELP:	H(ELP) <cr>
PASSCODE:	P(ASSCODE) <cr>
QUIT:	{Q(UIT) EX(IT) <cr>
RESTART:	R(ESTART) <cr>
SYMBOLS:	SY(MBOLS) [<symbolname>] <cr>
TERMINAL:	T(ERMINAL) <cr> <current format> CHANGE ? [Y| N] <cr>
VERIFY:	V(ERIFY) [<symbol group name>] <cr>

5-2.5. Help. The user may print out a list of available commands by asking for HELP. From either the MONITOR mode, or one of the extended modes, a HELP request will print the list of commands available at that mode only.For example, if in the ENGINEERING mode, only the ENGINEERING mode commands, and not the MONITOR mode commands, will be listed.

5-3. MONITOR COMMANDS

The MONITOR level commands are intended to be the primary tools used in interrogating the drive control and monitoring its status. The instruction set provides display and modify functions, the ability to reset the control, and the power to direct variables to analog outputs for strip chart monitoring, among other things. Table 5-2 shows the complete list of instructions and their correct syntax. The commands are defined in their single line entry format, although the "prompt-through" type sequence (as discussed in section 2.3) is also valid.(See Table 5-2)

5-3.1. ASSIGN. Function: To designate that a particular variable be assigned to a circular list and/or D/A converter.

Using the ASSIGN function, the user may modify the configuration of the internal circular lists by changing the number of entries (maximum of 8) to be stored at each sample interval and/or selecting which entries are being captured. A total of 2K bytes of memory are available for storage of the diagnostic list information. This memory is evenly divided among the parameters within the list. As an example, if 4 word parameters are monitored in the circular a total of 256 events will be recorded.

The list may be triggered either by a level trigger or mask trigger. In either case the trigger is compared against the variable selected in list position no. 1.. In a level (>, <, =). When the bit trigger mode is selected, any bit of variable 1 becoming true causes the list to be taken. Bits of variable 1 may be masked using the mask variable option of the list. The list can only be triggered by the level trigger or the bit trigger. When more than one trigger has been activated only the last will be accepted by the list function.

The user may also direct drive variable contents to a D/A converter for observation on an oscilloscope or chart recorder. The analog outputs are available from test points on the front edge of the printed circuit board HLCB. When assigning a variable to a D/A output, the operator selects the two parameters, SCALE and OFFSET. Since the possible range of a variable often exceeds the range of the D/A converter, a function is provided to scale the variable contents before sending data to the D/A converter. The following table gives the possible SCALE parameters; beside each one is the value by which the variable contents are multiplied before being converted if that SCALE factor is selected.

Table 5-3. D/A Testpoint Scaling

SCALE	MULTIPLIER	SCALE	MULTIPLIER
15	1/32768	0	1
14	1/16384	-1	2
13	1/8192	-2	4
12	1/4096	-3	8
11	1/2048	-4	16
10	1/1024	-5	32
9	1/512	-6	64
8	1/256	-7	128
7	1/128	-8	256
6	1/64	-9	512
5	1/32	-10	1024
4	1/16	-11	2048
3	1/8	-12	4096
2	1/4	-13	8192
1	1/2	-14	16384
0	1	-15	32768

An OFFSET may also be selected during the ASSIGN process -- this value is subtracted from the variable contents before the scaling process. This function may be used when the slight variations around a fixed value need to be examined (see example below).

Examples:

```
*<ASSIGN LIST

Settings:
1 — samples                    ; selects variables to be sampled
2 — trig level                 ; sets up level triggered list
3 — bit trig                   ; sets up bit triggered list
4 — site                       ; selects site where data is taken

Choice? 1=
NO OF ENTRIES=1                ; previously was set to capture 1 parameter
NEW VALUE ? 3<cr>              ; set to capture 3 parameters
ENTRY 1 = FLUXFB               ; previously set to capture FLUXFB
NEW ENTRY? SPDREF<cr>          ; speed reference (no. 1)
ENTRY 2 = 0000H                ; null value
NEW ENTRY? SPDFB<cr>           ; speed feedback (no. 2)
ENTRY 3 = 0000H                ; null value
NEW ENTRY? IAFB<cr>            ; armature current feedback (no. 3)

*>ASSIGN LIST
Settings:
1 — samples
2 — trig
3 — bit trig
4 — site
Choice? 2
TRIGGER LEVEL = 0
NEW VALUE? 200                 ; level of trigger in counts
TRIGGER CONDITION? OFF
NEW VALUE? <                   ; choices are <, >, =
*>ASSIGN LIST

  Settings:
1 — samples
2 — trig
3 — bit trig
4 — site
Choice? 4
TRIGGER SOURCE = OFF           ; present data site
CHOICES (1-5):                 ; available data sites
  SAC, SPLL, LAC, LPLL, BG
```

```
TRIGGER LEVEL = 0
NEW VALUE? 1                    ; selects SAC (0 disables list

*>ASSIGN LIST

Settings:
  1 — Samples
2 — Trig Level
3 — Trig Mask
4 — Site
Choice? 3
PRESENT MASK FF                 ;value must be in hexadecimal
NEW VALUE? OF
*>

*> ASSIGN DA1 <cr>             ; choose DA1 (TP2) on HLCB (slot 2E)
ENTRY = SPDSTR                  ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = ISRC <cr>               ; set new parameters -- I feedback
SCALE = 6 <cr>                  ; divide by 64 to get in range
OFFSET = <cr>                   ; leave the offset the same (at 0)
*>

*> AS DA2 <cr>                 ; choose the DA2 (TP3) on HLCB (slot 2E)
ENTRY = SPEED                   ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = <cr>                    ; still look at speed feedback
SCALE = -2 <cr>                 ; multiply 4 to accent variations
OFFSET = 1000 <cr>              ; look at fluctuations around 1000
*>
```

The following notes on using the diagnostic list apply.

1. When using the bit trigger option the mask must always be entered in hexadecimal. The bit (s) to trigger the list must be set to a "1" in the appropriate bit location.
2. In either the bit trigger or magnitude trigger mode, entry "1" in the variable selections is compared against either the magnitude or bit trigger word.
3. When using the bit trigger option, the word must be re-entered, wherever selection 3 of "ASSIGN LIST" is accessed, otherwise the mask is reset to "0".
4. The list has completed taking data when the trigger source has returned to "OFF" after entering another selection.
5. List sites must be selected with care. The list may not necessarily fill up if a site is selected which doesn't operate during the event to be captured. For example, LPLL is disabled when the drive "RUN" command is removed.

5-3.2. DISPLAY. Function: to display the contents of select drive control variables and circular lists.

Several options are allowed in the DISPLAY command:

1. DISPLAY LIST, to display the circular list information.

2. DISPLAY FAULTS, to display the status of the fault and alarm logic.

3. DISPLAY RELAYS, to display a concise list of the drive's background relay status.

The DISPLAY FAULTS command will produce a list of the most recent STATUS messages (stored in a circular "fault log"). This list includes current status indications (shown also with the DISPLAY STATUS command).

Drive variables are displayed in their respective engineering units. Generally they are in per unit (of flux, current, speed,etc...). The general format in which variable contents are displayed is:

<varname> = <value>

A particular variable may be displayed repetitively by simply typing a "C" (for continuous) following the variable name on the command line.

Some DISPLAY command options may be abbreviated to two letters. These abbreviations and their equivalents are listed below:

DF ::= DISPLAY FAULTS
DL ::= DISPLAY LIST
DR ::= DISPLAY RELAYS

Examples:

*> DISP SPDREF <cr>
SPDREF = 472
*> D <cr>
NAME? SPDREF <cr>
SPDREF = 472
*>

*> D STATUS <cr>
<drive relay status list>
*>

*> DISPLAY LIST <cr>
<listing of circular list>
*>

*> DISPLAY FAULTS <cr>
<listing of circular fault log>
*>

5-3.3. HELP. Function: to display at the terminal a list of the commands available in the current mode of monitor operation.

Examples:

*> H <cr>
<list of available commands>
*>
*> HELP <cr>
<list of available commands>
*>

5-3.4. PASSCODE. Function: to enable the monitor's privileged functions.

*** NOTE ****

For security, the characters are not echoed when entering the password at the terminal.

Examples:

***> PASSCODE <cr>**
PASSWORD : <password> ; password not echoed
PRIVILEGES ENABLED
***>**

5-3.5. QUIT. Function: to terminate the current terminal session. Upon execution of the QUIT command, the monitor will print a logout message and then begin idling, waiting for the carriage return inputs that indicate the start of a new terminal session.

Examples:

*> Q <cr>
MONITOR OFF
*> EX <cr>
MONITOR OFF

5-3.6. RESTART. Function: to execute a software reset of the microprocessor.

Executing the RESTART command is essentially a soft reset. The diagnostic checks are run the same as during power-up, and the drive control is re-initialized to restore the hardware (counters,etc.) to its normal running state.

*** NOTE ****

To execute the RESTART command, the drive must be stopped. If the drive is running, the RESTART routines will not run, and a warning will be printed at the terminal.

Examples:

*> RES <cr>
<selftest results messages>
*> RESTART <cr>
<selftest results messages>

5-3.7. SYMBOLS. Function: to display at the diagnostic terminal the symbolic names that may be used to access drive control variables. If no qualifier is given, the complete list will be printed; if a partial name is specified, however, all symbols that match that partial name are listed.

Examples:

*> SY <cr>
<complete symbol list>
*>
*> SYMBOLS VR <cr>
VRGAIN
VRTIME
*>
*> SYM SPDREF <cr>
SPDREF
*>

5-3.8. TERMINAL. Function: to examine and possibly modify the terminal format. The monitor will display the current format (either "handheld" or- "standard") and then ask the operator whether a change is required. Entering Y(ES) will cause the monitor to toggle to the other terminal format. Any other entry will cause no change.

Examples:

*> TERMINAL <cr>
HNDHLD-CHANGE? Y <cr>
STANDARD FORMAT
*>

*> TER <cr>
STD - CHANGE? <cr>
STANDARD FORMAT
*>

5-3.9. VERIFY. Function: displays the current state of all drive EEPROM settings and adjustments to allow the user to verify factory settings. Individual groups of variables may be displayed for verification by appending one or more group names to the VERIFY command (see example). The groups will be listed in the order specified. Following are the possible group names:

CR - Current Regulator
FI - Field Regulator
FL - Flux Reference
FR - Flux Regulator
MI - Misc. Variables
OV - Overvoltage Regulator
PR - Protection Variables
PT - Printer setup
RR - Speed Ramp Rate Conditioning
SQ - General Sequencing
SR - Speed Regulator

Examples:
*> VERIFY <cr>
<grouped list of all software jumpers and adjustments>
*>

*>V SR, FR, SQ <cr>
(list of speed reg. software jumpers and adjustments>
(list of flux reg. software jumpers and adjustments>
(list of software sequencing jumpers>
*>

5-4. EXTENDED MODES

5-4.1. ENGINEERING. The ENGINEERING mode provides primitive functions like examining and modifying memory and input and output ports. Absolute addresses, rather than symbolic names, are used to access memory locations and I/O ports. Table 5-3 summarizes the command syntax for this mode of monitor operation.

Table 5-5 Syntax Definitions for ENGINEERING Level Commands

DISPLAY:	D[W] <start addr>[,<end addr>] <cr>
HELP:	H <cr>
INPUT:	I[W] <port addr>,[, ...] <cr>
PUT:	P[W] <port addr>,[, ...] <cr>
MOVE:	M <start addr>, <end addr>, <dest addr> <cr>
OUTPUT:	O[W] <port addr>, <data>[,<data> ...] <cr>
QUIT:	{Q |EX} <cr>
SUBSTITUTE:	S[W] <addr>,[<new value>, ...] <cr>

<start addr>	::=	<pointer>
<end addr>	::=	<pointer>
<dest addr>	::=	<pointer>
<addr>	::=	<hex no>
<port addr>	::=	<pointer>
<new value>	::=	<hex no>
<data>	::=	<hex no>

5-4.2. DISPLAY . Function: to display the contents of a memory location or a block of memory. The current value of each byte (or word) is listed in hexadecimal form. Additionally, the ASCII equivalent of the memory shown on a line is displayed in a field to the right of the standard display. Only printable ASCII characters are shown --others are displayed as a period (.). The ASCII display function is suppressed when operating in the handheld terminal format.

Format:

D[W] <start addr>[,<end addr>] <cr>

Examples:

```
E> D DS:800 <cr>
F800:0800 41                A
A>

E> DW 200, 210 <cr>
0000:0200 55AA 00FF 1233 5256 4147 4E49
    2020 0020     .U..3.VRGAIN .
0000:0210 4243                      CB
E>
```

5-4.3. HELP -. Function: to list the available command set for the ENGINEERING mode.

Examples:

```
E> H <cr>
<list of available commands>
E>
```

5-4.4. INPUT . Function: to display a byte or word at an input port.

Format:

I[W] <port addr>, [, ...] <cr>

Examples:

```
E> I F000 <cr>
FE ,
FE ,
FF <cr>
E>

E> IW F080 <cr>
1080 <cr>
E>
```

5-4.5. PUT . Function: to modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the PUT mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

P[W] <port addr>,[,...]<cr>

Examples:

E> P DS:A40 <cr>
F800:0A40 A5- ,
F800:0A41 5A- AA <cr>
E>

E> PW DS:50 <cr>
F800:0050 0FFF- 7FF,
F800:0051 5A5A- <cr>
E>

5-4.6. MOVE . Function: to move a block of data from one area of memory to another. Memory contents are transferred sequentially from low memory to high.

Format:

M <start addr>, <end addr>, <dest addr> <cr>

Examples:

E> M DS:50,59,CS:00 <cr>
E>

E> M 800,900,DS:42 <cr>
E>

5-4.7. OUTPUT . Function: to output a byte or word to an output port.

Format:

O[W] <port addr>, <data>[,<data> ...] <cr>

Examples:

E>O 8010, FF <cr>
- F8,
- F0 <cr>
E>

5-4.8. QUIT . Function: to return from the ENGINEERING mode to the MONITOR mode.

Examples:

E> Q <cr>
*>
E> EX <cr>
*>

5-4.9. SUBSTITUTE. Function: to exam and modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the SUBSTITUTE mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

S[W] <addr>[<new value>, ...] <cr>

Examples:

E> S DS:A40 <cr>
F800:0A40 A5- ,
F800:0A41 5A- AA <cr>
E>

E> SW DS:50 <cr>
F800:0050 0FFF- 7FF,
F800:0051 5A5A- <cr>
E>

5-4.10. TUNEUP. In the TUNEUP mode of monitor operation, the user is able to adjust the performance of the drive by modifying regulator parameters and configurations. A step-by-step tuneup procedure is contained in Chapter 6 . Following are descriptions of the commands that are used as part of this procedure. Table 5-4 summarizes the command syntax for the TUNEUP mode.

For on -line tuneup of any regulator, the background sequencing RUN logic must be satisfied. Monitor commands alone will not allow the drive to run. This mode is intended to be used primarily when the drive is initially installed.

Table 5-6 Syntax Definitions for TUNEUP Level Commands

APPLY:	AP(PLY) <cr>
ASSIGN: *	AS(SIGN) {LIST | DA}
DISPLAY: *	{D(ISPLAY) { <varname> [C]| STATUS | LIST <listname> | } | DS | DL <listname>} <cr>
HELP:	H <cr>
MODIFY:	M(ODIFY) <varname> <cr> <old value message> <new value prompt> [I | D | <new value>] <cr>
MODIFY REFERENCE:	M(ODIFY) REF(ERENCE) <cr>
QUIT:	{Q | EX } <cr>
REMOVE:	REM(OVE) <cr>
SAVE:	SAV(E) <cr> <verify query> [Y | N] <cr>
SAVE D/A	(SDA) <verify query> [Y | N] <cr>
SET PASSWORD	Set (PASSWORD) <cr>
SYMBOLS: *	SY(MBOLS) [<symbolname qualifier>] <cr>
TERMINAL: *	T(ERMINAL) <cr> <current format> CHANGE ? [Y | N] <cr>
TUNEUP SYMBOLS:	TSY(MBOLS) <cr>
TUNEUP:	TUN(EUP) {F(IELD) | A(RMATURE)} <cr>
VERIFY: *	V(ERIFY) [<symbol group name>] <cr>

* - Also available in MONITOR mode -- see section 5-3 for descriptions.

5-4.11. Apply -. Function: to apply the test generator reference to the regulator being tuned up.

Examples:

T> AP <cr>
T>

5-4.12. Help -. Function: to list the available command set for the TUNEUP mode.

Examples:

T> H <cr>
<list of available commands>
T>

5-4.13. Modify -. Function: to allow the operator to modify the contents of select drive control variables. A new value may be entered, or the current value may be incremented or decremented repeatedly. Also,the increment (or decrement) interval may be selected by placing a new interval after the I or D entry (see example below). If none is specified, the increment interval will be one unit of the least significant digit. Values are displayed and entered in appropriate engineering units.

*** NOTE ***

Because modifying certain memory locations could have adverse effects on the drive control operation, the MODIFY command may only be executed after completing the privilege enabling sequence.

Examples:

```
T> MOD VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? 15 <cr>
VRGAIN = 15.0
NEW VALUE ? <cr>
T>
T> M VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? I<cr>
VRGAIN = 1.2
NEW VALUE ? I .3<cr>
VRGAIN = 1.5
NEW VALUE ? I<cr>
VRGAIN = 1.8
NEW VALUE ? D<cr>
VRGAIN = 1.5
NEW VALUE ? <cr>
T>
```

** *NOTE* **

When a tuneup case is selected (section 5-4.20),a numbered list of symbols associated with that case is displayed at the terminal. While running that particular test, a symbol's NUMBER (from the list) rather than its name may be used as the argument in the DISPLAY and MODIFY commands. This eliminates the need to laboriously type out long symbol names at the terminal during a tuneup session.

Examples:

```
T> MOD 3 <cr>
FRGAIN = 1.1
NEW VALUE ? 1.3 <cr> ;change gain from 1.1 to
                       1.3
FRGAIN = 1.3
NEW VALUE ? <cr>
T>
```

5-4.14. Modify Reference -. Function: to modify (customize) an internal reference generator. When a tuneup case is selected, its corresponding reference generator is automatically selected (see Section 5-4.20). The user then has the option to modify the characteristics of the reference generator as shown in the following examples. A reference generator is a series of two steps, each with selectable amplitude and duration. Step durations are entered in seconds (i.e., .05 for 50 milliseconds) and reference levels are specified in per unit levels (i.e., for a current test, the level is in per unit current and for a speed test in per unit speed, etc.).

After the steps are fully defined and the test generator is enabled by the APPLY command, the regulator specified by the TUNEUP command will be excited with a periodic signal corresponding to the test reference levels defined. A fixed constant level may be obtained by setting the amplitudes of both steps to the same value.The REMOVE command disables the test generator and removes all excitation to the specified regulator.

Examples:

(Fixed Reference)

```
T> MOD REF <cr>
STEP      LEVEL(PU)      TIME(SEC)
  1         0.100            1.00
  2         0.000            1.00

STEP 1 LEVEL = <cr>

STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.10 <cr>  ;make levels agree
STEP 2 DURATION = <cr>
T>
```

(Stepped Reference)

```
T> MOD REF <cr>
STEP      LEVEL(PU)      TIME(SEC)
  1         0.100            1.00
  2         0.100            1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.00 <cr>  ;set for 10% step
STEP 2 DURATION = <cr>
T>
```

5-4.15. Quit -. Function: to return from the TUNEUP mode to the MONITOR mode.

Examples:

```
T> Q <cr>
*>

T> EX <cr>
*>
```

5-4.16. Remove -. Function: to remove the test generator signal from the regulator being tuned up.

Examples:

```
T> REM <cr>
T>
```

5-4.17. Save -. Function: to store the present values of the regulator variables in the system's non-volatile memory. After modifying system parameters (for example, gains and time constants during a tuneup session), the new values may be transferred into permanent storage using the SAVE command. The previously stored values, of course, are lost during this process; hence, the monitor will prompt the user for confirmation before actually executing the command.

** *NOTE* **

Privileges must be enabled to execute the SAVE command. Also, the hardware write enable switches on the DMPB cards must be in the "store enable" position before values may be stored in the nonvolatile memory.

Examples:

*> SAVE <cr>
ARE YOU SURE? Y<cr>
CONSTANTS SAVED

*>

*> SAVE <cr>
ARE YOU SURE? <cr>
CONSTANTS NOT SAVED
*>

5-4.18. Save D to A (SDA). Function: To store the present values of the D to A assignment variables in the system's nonvolotile memory. During tuneup mode, the D to A converters (1 and 2) are continually reassigned according to the default values required for the regulator being turned. If the D to A values were included in the global "SAVE" command, the intended setup values would be lost. A separate "SAVE" command is therefore included.

When it is required to modify, the D to A assignments permanently, the "ASSIGN DA" command used first be used to setup the required parameters.

The hardware write enable switches on the DMPB card must be in the "write enable" position before values may be stored in the nonvolatile memory.

Examples:

*> SDA <cr>
ARE YOU SURE? Y<cr>CONSTANTS SAVED
* >

*>SDA <cr>
ARE YOU SURE? <cr>
CONSTANTS NOT SAVED
*>

5-4.19. (Set) Password -. Function: Allows the password granting privilege to the tuneup mode to be changed.

** *NOTE* **

To change the password permanently requires a "SAVE" following the "SET".

Examples:

T> SET<cr>
OLD PASSWORD?
NEW PASSWORD?
NEW PASSWORD?
T >

5-4.20. TUNEUP SYMBOLS -. Function: to list all tuneup variables associated with the current tuneup case.Examples:

T> TSY <cr>
TUNEUP VARIABLES ARE:
<numbered symbol list> or NONE
T>

5-4.21. TUNEUP -. Function: to initiate a particular regulator tuneup sequence. The user must specify which of several regulator is to be tuned. He is then allowed to select the desired tuneup case. Examples of valid commands are:

CR - Current Regulator
FC - Speed Regulator
FL - Flux Reference
FR - Flux Regulator
MI - Misc. Variables
OV - Overvoltage Regulator
PR - Protection Variables
RR - Speed Ramp Rate Conditioning
SQ - General Sequencing
SR - Speed Regulator
PT - Printer Datalog Parameters

Example:

(Current Regulator Tuneup)
T>TUNE CR<cr>

AVAILABLE VARIABLES ARE:

1-CRALPH1 = 90
2-CRALPH2 = 86
1-CRGAIN = 338
4-CRTC = 0.020
5-CRMIN = 0.20
6-CRSTART = 0.20
T>

5-4.22. Monitor Commands -. The following main-level monitor commands are also available in the TUNEUP mode:

ASSIGN
DISPLAY
SYMBOLS
TERMINAL
VERIFY

The syntax definitions and functional descriptions for these commands are contained in section 3.

5-5. LOADING DISPLAY MESSAGES HEADERS

Customized display messages can be loaded into EEPROM to identify the user, drive, personnel to contact, actions to be taken, etc. Five (5) different messages can be loaded - starting at the locations below. Each message can be up to 64 characters long (minus 1 for end of message = 63).

Starting Address	Message type
3000:200	Header - message printed at power-up
3000:240	Repetition - message printed at beginning of "data logging"
3000:280	Requisition - message printed when "Status" P.B. used
3000:2C0	Fault - message printed at beginning of "Fault" messages
3000:300	Alarm - message printed at beginning of "Alarm" messages

Procedure (see example 5B-1):

1. Write the message out on Form 5B-1 (remember to leave 2 spaces) for <cr> <LF> (by word) if you don't need to force a new line.
2. Convert the characters to their ASCII equivalent using Table 5-7. Add after the last conversion 00 to signal end-of-message.
3. Using the monitor Put command, load the ASCII equivalents.

*** NOTE ***

You can use a ","instead of a <cr> after each ASCII code to increment to the next location - rather than reloading the address (faster).

Suggestions
Always include user's name and the drive/channel #. For the fault, alarm, and repetition messages, a reminder such as "Log items in maint book' may be useful. The default messages should be okay . It should only be necessary to add the user's name and drive I.D. to the beginning (if sufficient space) or to the end (will need to move "00" of the default message.

```
MONITOR OFF <cr>     ; Form 5B-1 or 5B-2 should
                         be filled out.

LCI MONITOR
*>PASSWORD
PASSWORD ? <cr>

PRIV. ENABLED
*> ENG<cr>
]
E>PUT 3000:200<cr>   ; Enable EEPROM write
                         switch

3000:200 ??- 53,
3000;201 ?? 43,      ; Continue till all characters
done    :
   :
3000:2?? ??- 00<cr>
```

Example 5B-1 : MESSAGE HEADERS

Message Type_____

Starting Address - 3000:_____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01___	___	33___	___
02___	___	34___	___
03___	___	35___	___
04___	___	36___	___
05___	___	37___	___
06___	___	38___	___
07___	___	39___	___
08___	___	40___	___
09___	___	41___	___
10___	___	42___	___
11___	___	43___	___
12___	___	44___	___
13___	___	45___	___
14___	___	46___	___
15___	___	47___	___
16___	___	48___	___
17___	___	49___	___
18___	___	50___	___
19___	___	51___	___
20___	___	52___	___
21___	___	53___	___
22___	___	54___	___
23___	___	55___	___
24___	___	56___	___
25___	___	57___	___
26___	___	58___	___
27___	___	59___	___
28___	___	60___	___
29___	___	61___	___
30___	___	62___	___
31___	___	63___	___
32___	___	64 < EOM >	00

FORM 5B-1: For Header Message

Message Type _____

Starting Address - 3000: _____

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01____	20_	33_d_	64_
02____	20_	34____	20_
03____	20_	35_i_	69_
04____	20_	36_t_	74_
05____	20_	37_e_	65_
06____	20_	38_m_	6D_
07____	20_	39____	20_
08____	20_	40_i_	61_
09____	20_	41_n_	6E_
10____	20_	42____	20_
11____	20_	43_L_	4C_
12____	20_	44_O_	4F_
13____	20_	45_G_	47_
14____	20_	46____	47_
15____	20_	47_b_	20_
16____	20_	48_o_	6F_
17____	20_	49_o_	6F_
18____	20_	50_k_	6B_
19____	20_	51 < EOM >	00_
20____	20_	52____	____
21____	20_	53____	____
22____	20_	54____	____
23____	20_	55____	____
24____	20_	56____	____
25____	20_	57____	____
26 < cr >	0D_	58____	____
27 < 1f >	0A_	59____	____
28_R_	52_	60____	____
29_e_	65_	61____	____
30_c_	63_	62____	____
31_o_	6F_	63____	____
32_r_	72_	64 < EOM >	00

FORM 5B-2: Default for Header Message

Character	ASCII	Character	ASCII
A	41	a	61
B	42	b	62
C	43	c	63
D	44	d	64
E	45	e	65
F	46	f	66
G	47	g	67
H	48	h	68
I	49	i	69
J	4A	j	6A
K	4B	k	6B
L	4C	l	6C
M	4D	m	6D
N	4E	n	6E
O	4F	o	6F
P	50	p	70
Q	51	q	71
R	52	r	72
S	53	s	73
T	54	t	74
U	55	u	75
V	56	v	76
W	57	w	77
X	58	x	78
Y	59	y	79
Z	60	z	80
0	30	<space>	20
1	31	<cr>	0D
3	32	<1f>	0A
3	33	#	23
4	34	&	26
5	35	'	2C
6	36	-	2D
7	37	.	2E
8	38	:	3A
9	39	@	40

TABLE 5-7: ASCII Character Set

Message Type_____ANY_____

Starting Address - 3000:___ANY

Message Character	ASCII (HEX)	Message Character	ASCII (HEX)
01_S_	53_	33_d_	64_
02_C_	43_	34____	20_
03_O_	4F_	35_i_	69_
04_T_	54_	36_t_	74_
05_T_	54_	37_e_	65_
06____	20_	38_m_	6D_
07_P_	50_	39____	20_
08_A_	41_	40_i_	61_
09_P_	50_	41_n_	6E_
10_E_	45_	42____	20_
11_R_	52_	43_L_	4C_
12<cr>	OD_	44_O_	4F_
13<lf>	OA_	45_G_	47_
14_D_	44_	46____	20_
15_r_	72_	47_b_	62_
16_i_	69_	48_o_	6F_
17_v_	76_	49_o_	6F_
18_e_	65_	50_k_	6B_
19_#_	23_	51<EOM>	00_
20_1_	31_	52____	____
21____	20_	53____	____
22_C_	43_	54____	____
23_H_	48_	55____	____
24_._	2E_	56____	____
25_A_	41_	57____	____
26<cr>	0D_	58____	____
27<lf>	0A_	59____	____
28_R_	52_	60____	____
29_e_	65_	61____	____
30_c_	63_	62____	____
31_o_	6F_	63____	____
32_r_	72_	64<EOM>	00

FORM 5B-2: Default for Header Message
EXAMPLE 5B-1: MESSAGE HEADERS

CHAPTER 6

INSTALLATION AND STARTUP GUIDE (OUTLINE)

6-1. INTRODUCTION

1. Keep detailed installation log for each drive
2. Install Equipment
   (Align motor: Set up: coupling to run uncoupled from driven device; Motor protectives - lubrication, thermocouples, RTDs, vibration detectors; Transformers - fault relays, liquid alarm, overtemperature alarm, CTs, RTDs; Source and motor breakers - close and trip circuits, manual trip, IOC/LOR)
3. Interconnect equipment
4. Check Wiring
5. Checkout source switchgear and load isolators
6. Preliminary Checkout
   Check Berg jumpers, daughter board components, burden resistors
7. Apply control power
   Check out starters and blowers (Rotation)
   Check power supplies - PSFF, P105, PS7
   Check relays and lights
   Check/adjust electronics calibration voltages
8. Gate test mode Operation
   Check source and load side converter firing
   Check process inputs
   Check process outputs
   Adjust dc offset for flux waveform at NLIB cards
9. Apply source voltage and Check attenuator signals
   Magnitude and scaling
   Phase sequence
   DC to ground
   Flux waves
   Zero crossings
   VABS, OV, UV
10. Exciter voltage control
    Measure voltage, Ph sequence, current
    Measure field signals at MEM
11. Perform crowbar test
12. Run motor uncoupled - check lube, cooling vibration
    Local
    Remote
    Accelerate, decelerate, stop, regenerate
    Check input, output signals and contacts
    Repeat all of the above for other channel (B) – if 12 pulse or standby system
13. Check reference DMC/LCI Data Link (if used)
    Verify ribbon connections
    Check for communications toggling
    Verify dual ported RAM addressing
14. Tachometer Hook-up
    Mounting Recommendations
    Interface with LCI-A and LCI-B (if 2 channel)
    Enabling/disabling PLL operation
    Setting up marker pulse
15. Couple motor and tune-up the drive
16. Evaluate performance

6-1.1. DETAILED INSTALLATION INSTRUCTIONS

. Preliminary Considerations

PWB Variable Components Refer to system elementary. Extract each PWB listed on elementary sheet 1DA and confirm that Berg jumpers are in the position indicated and that variable components are the values shown in the tables.

Megger Tests

Preparation - Temporarily disconnect ribbon cable and chassis connections as follows:

MVB:
Disconnect the ribbon cable to each HPTK card in the converter leg modules. Also disconnect the 'GND' connection to the HPTK and the XPTN in each leg module. Disconnect the load-side line filters at the fuses.

Megger Power Busses - Confirm that both the source-side circuit interrupter and the load-side circuit interrupter are open. (Some installations have only one interrupter and some have both.)

Disconnect motor leads (M1, M2, and M3) from the load side bridge. (Back panel entry may be required) Temporarily jumper the converter source-side terminals (L1, L2, L3), load-side terminals (M1, M2, M3), and the dc buses (P1, N1, P2, N2) together. Megger from any bus to ground (they are all tied together) preferably using at least a 1000V megger. Indicated resistance should be greater than 100 megohms.

Remove jumpers and reconnect the ribbon and chassis cables. Reconnect the load-side line filter.

Jumper the load-side circuit interrupter load-side terminals (T1, T2, T3) together and megger to ground. Indicated resistance should exceed 100 megohms. Remove jumpers.

Check continuity across cells
Using a multimeter, check across each cell in the bridge.

Measurement should indicate approximately 10K ohms. If greater, check the equalizing resistors in the snubber boxes. If less than 100 ohms, check for defective snubber capacitor.

Apply Auxiliary Power
Open the electronics power circuit breaker at the power supply. Close-in the 120 Vac control power (May be off 480 Vac).

Close-in the 480 Vac breaker at the motor control center for drive blowers and confirm that the blower(s) bring air through the converters front grill and out the top of the fan assembly. Also check direction of rotation.

Signal Input Power Supply Measure output of P105 volt power supply on the I/O module and confirm that voltage output is approximately 105 Vdc.

Electronics Power Supply
Pull all the cards in the SEM. Turn on the power supply and measure the bus voltages at the test rings on the small electronics modules (SEMs) header (top). Voltages should be as follows:

(See elementary sheet 1BB for trimpot adjustments)

P5 = + 5.0V (±.1V)
P28 = +28.0V (+2V)
P15 = +15.0V (±.15V)
N15 = -15.0V (±.15V)

Turn off electronics power supply, check that PROMS are in the HMPG and DMPB and reinsert the cards into the SEM. Turn on power supply and check the bus voltages again.

Gating Power Supply
Check that 120 Vac is present at the input of the P70 gating power supply. Power on the supply and check for output of approximately 85 Vdc. Also check at each HPTK board in the bridge (front edge pin "P70") for 85 Vdc. If absent check HPTK fuses.

Set-up Panel Printer - Consult the panel printer's manual to enable the battery backup (clock). Then set the time and date. Also set the internal time greater than 60 minutes.

Run LCI Selftest:
Turn-on (up) NLCB test switch. Reset HMPG by jumpering TP1(RST) to TP2(DCOM). Follow the indicated instruction and investigate any fault messages. Repeat till unit passes all tests.

*** NOTE ***

With EEPROM variable SQETST=F, field exciter test is not performed. Also, with SQRT=F, at the relay outputs are not tested.

Operating Set-up Adjustments and EEPROM Memory Initialization:
Measure and set the voltages for all the OV, SUPP, RTD, UVR, OSP test rings as tabulated in the system elementary sheet 1DC.

Reset SEM, using the HLCB RESET switch; the HMPG IMOK LED should come on. Jumper testpoint TINO on HLCB-2E to DCOM, turn-on(up) WREN switch on DMPB(HMPG), WREN LED on the DMPB card should come on and reset SEM . Remove the jumper and turn off WREN switch. Dump EEPROM memory (using VERIFY) on a printing terminal and compare to values shown on system elementary (sheet 1DE, 1DF, 1DG). See Chapter 5 for monitor commands.

Bridge Current Feedback Calibration (MVB):

Verify resistors and Berg jumpers on the XTFH against system elementary (sheet DA). Remove CT lead plugs JR and JT from the XTFH. Measure the resistance across each of the burden resistors on the XTFH; it should equal the "R BURDEN SETTING" value per the elementary (sheet EA). Adjust the corresponding pot to fine tune. Re-connect plugs JR and JT (Elementary sheet AC & AD).

Gate Test Mode Operation

Refer to the paragraph entitled GATING TESTS in Chapter 7 for a description of procedure.

Verify that the source-side circuit interrupt and load-side circuit interrupter are open.

Follow the procedure described in GATING TESTS.

Check Process Inputs
Verify that the source-side circuit interrupter and load side circuit interrupter are open and locked out.

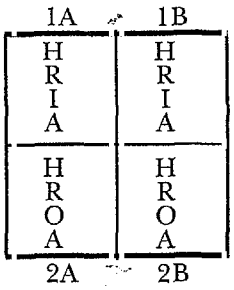
The purpose of the tests in this section is to operate each external input shown on system elementary sheets BD, BE into the SEM. Operate each of these inputs by the most convenient means and confirm that the correct HISA LED lights with a closed circuit in each case.

Check Process Outputs

Verify that the source contactor and load isolator are open and locked out.

Connect the hand terminal (or equivalent) to the HMPG and MODIFY EEPROMs variable (SQIOTST) to "T" to temporarily disable the interrupts. (See Chapter 5, section MONITOR - section EXTENDED MODES for procedure.)

Using the OUTPUT command, operate the output relays shown on system elementary sheet BG and BJ in accordance with the tables below. Confirm that the relays operate the proper device(s) per the system elementary. These leds are on the HRDA card in slot 2A. Also confirm that the proper "relay" lights on the HRDA cards in the I/O rack (lower back wall of control compartment) illuminate. Cards here are arranged as illustrated below.



Relays may be operated in any desired combination by adding the appropriate DATA values from Table 6-1.

Table 6-1. Slot 1A Relay Values

Address	Data	LED On
8040:	01:	0
8040:	02:	1
8040:	04:	2
8040:	08:	3
8040:	10:	4
8040:	20:	5
8040:	40:	6
8040:	80:	7
8044:	01:	8
8044:	02:	9
8044:	04:	A
8044:	08:	B
8044:	10:	C
8044:	20:	D
8044:	40:	E
8044:	80:	F

DC Offset Calibration

Adjust dc offset for flux waveform at NLIB cards. (Slots 1C and 1D)

This procedure should be performed whenever the NLIB card is replaced by a new one.

When the source power is off and the control is not in gate test mode, measure TP4, TP5 and TP6 at NLIB with a DVM in millivolts dc range. Adjust offset pots (blue) R1, R2 and R3 respectively on the NLIB mother board so that TP4, TP5 and TP6 read zero millivolt dc. Repeat procedure for other NLIB.

Set Up Analog Outputs

If a DS3820SIM module is provided, and one or more isolated output from the module are from the D to A port then use the "ASSIGN DA" command to assign the correct variable, then use "SDA" command to save the variable in the nonvolatile memory. See elementary sheet CQ.

Re-MODIFY SQIOTST to original value (F).

No-Load Full-Voltage Tests

The tests in this section require that the source power be connected and available. Open and lock-out the load side circuit interrupter. (If no interrupter is available disconnect leads to motor.) Make sure that switchgear relay can trip incoming switchgear when a fault is detected by the control.

Apply the control power, reset the SEM and check that all the IMOK LEDs are on. Check for any fault indications. Investigate and clear any indicated faults. Null all meters.

Close source side circuit interrupter. Observe that all source side bridge neon lamps are illuminated.

Voltage Phasing and Balancing

A. Connect an oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2C. Confirm that the phase sequence is 1-2-3 and that all these are approximately equal. (Check magnitude with DVM)

The actual system voltage = (ATTN RATIO) X V_{line} where "ATTN RATIO" is given on system elementary sheet 1EA ("Bridge voltage ATTEN RATIO") and V_{line} = (VAC+ VBA + VCB)/3

*** NOTE ***

Use V_{line} to obtain actual system voltage so that flux calibrations in next section will be balanced.

*** NOTE ***

Must measure between TP and ACOM testpoint on these cards.

Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9 (NSFC-2C). This value should be less than 5 millivolts.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FAB) which is on the DLIB (NLIB daughter board) to obtain voltage as follows:

$$FAB = \frac{\text{Actual System Voltage}}{\text{Nominal System Voltage}} \times 4.75 \text{ Vac rms}$$

"Actual system voltage" should have been calculated in Step A above and "nominal system voltage" can be found on system elementary Sh. 1AA (Secondary of transformer) or sheet 1EA (Table).

Now connect the voltmeter to TP2 (FBC) (NLIB-1C) and adjust R2(FBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FCA). Balance signals as accurately as possible (+5 millivolts). Check for dc offset at TP4, TP5 and TP6 of NLIB (< 100 millivolts).

C. Check for presence of zero crossing pulses at TP8 (SFT1) on the HLCB-2E.
This test point displays angle control information as well as zero crossing pulses. Look for narrow 2.5 Vdc pulses evenly spaced at 2.78 msec intervals (every 60° of the 60Hz source)

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the dc voltage level is approximately 2.229 X Vac at TP9 of NSFC-2C.

Set Overvoltage:
Connect the voltmeter to TP18 (OVR) (NLIB-1C daughter board) on the DLIB-1C. Turn R8 (OV) slowly counterclockwise until the "OV" LED comes on. The source side circuit interrupter should trip.

Observe the dc voltage on TP-18 and re-adjust R8 so that the voltage is as calculated in equation I. See elementary sheet DC for per unit source overvoltage.

Set Undervoltage.
Clear any faults and close the line contactor. Connect voltmeter to TP19 (UVR) of DLIB (NLIB-1C daughter board).

This test point is directly behind R1, R2, R3 (blue) pots on NLIB.

Connect an oscilloscope to pin 55-1C; this should be a logic zero. Turn R9 (UV) slowly(blue trim pot below UVR TP) CW until pin 55 just transitions high (may alternate high and low).

Observe TP19-1C and adjust UV(R9)-1C so that the measured value is as calculated in Equation 2. See system elementary sheet 1DC for per unit source undervoltage.

Load Side Bridge Voltage Tests with signals from Source Side Bridge
Trip source circuit interrupter. Disconnect ribbon cable JA from NSFC-2B and move the ribbon cable JA from NSFC-2C to NSFC-2B. Connect a test jumper from backplane pin 55 (NLIB-1C) to pin 1(NLIB-1C) (disables source undervoltage fault).

Voltage Signal Phasing and Balancing
Close source circuit interrupter.

A. Connect oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2B. Confirm that the phase sequence is 1-2-3 and all balanced. Also, ma y be desired to use DVM to measure RMS magnitudes. ("Base Voltage") The testpoints should measure the same as for the source-side. Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1D. Adjust R1 (FAB) which is on the DLIB (NLIB daughter board) to obtain voltage as calculated in Equation 3.

Rated frequency and nominal motor volts from elementary sheet 1EA.

The actual system voltage = (ATTN RATIO)x V_{line} where "ATTN RATIO " is given on system elementary sheet 1EA (" BRIDGE VOLTAGE ATTEN RATIO") and $V_{line} = (V_{AC} + V_{BA} + V_{CB})/3$.

Now connect the voltmeter to TP2 (FBC) and adjust R2 (FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (±5 millivolts). Check for dc offset at TP4(FBAR), TP5(FCBR), TP6(FACR) of NLIB-1D. (< 100 millivolts).

*** NOTE ***

Use V_{line} to obtain actual system voltage so that flux calibrations in next section will be balanced if > 100 mV refer to "DC offset calibration" procedure in preliminary set-up section.

C. Check for the presence of zero-crossing pulses at TP9 (SFT2) on the HLCB-2E. This testpoint displays other interrupt information as well as zero crossing pulses. Look for narrow 2.5 Vdc pulses evenly spaced at 2.78 msec intervals (every 60° of the 60 Hz. source).

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1D and check that the dc voltage level is the same as for the source-side 2.229 X VAC at TP9 of NSFC-2B

Set Overvoltage

Connect voltmeter to TP18 (OVR) on the NLIB-1D (Testpoint on daughter board - DL1B). This testpoint is directly behind R1, R2, and R3 (blue) trimpots on NL1B-IB. Turn R8 (blue trim pot below OVR TP)) slowly counterclockwise until the red "OVA" LED lights. If used, motor contactor in lineup should trip. Observe the dc voltage on TP18 and re-adjust R8 so that the voltage is as calculated in Equation 4.

Set Speed and Voltage Indicators

Since speed is derived from the motor voltage frequency, the source 60 Hz can be used to set the overspeed.

Set TP OSP NLCB-1E (testpoint on daughter board DLCB just below blue OSP trim pot) to -11.0 Vdc. Close in the source interrupter. Connect a voltmeter to Tp1 SPD NLCB-1E. Adjust SPD (R21) trimpot on DLCB (NLCB daughter board) until TP1 = 10.0 Vdc.

Speed Indication:

Adjust R4 (SPM)-1E so that the speed indicator on the control panel reads rated motor speed for 60 Hz.

Motor Voltage Indication:

Adjust VM (R7) -1D on DLIB so that the motor voltmeter reads actual line voltage.

Set Overspeed:

Connect voltmeter to testpoint OSP (TP6)-1E and slowly turn OSP (R2)-1E CCW until the OSP LED comes on. The voltage on TP6 should be +10 Vdc at the trip point. See elementary sheet DC for per unit overspeed.

Adjust OSP until TP6 = [per unit overspeed] X trip level.

Trip the source circuit interrupter.

Restore NSFC ribbon cables to correct configuration.

Remove jumpers.

Equation 1

$$\text{(NL1B-1C) "OVR" volts} = \frac{\text{Per unit source}}{\text{Overvoltage}} \times \text{(trip value)} \times \frac{\text{Nominal System Voltage}}{\text{Actual System Voltage}}$$

Equation 2

$$\text{(NL1B-1D) "UVR" volts} = \frac{\text{Per unit source}}{\text{undervoltage}} \times \text{(trip level)} \times \frac{\text{Nominal System Voltage}}{\text{Actual System Voltage}}$$

Equation 3

$$\text{FAB} = \frac{\text{Actual System Voltage}}{\text{Nominal Motor Voltage}} \times 4.75 \text{ Vac rms} \times \frac{\text{Rated Freq.}}{60}$$

Equation 4

$$\text{(NL1B-1C) "OVR" volts} = 1.2 \times \text{trip value} \times \frac{\text{Nominal Motor Voltage}}{\text{Actual System voltage}}$$

Equation 5

$$\text{TP IFB-2A} = \frac{\text{Rated exciter current}}{\text{Rated EVC output}} \text{ 5 Vdc.}$$

(See elementary Sh. 1EA)

Exciter Voltage Control

Preliminary Checkout:

Field Exciter Current Feedback Calibration
Remove CT lead plugs JB and JC from the NHVL. (Inside EVC Compartment) Measure the resistance across the burden resistors (R70 and R71). It should equal the "current feedback burden setting" given on system elementary sheet 1EA. Adjust R (IFB) to fine tune. Reconnect JB and JC.

Phase Sequence Check:
Apply source voltage and 480 Vac. Connect an oscilloscope to simultaneously observe the waveforms on testpoints ATTEN-AI(TP1), ATTEN-BI(TP3), ATTEN-CI(TP5) on the NHVL and confirm that the phase sequence is 1-2-3 and are balanced. Close-in the source switchgear and using a oscilloscope, compare the TP VAC-2C (NSFC) against TP LAC-2A (NSFE). Measure # of degrees LVAC-2A leads (-) or lags (+) VAC-2C.

Record, modify and save FANGLAG to value. (Typical values of -30, 0, +30).

Power Check:
Attach a voltmeter across one phase of the exciter output. Attach a clamp-on ammeter around one of the output lines.

Control From SEM

1. Set local speed reference to minimum (drive control in local).
2. Open the flux regulator by setting SQFE to "T"rue (Example 6-1). The field contactor should close.
3. Using the local speed pot, which now controls the EVC output current, increase the exciter current. (Turning in clockwise direction). Stop when rated exciter current ("Rated EVC Current system elementary sheet 1EA) is achieved as measured with clamp-on ammeter. (Verify Equation 5.)
4. Verify all 3 phase currents and voltages are equal
5. Reduce exciter current to zero and SQFE to "F"alse. Local speed pot full counterclockwise.

```
MONITOR    OFF    <cr>  ; Adjust local      speed reference pot to minimum

LCI MONITOR
* >PASSWORD  <cr>
PASSWORD ?  <cr>

PRIV. ENABLED
* >   TUN  <cr>

T >   MOD SQFE <cr>
SQFE = F
VALUE ? = T <cr>
SQFE = T  ; Adjust local reference to get      desired
VALUE ? = F  <cr>   EVC current ; when      finished.
SQFE = F
VALUE ? =  <cr>
```

Example 6-1 : EVC TEST

Crowbar Current Tests
Crowbar Test Description The crowbar test permits controllable dc link current to be circulated without involving the motor windings. On the load side, the current circulates through one or more diametric cell pairs and thus bypasses the motor. The signals to be observed on the oscilloscope will indicate whether the source-side CTs are phased correctly. For the tests in this section, the load-side circuit interrupter should be open. For lineups without load side interrupter, disconnect , M1, M2, and M3 to isolate bridge from motor (May require back access)

Test Setup:

A. Connect one channel of the oscilloscope to firing pulse #1 (SIG1) on NSFC-2C. Connect second channel to testpont 11 (01A) on NLIB-1C.

Establish Initial Conditions (Example 6-2):

A. Reset SEM and apply primary power.

B. Connect the hand terminal (or crt) to the HMPG.

C. Enable crowbar mode;
Modify variable SQCB to "T"rue

D. Set minimum current limit to zero; Modify CRMIN to 0

E. Set starting current to zero. Modify CRSTART to 0

F. If there is a motor isolator, make sure it can be closed by RUN relay; or jumper TB4-5 to TB4-6 (sheet BD) to simulate closure.

Actual Test:

A. With LCI in "Local" Mode (Local/Remote Switch in "Local" position, initiate a "Run" sequence; i.e., sequence push the "Run" pushbutton and observe hardware "Run" relay picks up as well as software "Run") The current regulator should now be active but regulating zero current. Check that MODEFLG bit 0 or 1 is set. If any problems, consult drive elementary sheets BH and FF. Some jumpering at I/O module may be required here.

B. Using the modify command, set integer CRSTART to .1 PU (10% current command). Check TP7 (IFB NL1B-1C) on the NLIB for -0.5 Vdc ± 0.2 Vdc. Current IFB should appear stable as viewed on an oscilloscope.

C. Increase CRSTART to .2 (20% current command) using the MOD command. Current should now be continuous (not going to zero). Observe the test-point pairs given in Figure 6-1, and check for proper phasing. If waveforms don't per Figure 6-1, current transformer feedback wires must be "rolled" to correct. (See drive elementary sheet AC-XTFH board).

Note: When circulating current in the crowbar mode, current should be applied only for short periods of time (about 15 second duration and 25% duty) when operating above .2 PU current.

Take scope traces of all three currents as indicated in Figure 6-1 ; record peak volts. This also checks source-side CT phasing.

D. Ammeter Check

1. Apply .5 P.U. (CRSTART) and check TP7 (IFB NL1B-1C) for .5 P.U. current (approximately

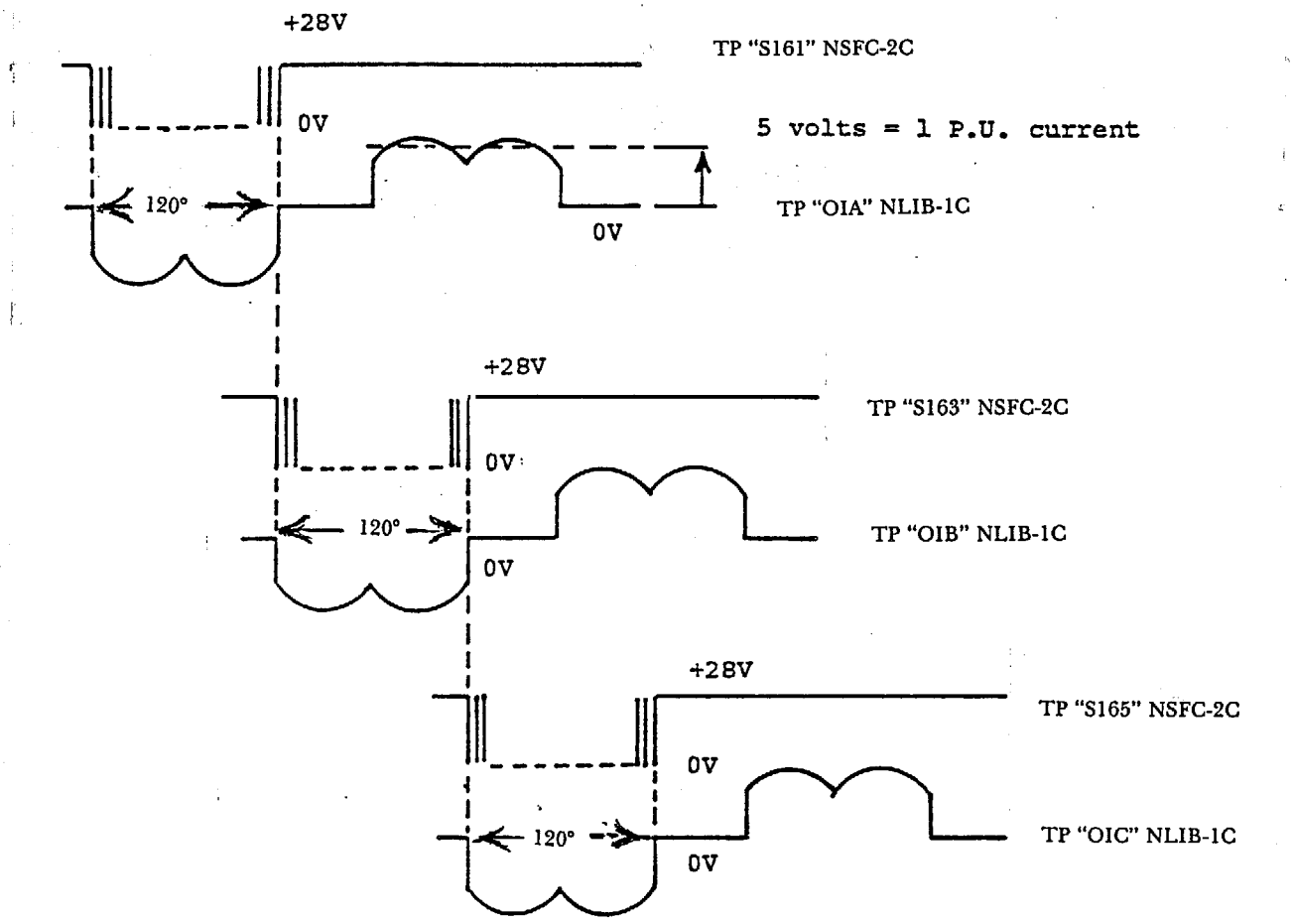


Figure 6-1. Firing Signals and Current Signals

2.5 Vdc +20%). Adjust ammeter to read 50% motor current . (See drive elementary sheet EA for this information).

E. Source Hardware Suppression

1. Adjust SUPP pot (NL1B-1C daughter board) to minimum (Fully cw)
2. Initiate current flow and adjust CRSTART up from .5 P.U. At .6 P.U. ±30%, the SUPPression LED on the NLIB-1C should come on and the line contactor should trip.
3. Readjust SUPP pot (NL1B-1C daughter board) for the value shown on system elementary sheet DC.

F. Modify variable SQCB, CRSTART and CRMIN to original value per system elementary sheets DE, DF, and DG, and reset SEM.

```
MONITOR OFF <cr>

LCI MONITOR
*> PASSWORD
PASSWORD ? <cr>

PRIV. ENABLED
* >TUN <cr>

T >MOD SQCB <cr>

SQCB = F
VALUE ? T <cr>

T> MOD CRMIN <cr>
CRMIN = 0.20
VALUE? 0 <cr>
CRMIN = 0.0
VALUE ? <cr>
VALUE ? <cr>

T >MOD CRSTART <cr>
CRSTART = 0.5
VALUE ? 0 <cr>
CRSTART = 0.0
VALUE ?<cr>
T >MOD CRSTART <cr>
CRSTART = 0.0          ;Start drive
VALUE ? 0 < cr>
CRSTART = 0.00         ;Check IFB
VALUE I .1 <cr>        ;Increase by 0.1
CRSTART = 0.1  ;Check IFB level and if continuous
VALUE ? I <cr>; Continue increasing I till
                       IFB = 0.5 pu
.
:
CCMD = 0.5     ;Check IFB and calibrate ammeter
VALUE ? I <cr> ;Adjust SUPP to min and
                  increase I till trip
               ;Approximately 0.6 pu
```

Example 6-2:

Run Motor Uncoupled

A. Test Connections

Considerations:
Motor bearing protection must be functioning at this time to protect the motor. Once the motor is started, it should be run-up to full speed and magnetic center checked, bearing temperature rise checked, bearing oil (sling) rings checked, and vibration checked.

1. Make the following connections to a six channel chart recorder:
   CH1 - DA1 TP2 NLCB-2E
   CH2 - 0IA TP11 NL1B-1D
   CH3 - FCBR TP5 NL1B-1D
   CH4 - 0IC TP9 NL1B-1D
   CH5 - IFB TP7 NL1B-1C
   CH6 - SPD TP1 NLCB-1E

2. Temporarily modify the following EEPROM variables:
   SRCLSC to 0.2
   SQOPT2 to 0
   SRMINR to .10
   SRGAIN to 3.5
   SRTC to 1.0
   RRMINL to .10
   MTRFMIN to .25 pu
   CRSTART to .4 pu
   FLALEN to "F"
   FLASEN to "F"

   FLPBRK1 to 1.1
   FLPBRK2 to 1.1
   FLPMAX1 to 0.5
   FLPMAX2 to 0.5

3. Assign LMODE to D/A #1
   (Scale = -5)
   (offset= 0

*** NOTE ***

Scale is minus 5 which means multiply LMODE by 32 before output to D/A port. (See 5-6 for this procedure)

B. Start and Run Motor
   Check that auxiliary and control power is applied.

1. Reset the micro-processor (temporarily touch RST (TP1) to DCOM (TP2) and clear any faults and/or alarms.
   Close source side circuit interrupter.
2. Select local speed reference (Local/Remote REF inside the door) and set the reference to 0.2 pu which equals 5600 counts (1 pu = 28000 counts)
3. Ensure SQOPT2=0 and SQPOS=F. (LCI will use PLL vs TACH feedback). Tach feedback is to be verified later.
4. Initiate run. Motor should start in made "0"; switch to mode "1", switch to mode "2" at about 2 to 6 Hz and then switch to mode "3" at about 7.5% speed. Check motor rotation. If motor runs backward, check LCI-to-motor wiring; swap two motor leads if necessary.
5. Verify that IFB, OIA, OIB and OIC appear as shown per Figure 6-2

If motor fails to start, observe the memory variable FLUXFBK. Then adjust the EEPROM variable MTRFMIN down until a start is obtained, or see Section B of "O"perate with motor coupled.

Running mode is determined by the voltage level out of the DALO testpoint of the DA1-2E (channel 1 of recorder) as follows:

With no tach

LMODE	DA1 output level	
0	0 Vdc	IDLE (No firing)
1	2.5 Vdc	Forced Commutated, PLL open
2	5.0 Vdc	Forced Commutated, PLL closed
3	7.5 Vdc	Self Commutated using PLL

with tach

0	0 Vdc	idle (no firing)
8	5.0 Vdc	forced commutation position open loop
7	4.4 Vdc	forced commutation position close loop
3	1.9 Vdc	self commutation using PLL for firing and tach feedback for regulation

6. Initiate stop and motor should coast to a stop.
7. Restart and check the load-side firing signals and current signals on NSFC(2B) and NLIB(1D) are like figure 6-1.

C. Speed Range Test

Start the drive and gradually increase the drive speed to rated value. If motor voltage fails to increase, reverse two field leads.

Monitor TP5 (SPD)-2E. Voltage should be -10 Vdc. Panel speed meter should read rated rpm.

D. Speed Ramp Test

Initiate a run from minimum speed to top speed by changing the reference as fast as possible, while recording the speed signal. Verify that the acceleration rate is correct.

Reduce reference to minimum as fast as possible while recording the speed signal. Verify that the deceleration rate is correct.

E. Tachometer Setup

*** NOTE ***

with SQOPT2=0, phase lock loop (PLL) is used as feedback to speed regulator (ie. zero crossings from motor CEMF to determine frequency which determines speed) with SQPOS=F, cell firing is calculated, again from PLL (zero crossings) of motor CEMF.

If PLL FDBK is used vs tach FDBK, the speed regulator and flux regulator must be "detuned" to 4RAD and 12RAD respectively. (See section on regulator tune-up procedures)

*** NOTE ***

If the tach feedback signals are not symmetric around zero (bipolar) make sure pin 10 of HTCA card is connected to pin 3 (to bias the non polar signals to bipolar).

a. Run motor at 10% - 20% speed, adjust tachometer so that B-HTCA (TP2-IF) lags A-HTCA (TP4-IF) by 90 degrees. Also the signals should be symmetric square waves. If not, check spacing between pick up and tooth wheel to manufacturer's specification.
b. Modify TUNEUP variable SQOPT2 to "4" to enable tach feedback calculation. Note, alarm "tach feedback or HTCA problem" may display. Reset alarm by pushing RESET PB on panel door.
c. Display variables speed and TSPD continuously (DC command) to check tach speed calibration.

$$TSPD = \frac{SPEED}{1023} * \frac{FRQM}{FRQMM} * 28000$$

TSPD should be ±% as calculated (±140 count).

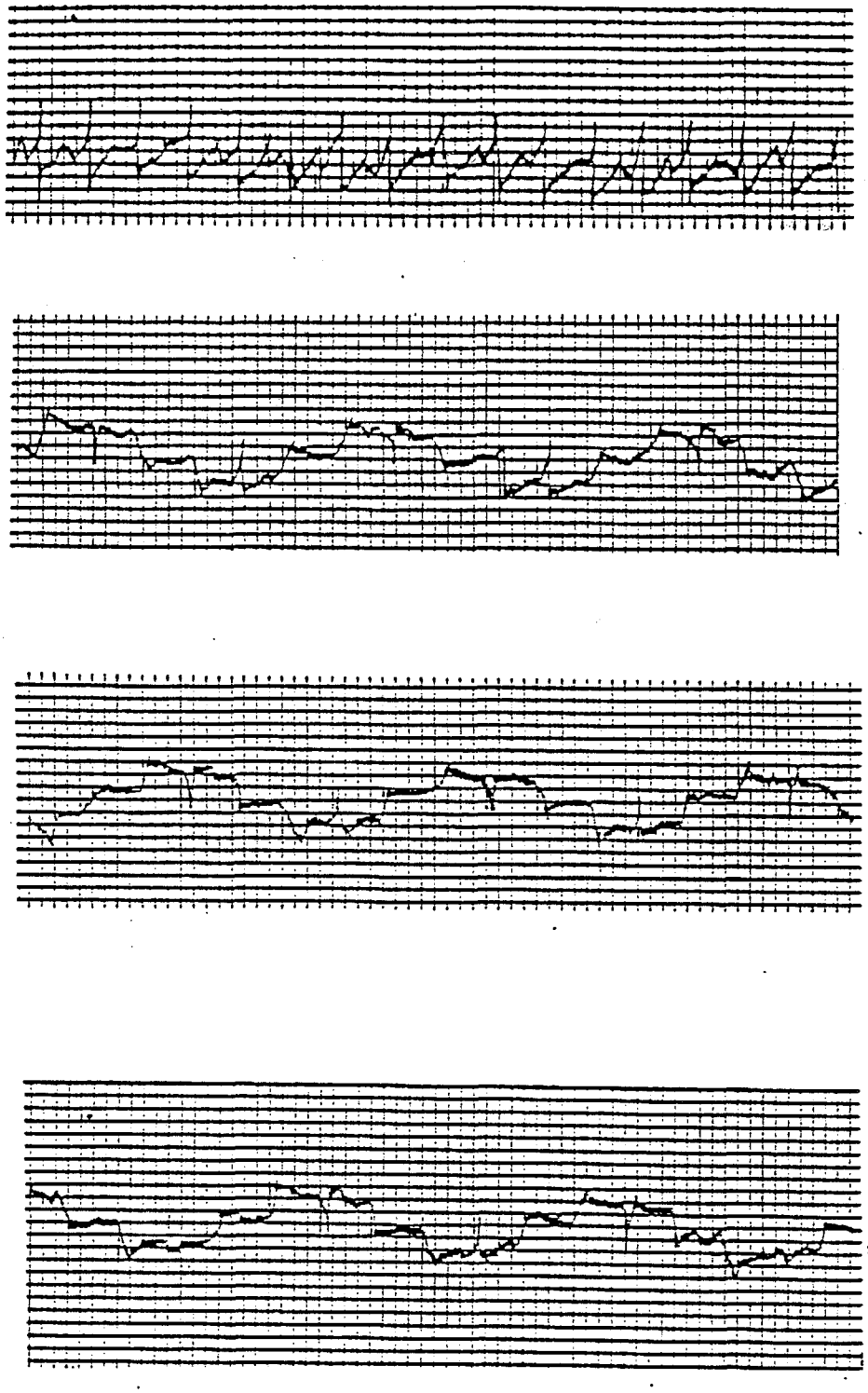


Figure 6-2. Phase Current Relationships

TSPD should be positive, negative means feedbacks are 180° out of phase.

d. Run motor up to top speed gradually, check TSPD calculation, check waveshape and relationship of A-HTCA and B-HTCA as speed increased.

e. MARKER PULSE CALCULATION (for tachometer with marker pulse only). The pulse tachometer marker pulse location must be known relative to the motor voltage. Since it is inconvenient to mechanically align the tachometer to the machine voltage, the drive software has been written to accommodate a random location of the marker pulse relative to the motor A to neutral voltage (– to + transition). The following initial conditions are required to perform the calculation.

1 Run the drive in LACMODE 3 by setting SQPOS false.
2. Run the drive at as high a speed as possible to minimize to NLIB integrator error. Run at least 40 Hz but avoid beat frequencies.
3. Run the drive as near zero load as possible.
4. Keep the flux at 1 per unit.

Initiate the test by setting monitor variable, CMFLG, true. The calculation procedure will operate for 8 load side calculations and then set CMFLG false again. The calculated value will overwrite MKPHASE a tuneup variable. To permanently store the value, a monitor "SAVE" command must be issued from tuneup mode.

While running motor at top speed, assign DA1 to TSPD (scale = 4, offset = 0). Modify SRNFTC to 0.1, connect brush to DA1-2E with zero on center line, and 25 V full scale. Start brush (5 mm/sec) and initiate a STOP. Observe TSPD on brush, verifying that coast down trace is smooth from top speed till motor stops, with no spikes (except at D/A "wrap around").

If this is a 12-pulse drive, check tach and MKPHASE setting separately for the other channel before running 12-pulse.

F. 12-Pulse Checkout

Preliminary: Once one channel of the 12-pulse drive has been run, the EEPROM in the idle channel should be updated to that of the previously run channel. Both channel's EEPROM must be identical before the drive can be run as a 12-pulse - except for SQMCH (Channel A should be "T" and channel B should be "F"), MKPHASE and FANGLAG.

Uncoupled: After checking out and running the motor uncoupled from both channels (one-at-a-time 6-pulse), test the 12 pulse link. (See elementary sheets 1BF and 2BF) Start Channel A (Channel B remains stopped) and run to minimum speed. Verify DA4-2E (TP5-HLCB) on Channel A equals 12P-1E (TP2-NLCB) on Channel B.

Verify that channel B is slave (SQMCH=F) and 12 pulse. (TWPLSE) Start channel B and see that the currents balance. Run the drive up near full speed (using as local speed pot), checking that currents, voltages, and flux are balanced.

Reduce speed to minimum, verify channel B's speed pot is fully counterclockwise and then stop A. Upon stopping A, it's EVC field contactor should open, and then channel B's field contactor should close to maintain field current while not upsetting motor operation. LCI-B should maintain motor operation with speed control following LCI-B's local pot.

Once this has been tested successfully, restart A. Upon starting A, the reverse EVC sequence (EVC'B de-energized, EVC-A energized) should occur.

Stop both channels and then start both channels simultaneously. If starting is too fast, reduce CRSTART, SRCLFC, CRALPHA1, and CRALPHA2. on both channels.

When tests are complete modify variables changed for the uncoupled tests back to their default values. These variables include:

RRMINL, SRMINR, CRSTART, SRGAIN, SRTC, MTRFMIN, CRMIN, CRALPH1, CRALPH2, SRCLSC.

Operate with Motor Coupled (6-Pulse)

A. Test Connections

Make the following connections to a six channel chart recorder.

CH1 - DA1 TP1-HLLB-2E
CH2 - 0IA TP11-NLIB-1D
CH3 - FCBR TP5- NLIB-1D
CH4 - 0IC TP9-NLIB-1D
CH5 - IFB TP7-NLIB-1C
CH6 - SPD TP5-NLCB-1E

B. Starting Tune-up

The start-up of the LCI should be optimized by varying EEPROM variables MTRFMIN. The procedure is as follows:

1. Connect terminal to the HMPG.

2. Attempt to start drive and monitor flux FCBR (TP5 NLIB-1D) on chart recorder. If motor fails to start, stop and start drive several times - adjusting STIM to try to maximize flux during the first few firings. If motor still fails to start then modify MTRFMIN down until drive starts. Modify and save .8 x the value of MTRFMIN

into EEPROM. Modify and save FLUXs to .9x and value of MTRFMIN. Record.

3. Tune up current regulator in SEG FIRE. Modify EEPROM variable as follows:

   SRMINR to .05 pu
   RRMINL to 0.05 pu

   This will allow the drive to operate continuously in SEG FIRE with the speed reference at zero. The purpose of this tune-up is to minimize rise time and overshoot of the current regulator during segment firing. Figure 6-3 shows an idealized trace of IFB (TP7 NLIB-1C).
4. Stop and start the drive several times from a dead stop. If it fails to start quickly every time, monitor flux feedback signal FCBR (TP5-NLIB-1D), on a chart recorder. Then stop the drive and lower MTRFMIN by .01 steps to a minimum of .04. Repeat as necessary, trying to maximize the flux signal during the first few pulses when the motor is trying to start. Save and record the final value used.

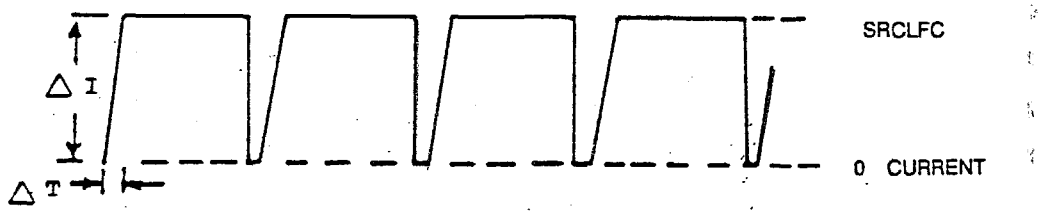


Figure 6-3. IFB during SEGFIRE

The variables to be set up are CRALPH1 and CRALPH2. CRALPH1 initializes the first source-side firing after a current zero to a value that should force link current to equal the setting of CRMIN. CRALPH2 then initializes the current regulator to call for a value of firing which will sustain this current. On the third and subsequent firings, the current regulator becomes active.

Start-up the drive and run at 5% speed. Confirm that the drive is in SEG FIRE. Run the chart recorder at 125 mm/sec and obtain a trace of IFB. Compare the trace obtained with the cases shown in Figure 6-4 and take appropriate tune-up action. Zap final values of CRALPH1 and CRALPH2 into EEPROM. Record for factory records.

C. Motor Flux Check

1. Start the drive and run up to some speed between 50% and 75%. Connect oscilloscope to testpoint "VAC"-NSFC-2B. The voltage at testpoints FBAR, FCBR, and FACR should appear as in the waveform shown in Figure 6-6.

It is important that these be balanced as closely as possible.
Calculate motor volts/Hz (flux).

$$\overline{V/Hz} = \frac{\text{(attn ratio) (Vpk/1.414)/f}}{\text{(rated motor volts)/60Hz}}$$

The flux integrators on 1D should then have peak voltages of:

$$Vf = (\overline{V/Hz}) * 6.718 = _____ Vpk$$

D. Current Regulator

The current regulator for the LCI is calculated for a 250 (± 50) radian response. The parameters involved in this calculation are fairly well known and measurement of the response is impractical; however; the stability of the regulator can be checked by the following procedure:

Run drive to 55%speed and connect the brush recorder to IFB (TP7 NLIB-1C).

Apply a 2-5% step for .1-.5 seconds to the current regulator using the monitor commands - MODIFY REFERENCE, APPLY and REMOVE (TUNEUP MODE-CURRENT REGULATOR). See Example 6-3.

The brush recorder should be running at its fastest speed.

It is important that these be balanced as closely as possible

Verify IFB-1C does not overshoot or oscillate on both the up-step to a higher current and speed and the return step to the lower current and speed.

*>**TUN** <cr>
T >**TUN CR** <cr>

Available Variables are:
(List of Variables)

T >**MOD REF** <cr>

Step	**Level(PU)**	**Time(sec)**
1		
2		

```
Step 1.      Level. =        0.02 <cr>
Step 1       Time   =        .1 <cr>
Step 2       Level  =        0 <cr>
Step 2       Time   =        .1 <cr>
T > APPLY <cr>          ; take brush recording of
                          up + down response
T > REMOVE <cr>         ; remove step function
```

Example 6-3

*** NOTE ***

DA1 has been temporarily assigned to current feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.
DA2 has been temporarily assigned to source side firing angle, for reference only.
100

E. Flux Regulator

1. Check flux vs. speed profile over the full speed range. Refer to elementary sheet 1EA for the desired profile. Connect a DVM to testpoint "FCBR" on NLIB 1D. Operate the drive from minimum speed to top speed, measuring the flux amplitude at selected speeds to verify the flux is following the profile shown on the Drive Data sheet of the system elementary diagram. Rated flux = 4.75 vrms at "FCBR".

2. Connect "VABS" on the NLIB in slot 1D to a chart recorder. This is a fairly clean dc signal which will be proportional to flux if the motor speed does not change during the following test.

3. Start the drive and run it up to approximately 55% speed. Apply a 5-10% step for 3-5 seconds to the flux regulator using the monitor (in the TUNEUP MODE FLUX REGULATOR case) MODIFY REFERENCE, APPLY and REMOVE commands to step the flux regulator reference.

4. Observe the response on the chart recorder. It should be critically damped at approximately 25 radians with tach being used) or approximately 12 radians (without tach operational). Tune per Figure 6-7.

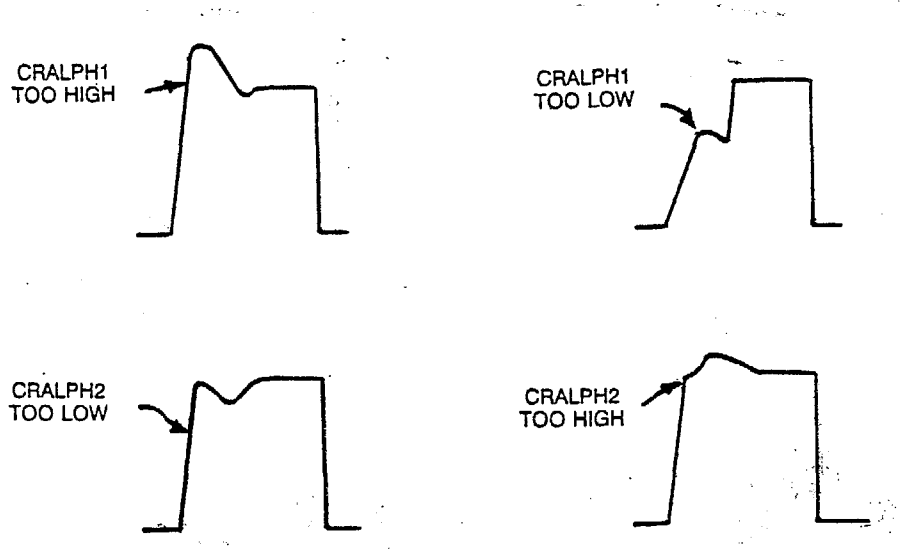


Figure 6-4. CRALPH1/2 TUNE-UP

5. If the response is too fast (or too slow), decrease (or increase) EEPROM variable FRTC until the nearest desired response is obtained.
6. If the response is overdamped or underdamped, increase or decrease the integral gain FRGAIN to obtain critical damping. If damping is now correct but response is not, return to step 5.

*** NOTE ***

DA1 has been temporarily assigned to flux feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.

```
*TUN <cr>
T> TUN <cr>

Available Variables Are:
(List of Variables)

T> MOD REF <cr>

Step        Level(PU)          Time(sec)
1
2
Step 1.     Level(pu)  =       0.02 <cr>
Step 1      Time       =        5 <cr>
Step 2      Level      =        0 <cr>
Step 2      Time       =        5 <cr>

t> APPLY <cr>     :Take brush of up + down
                                response
T> REMOVE <cr>    ;Remove step function
```

Example 6-4

G. Speed Regulator

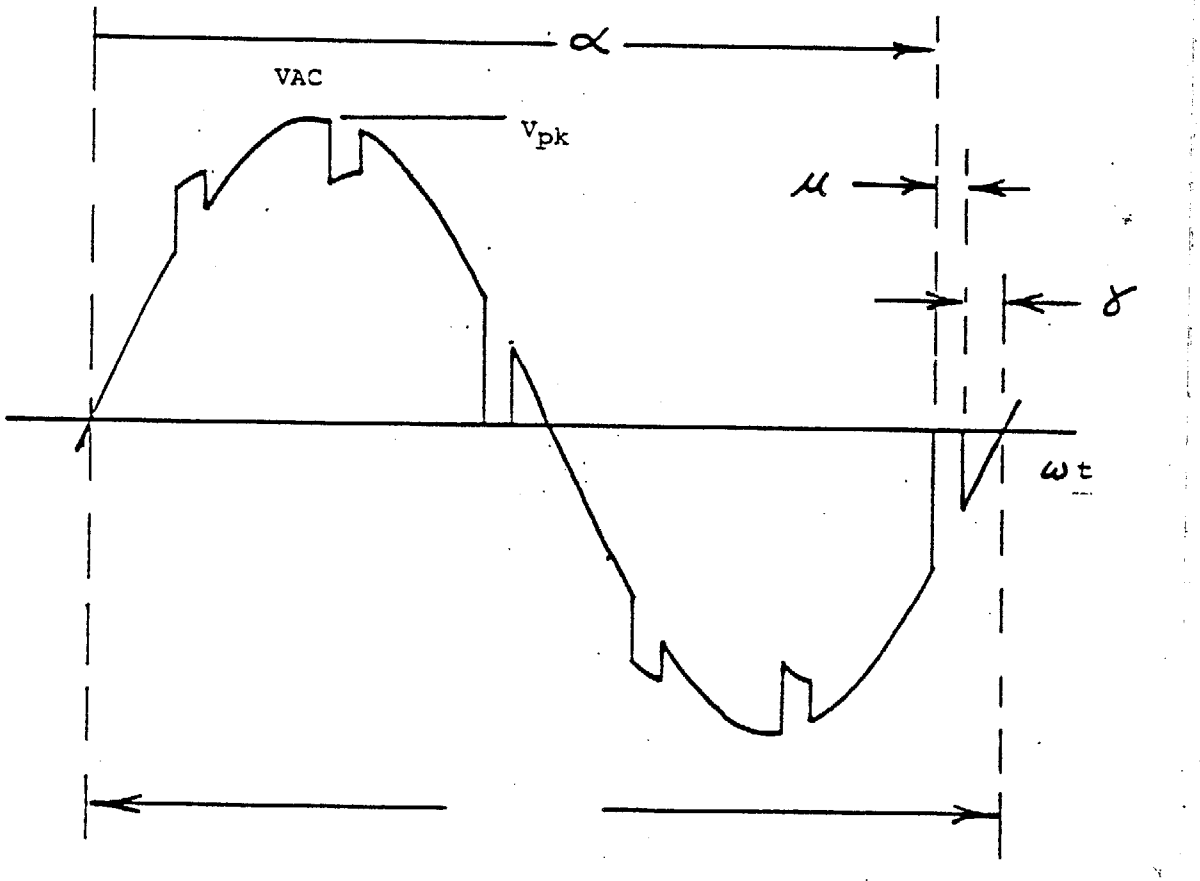


Figure 6-5. Motor Line to Line Voltage Waveform

Run the drive speed up to 75% speed. Verify that CCMD is greater than CURLL and that ALPHSRCO = 0. Apply a step size (2-5% for 5 seconds) to obtain as large a step as possible without hitting current limit (observe IFB on chart recorder). (Using TUNEUP mode-SPDREG: commands MODIFY REFERENCE, APPLY, REMOVE) See Example 6-5.

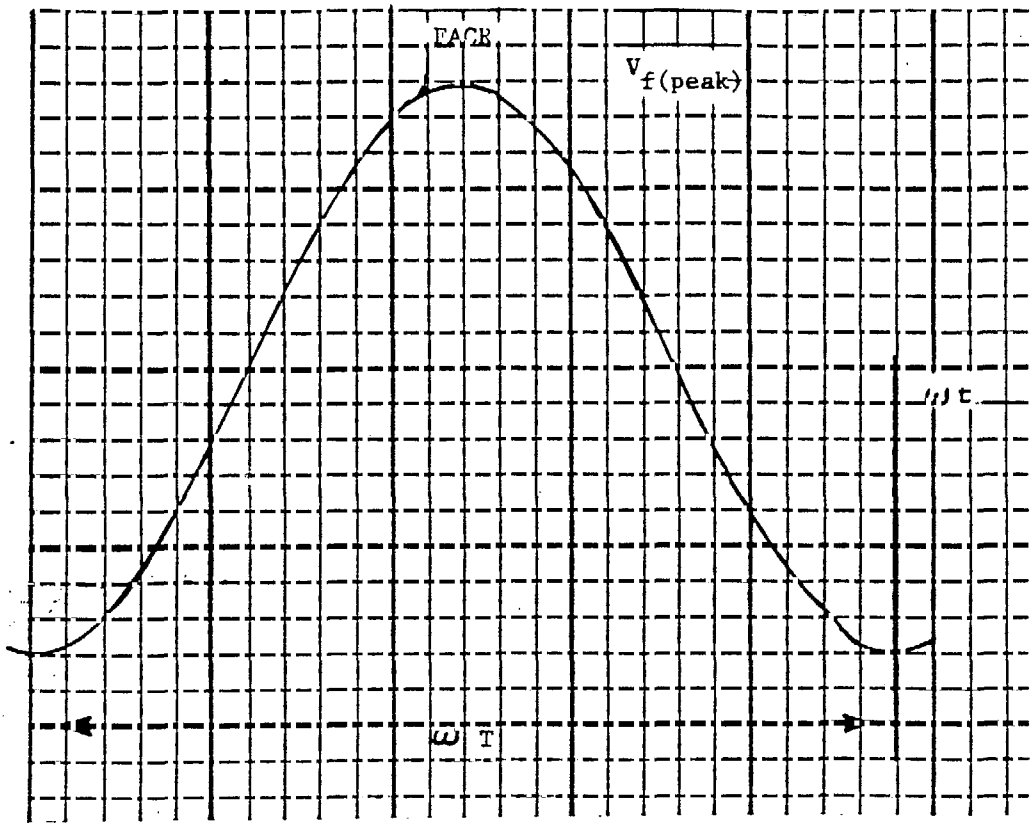


Figure 6-6. Ideal Flux Wave

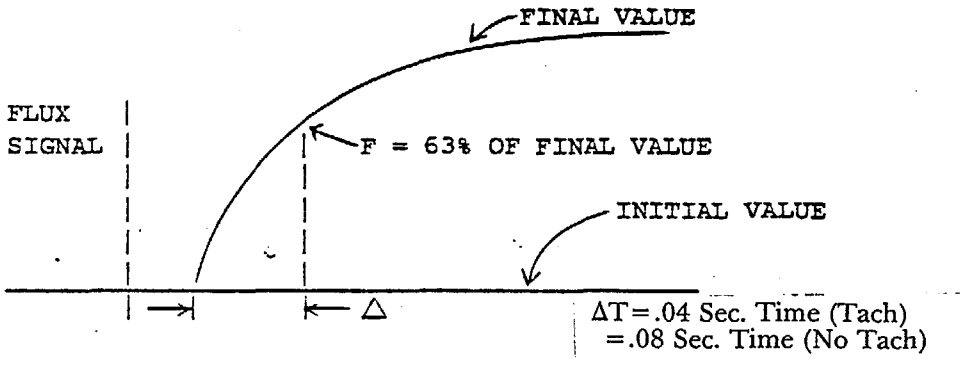


Figure 6-7. Flux Regulator Response

** *NOTE* **

It may be necessary to buck-out the speed signal with a battery voltage to obtain enough amplification. If step down response is different from step up, use step down response.

Observe the step response and adjust SRGAIN and SRTC for a 10 radian (with tach) or 4 radian (no tach). Response as shown in Figure 6-8.

Save these values for SRGAIN, SRTC, into EEPROM. Record.

** *NOTE* **

DA1 has been temporarily assigned to speed feedback, use "ASSIGN DA" command to offset the output so that the step response could be examined.
DA2 has been temporarily assigned to current feedback, monitor the signal so that during the step to make sure the current is not in current limit.

List of Variables are:

*> TUN <cr>
T> TUN SR <cr>

T> MOD REF <**cr**>

Step	Level(PU)	Time(sec)
1		
2		
Step 1.	Level(pu) =	.05 <**cr**>
Step 1	Time =	2 <**cr**>
Step 2	Level =	0 <**cr**>
Step 2	Time =	2 <**cr**>

t> APPLY <**cr**> :Take brush of up + down response

T> REMOVE <**cr**> ;Remove step function

Example 6-5

H. Load-side Fire-As-Late-As-Possible (FALAP)

The load-side converter ordinarily operates (when under load) so that firing occurs at a time which permits a 20 degree commutation margin. Since the commutation time is a function of current and volts/Hz, it is necessary to adapt the firing angle to these parameters. It is the purpose of the load-side fire-as-late-as-possible routine to adjust the firing as a function of load current such that commutation is complete at alpha = 160 degrees. It should be noted that other routines may affect the load-side firing, especially when operating at very light loads.

It is desirable to tune-up under the heaviest steady load possible. Run the drive to any convenient speed and check variables CURLL, CCMD, ALPHSRCO. If CCMD is greater than CURLL and ALPHSRCO equals zero, then the drive should be operating in the fire-as-late-as-possible mode. Verify FLALEN and FLASEN are "F".

Connect a scope to "VAC"-2B testpoint on load NSFC. The waveform obtained should be similar to that shown in Figure 6-9.

γ = MARGIN ANGLE. Proper drive adjustment keeps this angle constant, at a minimum value (after transients have died down), over the entire process range.

μ = COMMUTATION ANGLE

ß = ANGLE OF ADVANCE

$$X_c = \frac{\frac{3}{\pi}}{I_D} * \overline{V/H} * (\cos\gamma - \cos ß);$$

Xc = per-unit commutating reactance,

I_D = per-unit current = I/5V.

LIMARG = $\overline{Xc}$

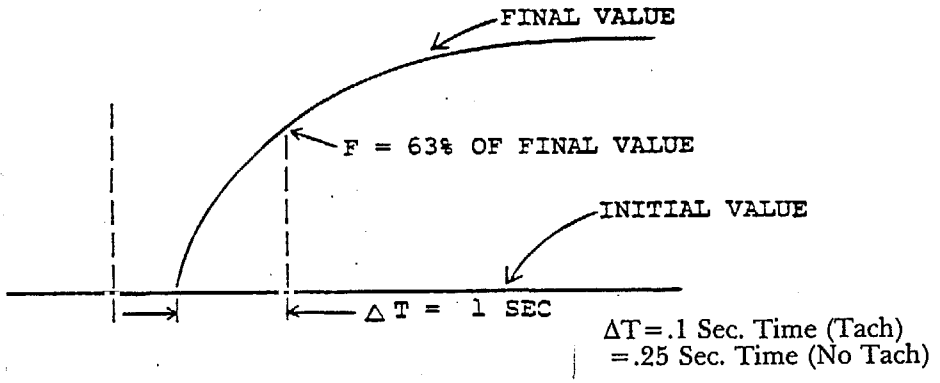


Figure 6-8. Speed Regulator

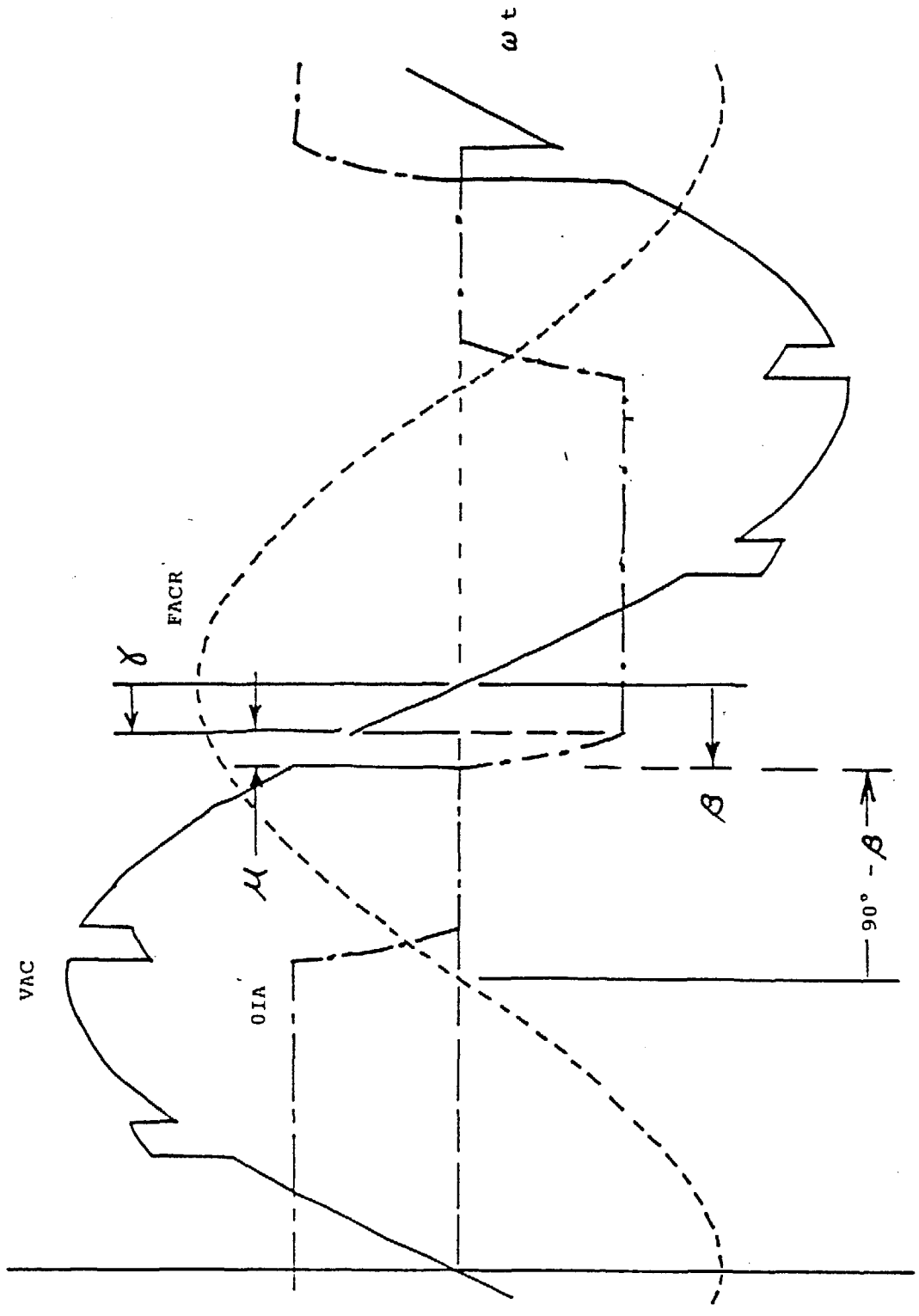


Figure 6-9. Load Side FALAP

EEPROM variable LIMARG controls the amount of advance of the firing angle with load current. If the margin angle is less than 18 degrees, increase the value of LIMARG. If the margin is greater than 22 degrees, decrease LIMARG. The margin angle should remain constant regardless of current if LIMARG is tuned correctly. If change in original value SAVE into EEPROM.

This waveform also shows certain performance parameters. The angle (90° - ß) shows the transition from motoring to regenerating. When the angle is positive (as illustrated) the motor is driving the load in the normal manner. When the angle goes to zero and beyond (negative), the system is regenerating, with the load driving the motor as a generator. The current waveform and the flux wave can be observed also, but the voltage wave by itself shows all the information. The voltage wave is preferred, for accurate firing angle measurement.

If it can be assumed that the voltage behind commutating reactance is an undistorted sine wave, then a reasonably accurate calculation of power can be made by multiplying the line-to-line voltage behind commutating fundamental inverter power factor, by $\sqrt{3}$, and by the motor efficiency. The RMS line-to-line voltage is Vpk * Attenuation ratio divided by $\sqrt{2}$. The current can be read from the panel ammeter; it is also I_D * motor nameplate current (per channel).

The fundamental inverter power factor is:

$$\frac{\cos(ß - \mu) + \cos ß}{2}$$

I. Source-side Line Impedance

Normally the source-side operates in the rectifying quadrant. However, when the drive is forced to slow down, the source converter may have to regenerate (invert). To avoid commutation failures, the angle of retard must be programmed as a function of current. Since a regenerating condition is difficult to hold steady-state, a measurement of system impedance will be made and this value will be programmed into EEPROM.

Run the drive to some convenient speed (the higher the better). Connect channel 1 of the oscilloscope to testpoint "VAC"-1C. Connect channel 2 to testpoint "IA"-1C. The signal should be similar to those shown in figure 6-10.

$\overline{V1\text{-}1}$ = (attn ratio) (Vpk/1.414)/Vnom

where Vnom = source-side nominal voltage

$\overline{I_d} = I/5$

$\overline{X_c} = (\overline{V1\text{-}1}/\overline{Id})[\cos \propto - \cos(\propto + \mu)]$

SIMARG = $(\overline{Xc})$ = _____

Round-up to nearest integer and SAVE into EEPROM-. Take drive to top speed then decelerate as fast as possible. If a software overcurrent occurs, then increase SIMARG.

J. 12-Pulse Coupled Tests

Start channel A and run up to maximum 6-pulse speed. Load channel A to rated current, if possible. (At China Steel, full loading is not possible without rolling the mill).

Start channel B and observe that current balances in 3-5 seconds. Run drive up and down checking current, voltage, and flux balance. Run the drive to maximum 6-pulse speed, stop A, and then restart A. (Remember the EVC transfer sequencing discussed in "12-pulse checkout").

Reduce load and run the drive up to near rated speed. Drop off channel A and see that channel B stays on.

Repeat for channel B.

Stop the motor, then start both channels at once. Verify simultaneous starting current, following rotor position determination at standstill.

If starting is too rapid, reduce SRCLSC, SRCLFC, and retune CRALPH1 and CRALPH2 as necessary.

*** NOTE ***

When in 12-pulse operation, additional commutation notches are present in the voltage waveforms. These notches are not compensated for in the flux wave reconstruction, and cause a phase shift in the direction which increases the margin angle. This 12-pulse waveform is shown below. Also, the current waveform in 6-pulse and 12-pulse operation is not the ideal flat-topped waveshape. It contains a "double-hump" ripple at both the power system frequency; and the motor frequency, in addition to other, smaller irregularities.

K. Source-side Fire-As-Soon-As-Possible

1. Modify and save FLALEN and FLASEN to value listed on drive system elementary sheets 1DE - 1DG.
2. Rerun drive - verifying that there are no performance problems (trips, excessive swings in current, etc.).

1. Motor power factor and power readings (per channel) use Figure 6-9

1. Power factor (PF)

a. Measure μ

b. Measure ß

$$\text{Power factor} = \frac{\cos(\beta - \mu) + \cos\beta}{2}$$

2.I Kilowatts (KW)

a. Measure power factor (PF).

b. Measure OIA (peak average) (OIA).

c. Measure peak voltage of VAC (Vacpk).

$$\text{Kilowatts} = \frac{\text{Vacpk} * \text{attn ratio}}{1000 * \sqrt{2}} * \frac{\text{OIA}}{5\text{V/unit}}$$

*rated current per winding * PF.

2.II Alternative to 2 (eliminates P.F reading)

a. If VDC (NSFC) not being used, turn to min (full CCW).

b. Measure VDC

c. Measure IFB

$$\text{Kilowatts} = \frac{\text{IFB}}{5\text{V/volt}} * \text{rated current} * 1.2825 * \text{per winding}$$

$$\frac{\text{VDC} * \text{ATTN RATIO}}{1000} * 2/3$$

Eff-efficiency can be estimated at .97 for large motors See motor data sheets for accurate data).

3. Horsepower

a. Measure kilowatts

$$114 \quad \text{Horsepower} = \frac{\text{kilowatts} \cdot \text{Eff.}}{.7457}$$

4. Torque

a. Measure horsepower

b. Measure SPD (RPM)

Put oscilloscope's time base on calibrated for this entire measurement. Use as much as possible of the screen to display a half-cycle. Measure the time of this half-cycle (T).

SPD = [(1/rated frequency) * rated RPM at rated frequency)/(2*t)]

$$\text{Torque} = \frac{5250 * \text{HP}}{\text{SPD}}$$

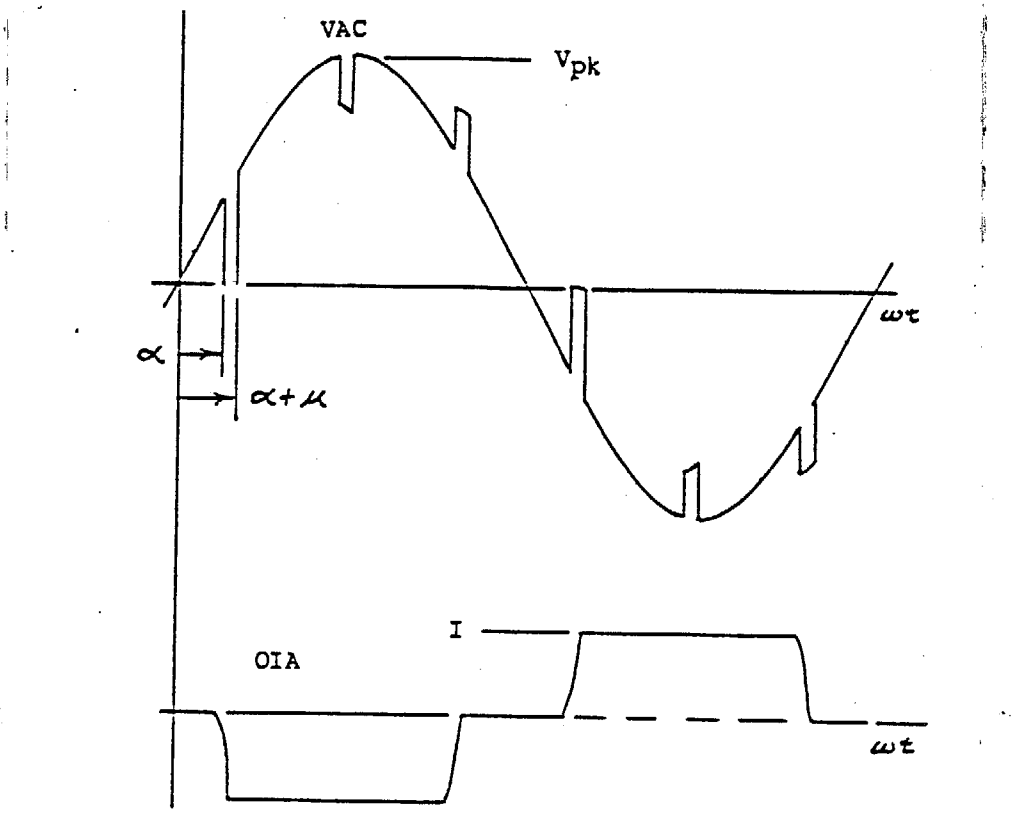


Figure 6-10. Source Side Firing Angle

CHAPTER 7

DIAGNOSTICS AND TROUBLESHOOTING

7-1. SELFTEST

7-1.1. Basic Structure. The Selftest routines are PLM86 procedures which may be executed by the HMPG resident in the Structured LCI's control Small Electronics Module (SEM) to test individual PWB's and functions. These routines are executed upon power-up if the TEST switch on the NLCB is in the up position. The function of the Selftest is to test the entire control panel for the Structured LCI, with great emphasis placed on the testing of the PWB's in the SEM.

7-1.2. Progression of Testing/Philosophy. Testing of the panel will begin by performing tests of the hardware elements within a neighborhood of the HMPG's microprocessor, and continue to tests of hardware in increasingly remote vacinities. All of the tests incorporated within the selftest may be classed into 3 groups:

1. PWB-Specific tests
2. Paired PWB tests
3. Panel-level test

7-1.2.1. PWB-SPECIFIC TESTS. A great amount of effort has been placed on performing tests that will pinpoint problems on specific PWB's first. In this manner, if any easily diagnosed problem exists, it will be clearly spelled out to the operator, instead of vaguely defined by a list of possible solutions. Tests such as these comprise the PWB-Specific test group.

7-1.2.2. PAIRED PWB TESTS. Tests which depend on the operation of several PWBs are classed within the Paired—PWB group. Diagnostic messages for failures of these tests will indicate a functional problem and also a list of possible solutions which may be somewhat general in nature.

7-1.2.3. PANEL-LEVEL TESTS. Panel-level tests also require correct operation of several PWB's or modules, however, these tests are segregated to allow for the user to perform either a module-level test (involving only the control SEM) or a complete panel-level test which would include this last category of procedures.

From a practical point of view, the sequence of testing within the control SEM will progress in the following manner:

1. HMPG
2. HLCB
3. HAIA
4. NSFC (source)
5. NSFC (load)
6. NSFE
7. NLCB
8. NLIB (source)
9. NLIB (load)
10. HISA
11. HRDA
12. HTCA
13. HXMA (Optional board)

Further tests which include PWB's outside the SEM, progress in the following sequence:

1. Cables leading from the SEM.
2. the I/O Module.
3. the Field Exciter Module.

7-1.3. Executing the Tests. The selftest routine may be initiated by resetting the HMPG with the TEST switch (on the NLCB) in the UP position. The entire test is completed in approximately 20 seconds. At its completion, status information will be printed out at the door-mounted printer. In the present Structured LCI product, the entire self test routine is executed however provisions for executing small sections of the code are in place.

Upon completion of the self test the drive will operate in the gate test mode if the drive permissives are met. To return to normal operation, the NLCB's test switch must be returned to the normal position and the HMPG must be reset.

7-1.4. Optional Testing. The drive configuration may preclude executing the complete self test. For example, if the self test were operated on one channel of a twelve pulse converter while the other was in service, false indications of activity might be received by the active channel resulting in a total drive shutdown. Likewise, the exciter package requires a load for thorough testing which may not be available while the other channel is in operation. To allow flexibility and some test capability in all conditions, some of the self test capability may be bypassed using the drive monitor. This is done as follows:

1. Set the appropriate variable with the drive monitor.
2. Save the variables.

3. Execute the self test.

Setting variable, SQETST, "F"alse disables the exciter test. Setting variable, SQRT, "F"alse disables the testing of the relay outputs. In both cases, a reminder no test message is printed when the self test is executed.

Testing of the overspeed circuit on the NLCB board may be skipped as a function of the setting of the motor frequency variable, FRQMM. This is because the test oscillator frequency if fixed at 55 Hz. When the drive motor frequency is set less than this value, an overspeed fault would result during the test which would cause a failure of the test and hardware failure message. To prevent this condition the overspeed circuit is not tested unless the motor rated frequency is above 55 Hz. Since the data link board (HXMA) is optional, its self-test may be disabled by setting the tune-up variable SQDL to FALSE.

7-1.5. Diagnosing Failures. As mentioned above, failure of any of the self test procedures will produce a diagnostic message on the door-mounted printer. The format of the diagnostic message is:

```
DATE        TIME
      FAILED

(aabb) DURING          aa = TEST NUMBER
  TEST OF:
description of test    bb = SUB-TEST
                          NUMBER

PROBLEM:
problem statement

POSSIBLE SOLUTIONS:
solution 1
solution 2
solution 3
```

Usually, a failure can easily be diagnosed by the problem statement. At least one solution will be suggested in the print out. It is suggested that troubleshooting be performed at the PWB level (Replace the PWB not its component). Although the diagnostic messages may indicate more specific failures, it is not recommended that PWB components be replaced or repaired (always write or attach the diagnosis to the card).

The TEST NUMBER and SUB-TEST NUMBER uniquely define a failure within the Selftest procedure. The TEST NUMBER is incremented from 1 to 31 (hex) as the tests progress. For different test numbers, the SUB-TEST parameter will vary form 1 up to the number of sub-tests within that major test. The Description of Tests section that follows details the complete Selftest procedure according to the TEST NUMBER and SUB-TEST NUMBER parameters.

7-1.6. DESCRIPTION OF TESTS. The following lists all tests available with sub-test information:

TEST DESCRIPTION

0. HMPGROMSTST: Reads all of ROM memory (F000:0 to F000:FFFF). Possible failures are:
   .0 - BusHog error
   .1 - Read error

1. HMPGSRAMSTST: Reads all of RAM memory (0:0 to 0:3FFF). Possible failures are:
   .0 - BusHog error
   .1 - Read error
   .2 - Write error

2. HMPGINTRSTST
   .1 - Verifies control and data lines to HMPGs 8259

3. HMPG_TMR
   .1 - Verifies addr lines & data bits 6 & 7
   .2 - Verifies addr lines & data bits 6 & 7
   .3 - Verifies addr lines & data bits 6 & 7
   .4 -Verifies data bits 5-0 with walking 1 pattern
   .5 - Verifies that all outputs are free to toggle.
   .6 - FOR TMR0: Checks for null count = 1
   .7 - FOR TMR0: Loads TMR; checks for null count = 0
   .8 - FOR TMR0: Verify TMR time out
   .9 - FOR TMR0: Verify TMR reloads
   .A - FOR TMR1: Checks for null count = 1
   .B - FOR TMR1: Loads TMR; checks for null count = 0
   .C - FOR TMR1: Verify TMR time out
   .D - FOR TMR1: Verify TMR reloads
   .E - FOR TMR2: Checks for null count = 1
   .F - FOR TMR2: Loads TMR; checks for null count = 0
   .10 - FOR TMR2: Verify TMR time out
   .11 - FOR TMR2: Verify TMR reloads

4. HMPG_SIO_PRES
   .1 - Check that trans and rec buffers are empty

5. HMPG_STALL
   .1 - Verifies software reset of Stall circuit.
   .2 - Initiate stall timer; checks for no STALL after 100

.3 - Verifies STALL asserted after 1.5 seconds.

6. HMPG_INTR
   - .1 - Verify operation of BusHog interrupt.
   - .2 - Verify that Bus Hog does not occur for legitimate read.
   - .3 - Verify operation of Time Tic interrupt.

7. DMPB
   - .1 - Checks that Write Enable Switch is not asserted.
   - .2 - Checks for no BusHog during EEPROM reads.
   - .3 - Checks for expected pattern in EE-PROM.

8. HLCB_PRES
   - .1 - Enables IMOK; checks for 0IMOK=0.
   - .2 - Disables IMOK; checks for 0IMOK=1.

9. HLCB_DAK
   - .1 - Checks for no BusHog during reads of HLCB.

A. HLCB_TMR_U22
   - .1 - Verifies addr lines & data bits 6 & 7
   - .2 - Verifies addr lines & data bits 6 & 7
   - .3 - Verifies addr lines & data bits 6 & 7
   - .4 - Verifies data bits 5-0 with walking 1 pattern
   - .5 - Verifies that all outputs are free to toggle.
   - .6 - FOR TMR0: Checks for null count = 1
   - .7 - FOR TMR0: Loads TMR; checks for null count = 0
   - .8 - FOR TMR0: Verify TMR time out
   - .9 - FOR TMR0: Verify TMR reloads
   - .A - FOR TMR1: Checks for null count = 1
   - .B - FOR TMR1: Loads TMR; checks for null count = 0
   - .C - FOR TMR1: Verify TMR time out
   - .D - FOR TMR1: Verify TMR reloads
   - .E - FOR TMR2: Checks for null count = 1
   - .F - FOR TMR2: Loads TMR; checks for null count = 0
   - .10 - FOR TMR2: Verify TMR time out
   - .11 - FOR TMR2: Verify TMR reloads

B. HLCB_TMR_U23
   - .1 - Verifies addr lines & data bits 6 & 7
   - .2 - Verifies addr lines & data bits 6 & 7
   - .3 - Verifies addr lines & data bits 6 & 7
   - .4 - Verifies data bits 5-0 with walking 1 pattern
   - .5 - Verifies that all outputs are free to toggle.
   - . 6 - FOR TMR0: Checks for null count = 1
   - .7 - FOR TMR0: Loads TMR; checks for null count = 0
   - .8 - FOR TMR0: Verify TMR time out
   - .9 - FOR TMR0: Verify TMR reloads
   - .A - FOR TMR1: Checks for null count = 1
   - .B - FOR TMR1: Loads TMR; checks for null count = 0
   - .C - FOR TMR1: Verify TMR time out
   - .D - FOR TMR1: Verify TMR reloads
   - .E - FOR TMR2: Checks for null count = 1
   - .F - FOR TMR2: Loads TMR; checks for null count = 0
   - .10 - FOR TMR2: Verify TMR time out
   - .11 - FOR TMR2: Verify TMR reloads
   - .12 - FOR TMR2: Initialize to Mode 0; verify status.
   - .13 - FOR TMR2: Load with 10 ms delay; verify status.
   - .14 - FOR TMR2: After 4 ms, verify status.
   - .15 - FOR TMR2: After 15 ms, verify time out.
   - .16 - FOR TMR2: Check that count = 0.

C. HLCB_INTR_PRES
   - .1 - Verify addr and data lines to HLCBs 8259.

D. HLCB_SIO
   - .1 - Initialize and check status of HLCBs 8251.
   - .2 - Verify that single byte of data is send, by sensing status.
   - .3 - Verify transfer of data pattern.

E HAIA_PRES
   - .1 - Checks that HAIA is present by reading single bit.

F. HAIA_DAK
   - .1 - Checks for no BusHog during reads of HAIA.
   - .2 - Checks for no BusHog during writes to HAIA.

10. HAIA_TMR
    - .1 - Verifies addr lines & data bits 6 & 7
    - .2 - Verifies addr lines & data bits 6 & 7
    - .3 - Verifies addr lines & data bits 6 & 7
    - .4 - Verifies data bits 5-0 with walking 1 pattern
    - .5 - Verifies that all outputs are free to toggle.
    - .6 - FOR TMR0: Checks for null count = 1
    - .7 - FOR TMR0: Loads TMR; checks for null count = 0
    - .8 - FOR TMR0: Verify TMR time out

.9 - FOR TMR0: Verify TMR reloads
.A - FOR TMR1: Checks for null count = 1
.B - FOR TMR1: Loads TMR; checks for null count = 0
.C - FOR TMR1: Verify TMR time out
.D - FOR TMR1: Verify TMR reloads
.E - FOR TMR2: Checks for null count = 1
.F - FOR TMR2: Loads TMR; checks for null count = 0
.10 - FOR TMR2: Verify TMR time out
.11 - FOR TMR2: Verify TMR reloads

11. HAIA_CONVERSION
.1 - Start A/D conversion; check that BUSY=1.
.2 - Check that 1DAVAL = 0.
.3 - Wait for conversion to complete; check that BUSY=0.
.4 - Check that 1DAVAL = 1.

12. HAIA_PSUPPLY
.1 - Check that N15 supply is read correctly.
.2 - Check that P15 supply is read correctly.

13. NSFC_S_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

14. NSFC_S_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 -Verifies data bits 5-0 with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.

15. NSFC_S_TMODE
.1 - FOR TMR0: Checks for null count = 1
.2 - FOR TMR0: Loads TMR; checks for null count = 0
.3 - FOR TMR0: Verify TMR time out
.4 - FOR TMR0: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

16. NSFC_L_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

17. NSFC_L_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 - Verifies data bits 5-0 with walking 1 pattern
.5 - Verifies that all outputs are free to toggle.

18. NSFC_L_TMODE
.1 - FOR TMR0: Checks for null count = 1
.2 - FOR TMR0: Loads TMR; checks for null count = 0
.3 - FOR TMR0: Verify TMR time out
.4 - FOR TMR0: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same.
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

19. NSFE_PRES
.1 - Set IMOK true; read back IMOK signal.
.2 - Set IMOK false; read back IMOK signal.

1A NSFE_TMR_PRES
.1 - Verifies addr lines & data bits 6 & 7
.2 - Verifies addr lines & data bits 6 & 7
.3 - Verifies addr lines & data bits 6 & 7
.4 -Verifies data bits 5-0 with walking 1 pattern
.5 - Verifies that all outputs are free to

toggle.

1B NSFE_TMODE
.1 - FOR TMR0: Checks for null count = 1
.2 - FOR TMR0: Loads TMR; checks for null count = 0
.3 - FOR TMR0: Verify TMR time out
.4 - FOR TMR0: Verify TMR reloads
.5 - FOR TMR1: Checks for null count = 1
.6 - FOR TMR1: Loads TMR; checks for null count = 0
.7 - FOR TMR1: Verify TMR time out
.8 - FOR TMR1: Verify TMR reloads
.9 - Perform operations to set pulse train to 0; verify.
.A - In TEST Mode, write all on pattern, expect echo of same.
.B - In TEST Mode, walk 1 thru primary & firing latch.
.C - In TEST Mode, see if pulse train is stuck low.
.D - In TEST Mode, see if pulse train is stuck high.
.E - In TEST Mode, check that pulse train is toggling.

1C NSFE_DIAG
.1 - Tests for dormant field status information.

1D NSFC_S_PLL
.1 - FOR TMR2: Enables PLL; Checks for null count = 1
.2 - FOR TMR2: Loads TMR; checks for null count = 0
.3 - FOR TMR2: Verify TMR time out
.4 - FOR TMR2: Verify TMR reloads

1E NSFC_L_PLL
.1 - FOR TMR2: Enables PLL; Checks for null count = 1
.2 - FOR TMR2: Loads TMR; checks for null count = 0
.3 - FOR TMR2: Verify TMR time out
.4 - FOR TMR2: Verify TMR reloads

1F NSFE_PLL
.1 - FOR TMR2: Enables PLL; Checks for null count = 1
.2 - FOR TMR2: Loads TMR; checks for null count = 0
.3 - FOR TMR2: Verify TMR time out
.4 - FOR TMR2: Verify TMR reloads

20. ROLL_CALL
.1 - Test for ribbon cable connecting HAIA(JA) to SEM(JG).
.2 - Enable NLCB IMOK; check for IMOK true.
.3 - Disable NLCB IMOK; check for IMOK false.
.4 - Enable NLIBS IMOK; check for IMOK true.
.5 - Disable NLIBS IMOK; check for IMOK false.
.6 - Enable NLIBL IMOK; check for IMOK true.
.7 - Disable NLIBL IMOK; check for IMOK false.

21. NLCB_ANALOG_DORMANT
.1 - Test for 0 volts at REFO.
.2 - Test for 0 volts at TREFO.
.3 - Test for 0 volts at 12PREFO.
.4 - Inject current in REFO circuit; test for change in output.
.5 - Inject current in TREFO circuit; test for change in output.
.6 - Inject current in 12PREFO circuit; test for change in output.
.7 - Test for 0 volts at SP (speed).
.8 - Reset existing faults; test for 1OSP (overspeed) = 0.

22. NSFE_ANALOG
.1 - Test for EI (exciter current) = 0 volts.
.2 - Test for no current imbalance
.3 - Inject signal in current circuit; test for output change.
.4 - Again, test for no current imbalance.
.5 - Excite current circuit with 55 Hz inputs; verify that current imbalance is detected.
.6 - Wait 100 ms; verify no current imbalance condition.

23. HLCB_SQUARE_WAVES
.1 - Source side square wave tests.
.2 - Load side square wave tests.

24. NLIB_S_ANALOG
.1 - Reset existing faults; verify UV & not OV condition.
.2 - Verify not SUPPression condition.
.3 - Verify 1UV70=0.

.4 - Test C17 for time delay. (Overvoltage circuit)
.5 - Test C17 for time delay. (Overvoltage circuit)
.6 - Test for 0 volts at IFBS.
.7 - Test for 0 volts at FLUX BAS.
.8 - Inject current in source current condition circuit. Detect change in IFBS.
.9 - Detect change in FLUX BAS.

.A - For FLUX BAS: Check for 0 volts dc offset.
.B - For FLUX BAS: Check level of positive peak.
.C - For FLUX BAS: Check level of negative peak.

25. NLIB_L_ANALOG
.1 - Reset existing faults; verify UV & not OV condition.
.2 - Verify not SUPPression condition.
.3 - Verify 1UV70 = 0.
.4 - Test C17 for time delay. (Overvoltage circuit)
.5 - Test C17 for time delay. (Overvoltage circuit)
.6 - Test for 0 volts at IFBL.
.7 - Test for 0 volts at FLUX BAL.
.8 - Inject current in source current condition circuit.Detect change in IFBL.
.9 - Detect change in FLUX BAL.
.A - For FLUX BAS: Check for 0 volts dc offset.
.B - For FLUX BAS: Check level of positive peak.
.C - For FLUX BAS: Check level of negative peak.
.D -For FLUX CBS: Check for 0 volts dc offset.
.E - For FLUX CBS: Check level of positive peak.
.F - For FLUX CBS: Check level of negative peak.
.10 -For FLUX ACS: Check for 0 volts dc offset.
.11 - For FLUX ACS: Check level of positive peak.
.12 - For FLUX ACS: Check level of negative peak.

26. HLCB_INTR
.1 - Enable test oscillator; verify occurrence of source zero crossing interrupt (slave interrupt 0).
.2 - Verify frequency of source zero crossing interrupt.
.3 - Enable test oscillator; verify occurrence of load zero crossing interrupt (slave intr. 1).
.4 - Verify frequency of load zero crossing intr.
.5 - Verify occurrence of source firing interrupt.
.6 - Verify occurrence of load firing interrupt.
.7 - Verify occurrence of field firing interrupt.

27. NLCB_ANALOG_ACTIVE
.1 - Enable test oscillator; verify speed circuits reads 55 Hz.
.2 - Verify no overspeed condition.

28. HISA_PRES
.1 - Disable IMOK; verify by reading back bit.
.2 - Enable IMOK; verify by reading back bit.

29. CABLE_CONT
.1 - Tests for continuity in ribbon cable from SEM, JA to I/O Module.
.2 - Tests for continuity in ribbon cable from SEM, JK to I/O Module.
.3 - Tests for continuity in ribbon cable from SEM, JH to I/O Module.
.4 - Tests for continuity in ribbon cable from SEM, JB to Door Module.

2A. HISA_IO
.1 - Set all inputs from I/O Module high; verify by reading back inputs to HISA.
.2 - Set all inputs from I/O Module low; verify by reading back inputs to HISA.

2B P105_SUPPLY_CHECK
.1 - Check that 105 volt dc power supply is OK.

2C HROA_1A_IO:
In sequence, pick-up single relays on the HROA in slot 1A of the I/O module. Expect feedback signal to indicate change.
.1 -Force all HROA(1A) relays to drop-out; verify by feedback.
.2 -Force 105 Vdc relays to pick-up; verify by feedback.
.3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.

.4 -Pick-up single 28 V relay; verify by feedback.
.5 -Drop-out all relays; verify by feedback.
.6 -Pick-up single 28 V relay; verify by feedback.
.7 -Drop-out all relays; verify by feedback.
.8 -Pick-up single 28 V relay; verify by feedback.
.9 -Drop-out all relays; verify by feedback.
.A -Pick-up single 28 V relay; verify by feedback.
.B -Drop-out all relays; verify by feedback.

2D HROA_1B_IO:
In sequence, pick-up single relays on t he HROA in slot 1B of the I/O module. Expect feedback signal to indicate change.
.1 -Force all HROA(1A) relays to drop-out; verify by feedback.
.2 -Force 105 Vdc relays to pick-up; verify by feedback.
.3 -Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
.4 -Pick-up single 28 V relay; verify by feedback.
.5 -Drop-out all relays; verify by feedback.
.6 -Pick-up single 28 V relay; verify by feedback.
.7 -Drop-out all relays; verify by feedback.
.8 -Pick-up single 28 V relay; verify by feedback.
.9 -Drop-out all relays; verify by feedback.
.A -Pick-up single 28 V relay; verify by feedback.
.B -Drop-out all relays; verify by feedback.
.C -Pick-up single 28 V relay; verify by feedback.
.D -Drop-out all relays; verify by feedback.
.E -Pick-up single 28 V relay; verify by feedback.
.F -Drop-out all relays; verify by feedback.

2E NLCB_R_DVR
.1 - Verify presence of HROA in slot 1B of I/O Module.
.2 - Verify presence of HROA in slot 1A of I/O Module.
.3 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.
.4 - Force relay drivers on NLCB on; verify response from I/O Module via feedback.

2F DOOR_MOD_PRES
.1 - Test for CTS asserted.
.2 - Test for DSR asserted.

30. FLD_EXC_MOD_ATTEN
.1 - Check for no absence of ac on the SFE Module, via the cell state sensors.
.2 - Verify presence of ac on phase A, via the cell state sensors.
.3 - Verify presence of ac on phase B, via the cell state sensors.
.4 - Verify presence of ac on phase C, via the cell state sensors.
.5 - Check level of the ac input voltage.
.6 - Check level of the ac input voltage.
.7 - Check dc offset of the ac input voltage.
.8 - Check that exciter output voltage is equal to 0.
.9 - Check that exciter output voltage is equal to 0.
.A - Check that exciter current is equal to 0.
.B - Check that exciter current is equal to 0.
.C - While sampling cell state sensor patterns, verify that cell state sensors are operational.
.D - Verify correct cell state pattern is in sampled data.
.E - Verify phase sequence of SFE Module input voltage by finding correct second cell state pattern in sampled data.
.F - Verify phase sequence of SFE Module input voltage by finding correct third cell state pattern in sampled data. (Completion of phase reversal check.)

31. FLD_EXC_MOD_1PHASE
** Single phase operation of SFE module, A-B. **
.1 - Waiting for negative level of field exciter sync signal.
.2 - Waiting for positive edge of field exciter sync signal.
.3 - After delay, fire correct SCR pair; verify via cell state sensors.
.4 - Waiting for positive level of field exciter sync signal.
.5 - Waiting for negative edge of field exciter sync signal.
.6 - After delay, fire correct SCR pair; verify via cell state sensors.
.7 - After 5 cycles, verify that exciter current < 0.5 per unit.
** Single phase operation of SFE module, B-C. **

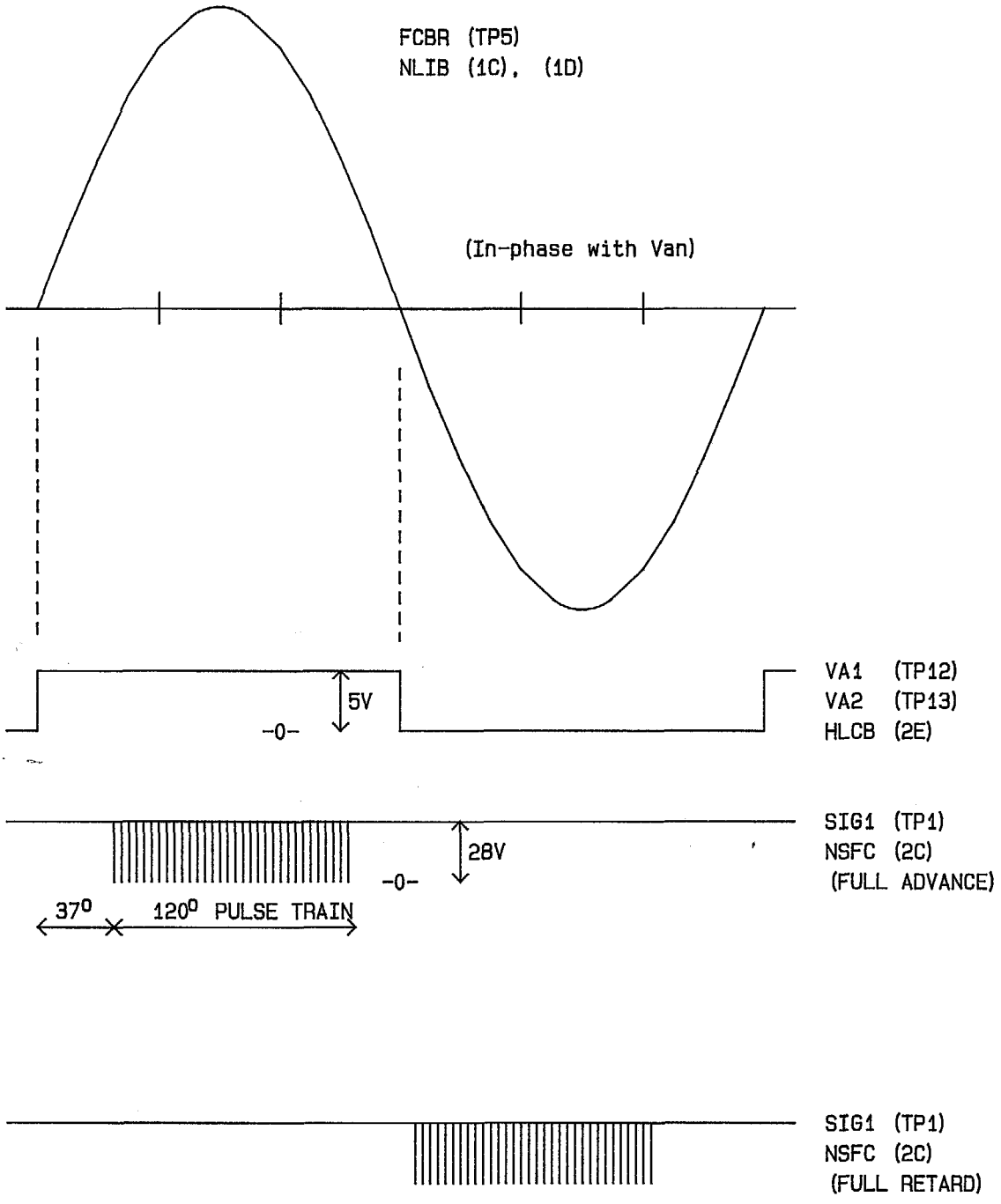


Figure 7B-1. Firing Angle Limits

.8 - Waiting for negative level of field exciter sync signal.
.9 - Waiting for positive edge of field exciter sync signal.
.A - After delay, fire correct SCR pair; verify via cell state sensors.
.B - Waiting for positive level of field exciter sync signal.
.C - Waiting for negative edge of field exciter sync signal.
.D - After delay, fire correct SCR pair; verify via cell state sensors.
.E - After 5 cycles, verify that exciter current < 0.5 per unit.

** Single phase operation of SFE module, C-A. **

.F - Waiting for negative level of field exciter sync signal.
.10 - Waiting for positive edge of field exciter sync signal.
.11 - After delay, fire correct SCR pair; verify via cell state sensors.
.12 - Waiting for positive level of field exciter sync signal.
.13 - Waiting for negative edge of field exciter sync signal.
.14 - After delay, fire correct SCR pair; verify via cell state sensors.
.15 - After 5 cycles, verify that exciter current < 0.5 per unit.

32. HTCA_PRES
.1 - HTCA presence tests

33. HTCA_8254_TST
.1 HTCA 8254 timer tests

34. HTCA_TACH_COUNT_TST
.1 HTCA position counter test

35. HTCA_PS12_TST
.1 - HTCA P12 power supply test

36. HXMA_PRES
.1 HTMA presence tests

37. HTCA_CABLE_TST
.1 HTCA cable continuity test

7-2. SECTION B - GATE-TEST-MODE

7-2.1. Gate-Test-Mode. The gate test mode allows off-line testing of much of the analog signal conditioning, microprocessor firing control, and gate pulse generation circuitry. In the gate-test-mode, a local test oscillator supplies 3 phase, approximately 60 Hz signals in place of the attenuated bus signals from the source and load bridges. The electronic control can then lock onto these signals and generate signals to "dry fire" the thyristors in both bridges. Thyristor gate pulses may then be examined for proper magnitude and phasing. Firing may be adjusted between advance and retard limit for each bridge. In the gate-test-mode, the control produces pulse train firing, thus it is necessary to run the converter ventilating fans for this test.To perform the following tests, a dual channel oscilloscope with voltage probes and a current probe is required.

7-2.2. Getting Into Gate-Test-Mode.

A. The drive should be stopped, primary switchgear and/or transformer isolation switch (if any), and load isolation device (if any) should be open. Check the software ladder diagram in the system elementary for any other conditions that must be met.

B. Put the NLCB (1E) "TEST" switch (SW1) in the test position.

C. Reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E or 2J).

*** NOTE ***

Always use dc coupling on the oscilloscope channels.

7-2.3. Gate-Test-Mode Checkout.

A. Connect channel one of the oscilloscope to NLIB (1C) "FCBR" (TP5) and channel two of HLCB (2E) "VA1" (TP12). Trigger the scope on channel one and verify the waveforms look like Figure 7B-1.

B. Connect channel one of the oscilloscope to NLIB (1D) "FCBR" (TP5) and channel two to HLCB (2E) (TP13). Trigger the scope on channel one and verify the waveforms look like Figure 7B-1.

C. Connect channel one of the oscilloscope to HLCB (2E) "VA1" (TP12) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e, 60 degrees per major division or 12 degrees per minor division).

D. One at a time, connect channel two to SIG1 thru SIG6 on the source side "A" NSFC (2C) (TP1 thru TP6) and verify that the gate pulses are positioned

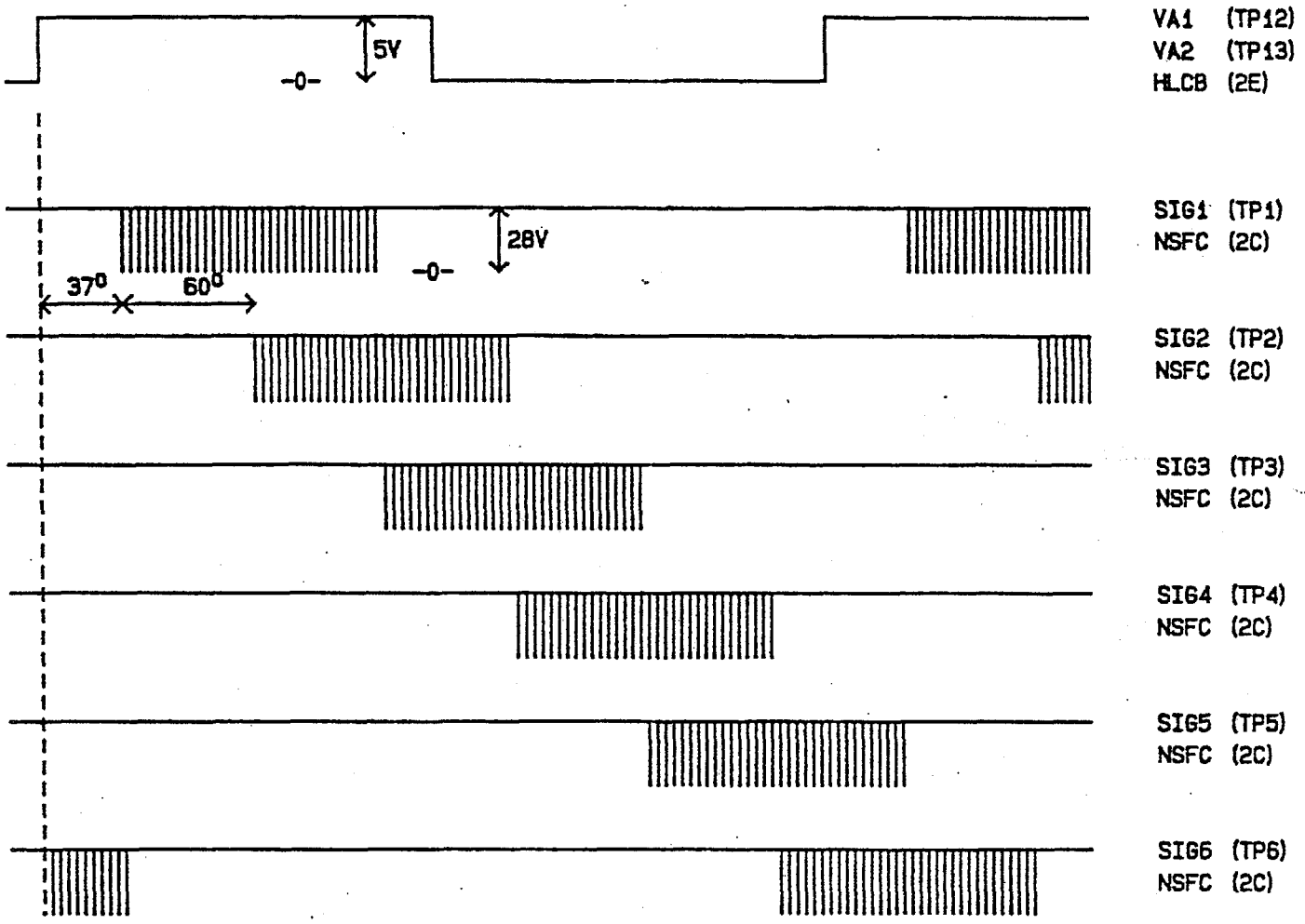


Figure 7B-2A. Source Firing - Full Advance

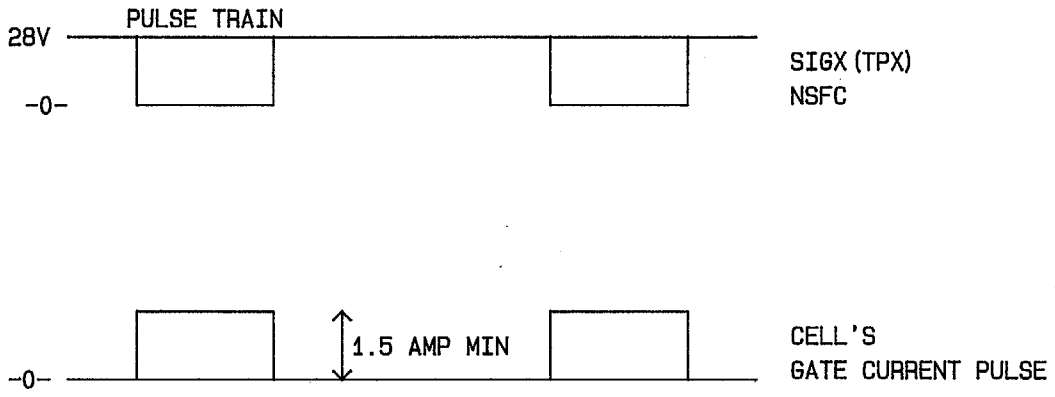


Figure 7B-4A. Typical Gate Pulse Train

as in Figure 7B-2A.
Note: The firing angle can be adjusted by varying the speed demand reference (place in "LOCAL" and adjust "LOCAL REF").

E. Connect channel one of the oscilloscope to HLCB (2E) "VA2" (TP13) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e., 60 degrees per major division or 12 degrees per minor division).

Connect a current probe to the scope and trigger on this channel. Hook the current probe to each of the gate leads (white wire) of the SCRs with the arrow on the current probe pointing away from the gate lead termination point. Verify that the waveform has the same characteristics as Figure 7B-4B. If the pulse fails the criterion, the HPTK card in that leg cell module is most likely the cause, or the NTDA connections may be reversed.

F. One at a time, connect channel two to SIG1 thru SIG6 on the load side NSFC (2B) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.

G. Verify that each of the red gate pulse LEDS on the HPTK cards in the leg cell modules is illuminated indicating that the cell is getting gate pulses.

H. With a current probe, look at the gate pulse of each thyristor while simultaneously monitoring the corresponding gate pulses on the NSFC card. The two signals should line up with each other as shown in Figure 7B-4A.

I. Put the NLCB (2E) "TEST" switch (SW1) in the OFF position and reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB(2E).

Table 7B-1

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR	CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2C)	STM1	SIG1 (TP1, 2B)	LTM1
SIG1(TP2, 2C)	STM2	SIG1 (TP2, 2B)	LTM2
SIG1(TP3, 2C)	STM3	SIG1 (TP3, 2B)	LTM3
SIG1(TP4, 2C)	STM4	SIG1 (TP4, 2B)	LTM4
SIG1(TP5, 2C)	STM5	SIG1 (TP5, 2B)	LTM5
SIG1(TP6, 2C)	STM6	SIG1 (TP6, 2B)	LTM6

** *NOTE* **

For silco bridge, HPTF will be used instead of HPTK.

7-3. SECTION D - TERMINALS

7-3.1. Serial Monitor and Hand Terminal (or other Diagnostic Terminal). Integrated into the firmware for both source and load electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of each microcomputer card. Communication with the microcomputer can be established by connecting the Silent 700 terminal supplied with the drive to the RS-232 port. This connection is made using the HNMB null modem (supplied). See Figure 7D2.

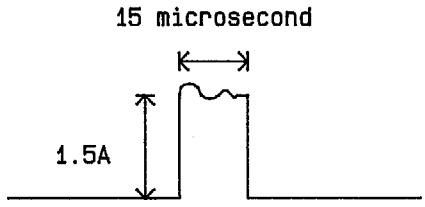


Figure 7B-4B. Typical Firing Pulse (Cell Gate Current Pulse)

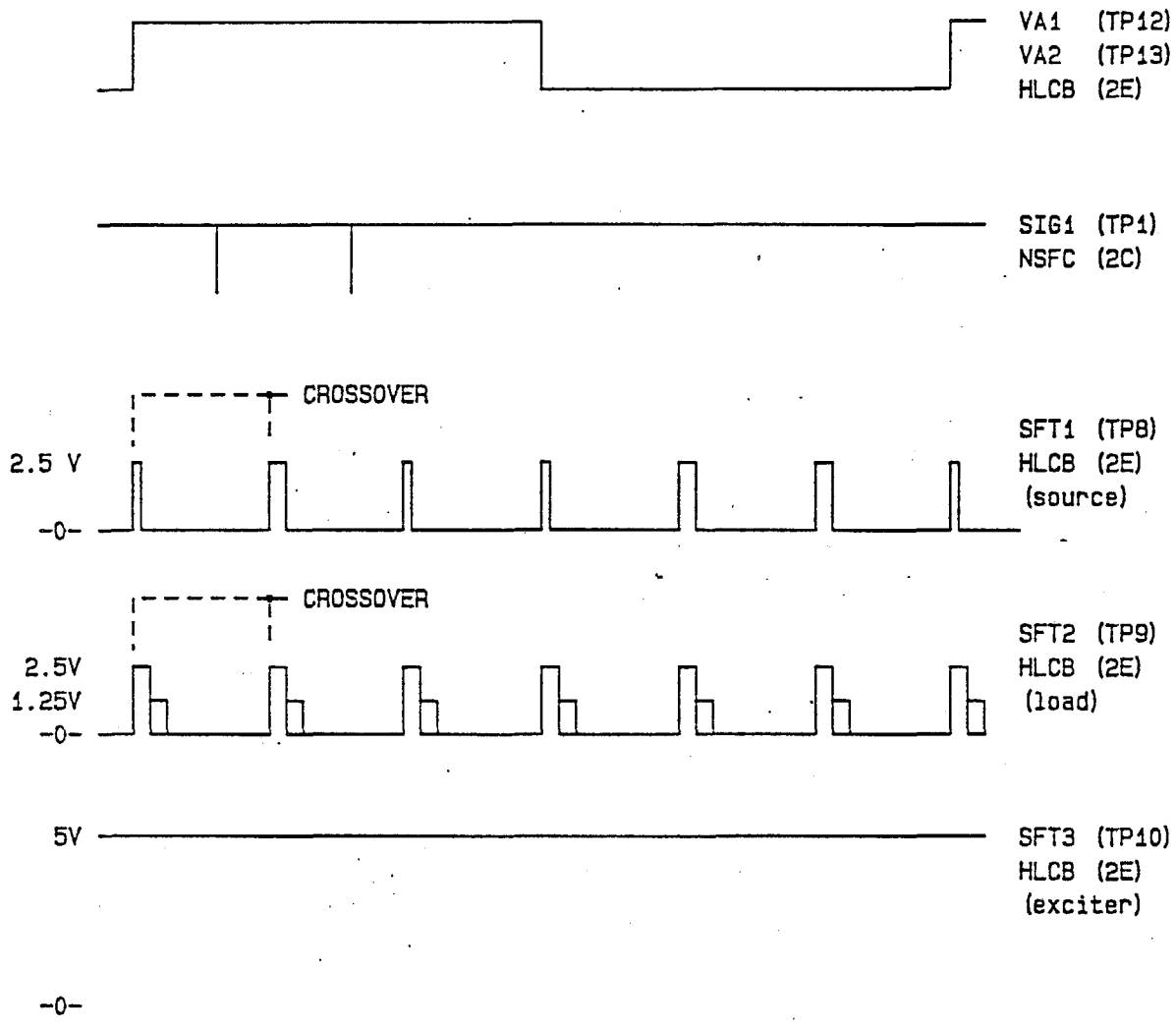


Figure 7B-6. Software Interrupts - Full Advance (Source)

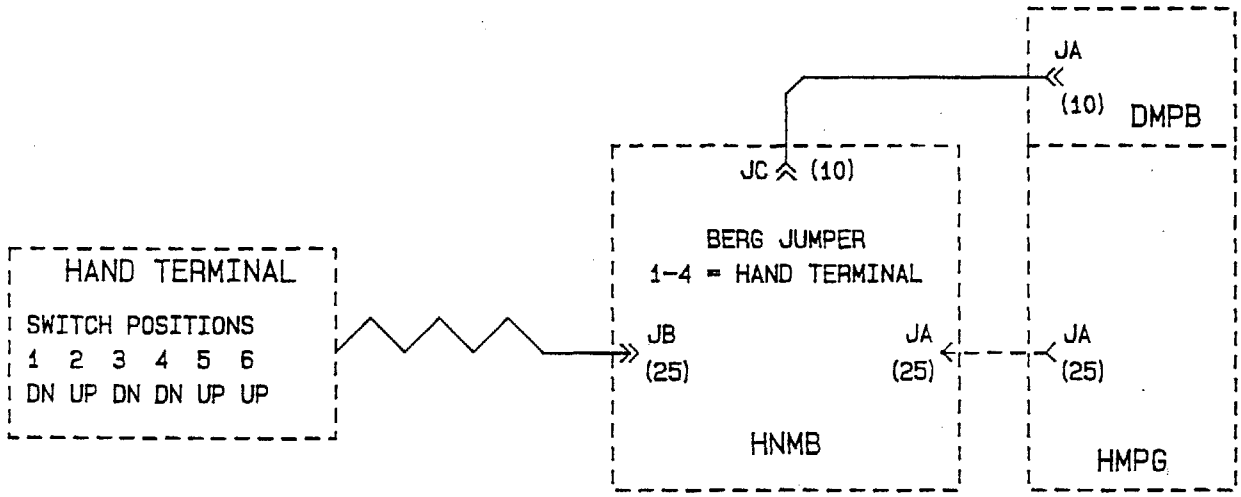


Figure 7D-1. Hand Terminal Connection

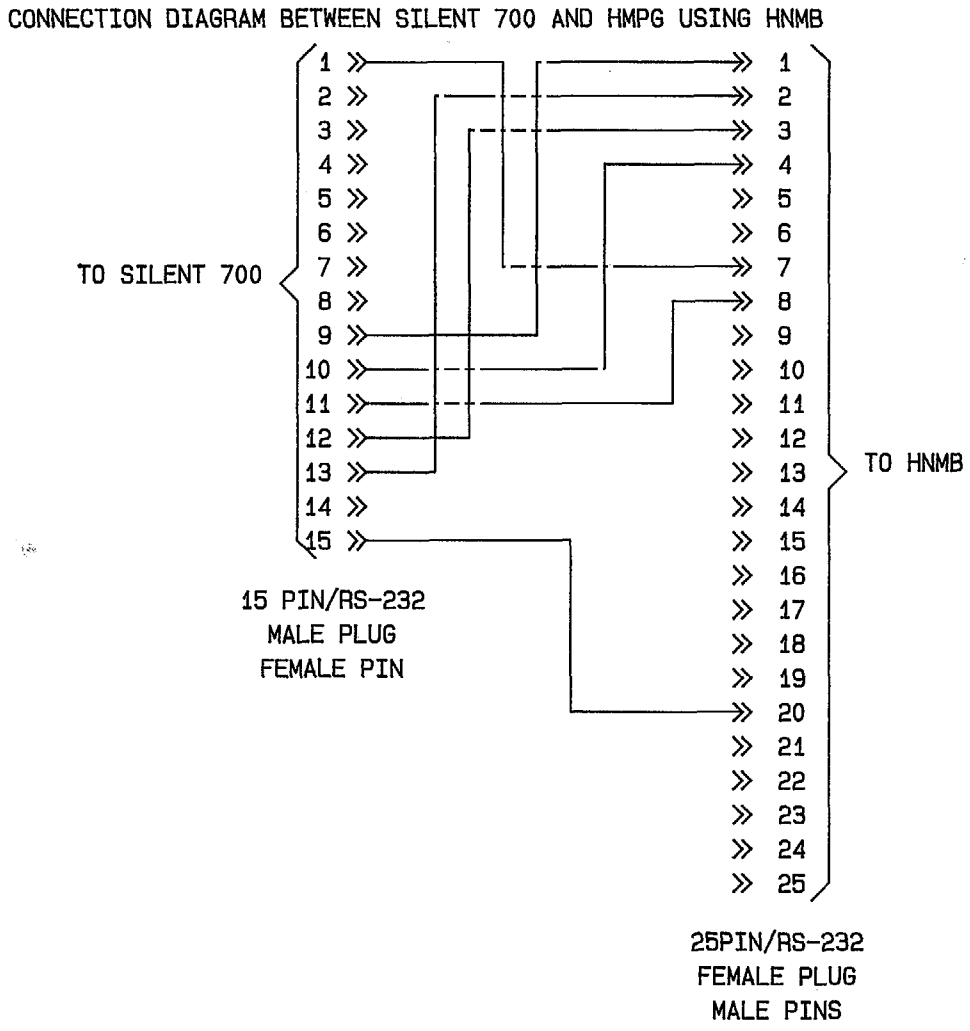

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG USING HNMB
TO SILENT 700
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
TO HNMB
25PIN/RS-232
FEMALE PLUG
MALE PINS

Figure 7D-2.

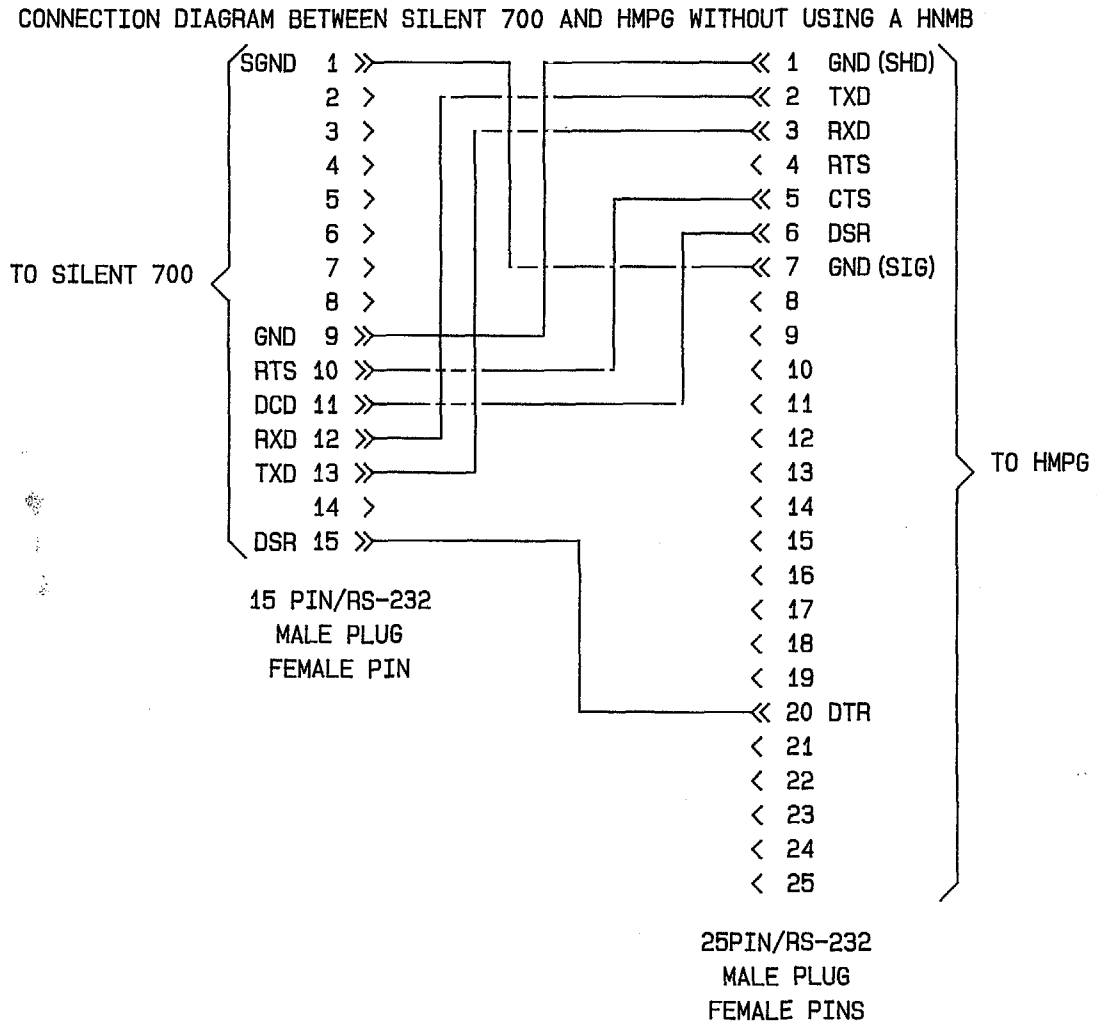

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-3.

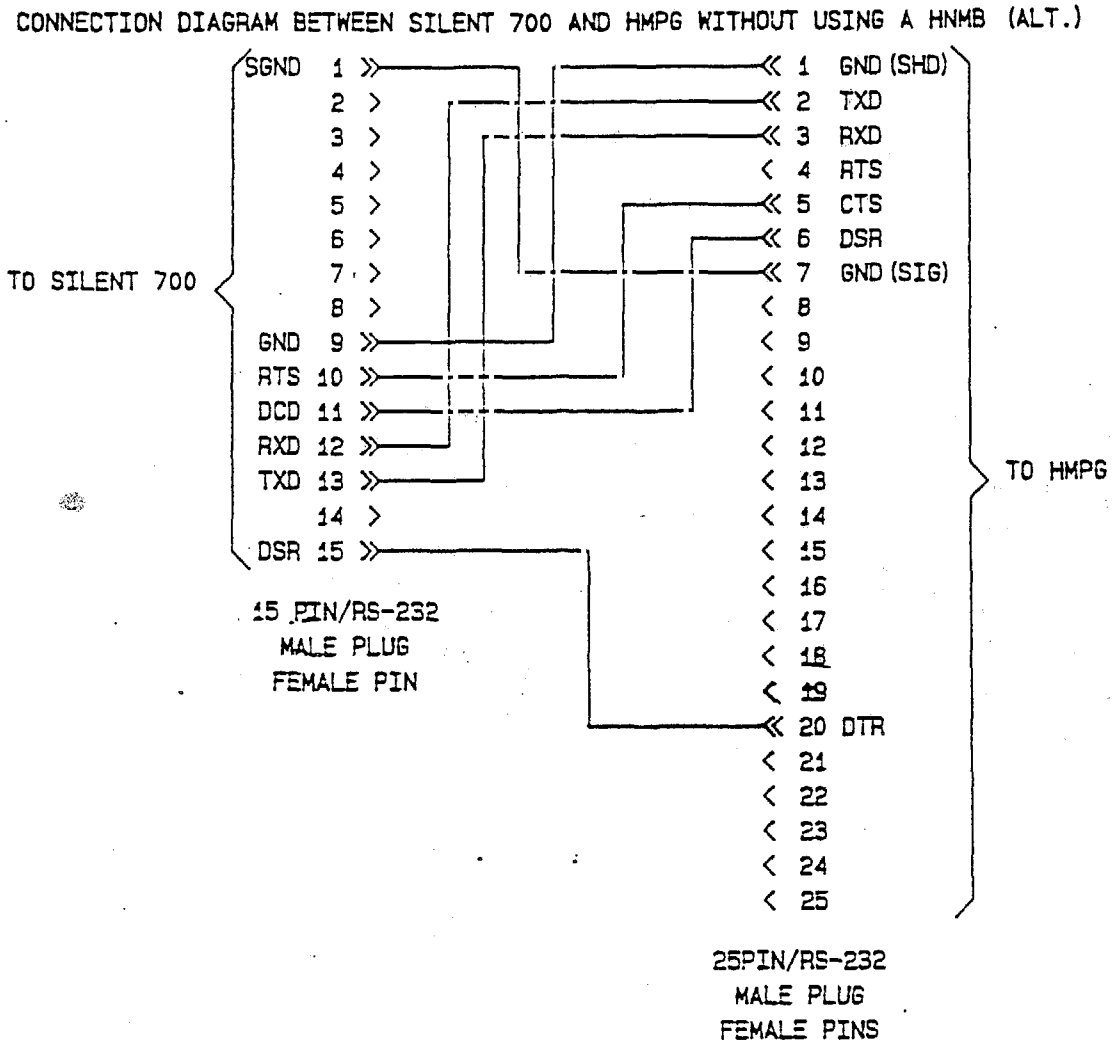

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB (ALT.)
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-4.